

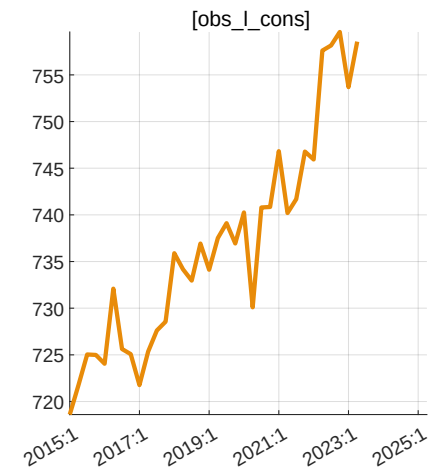
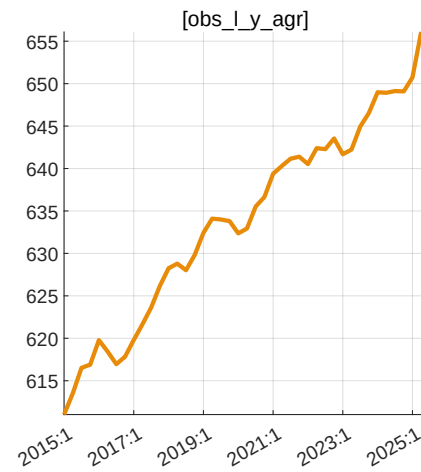
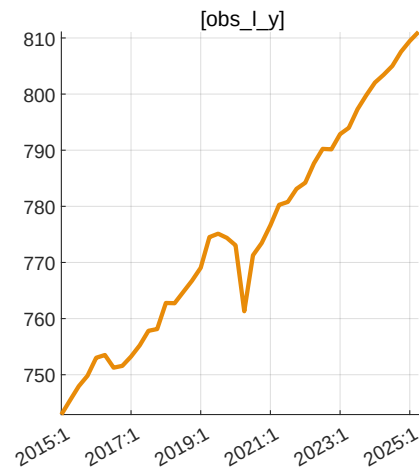
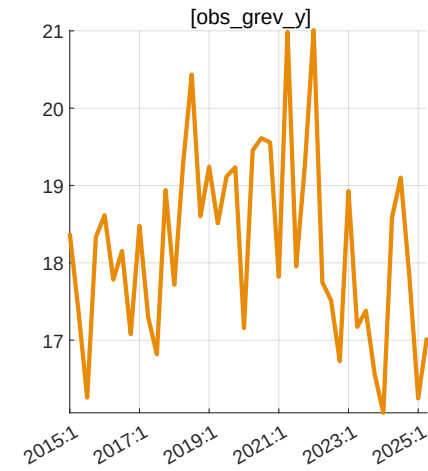
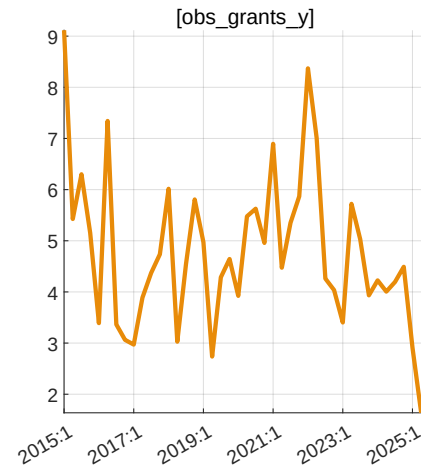
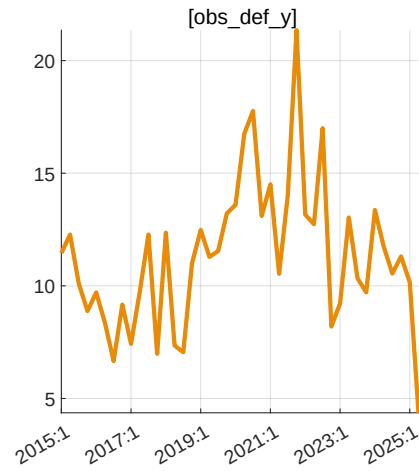
## Observed and filtered data

Round: 2025 Sept Forecast, time: 18-Nov-2025 19:59:54.

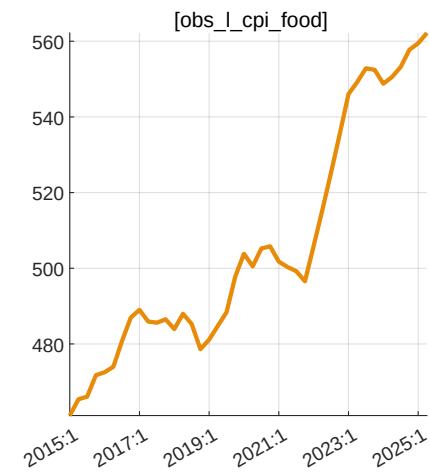
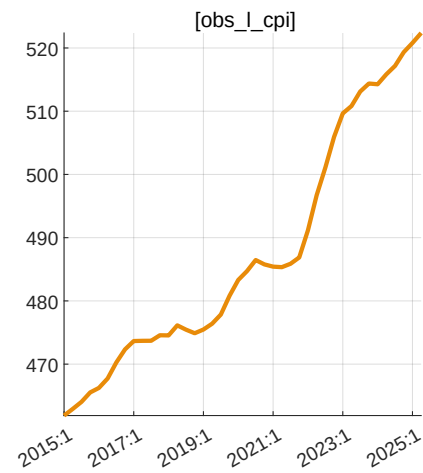
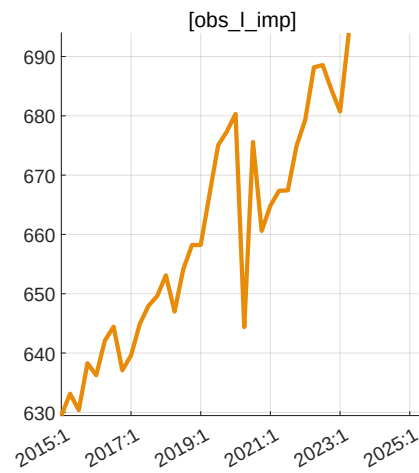
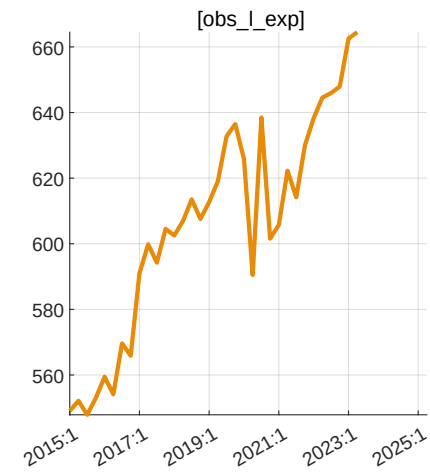
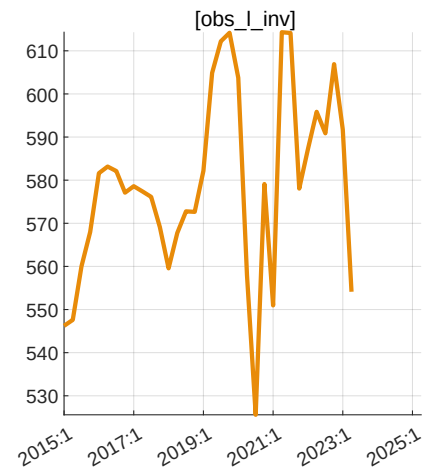
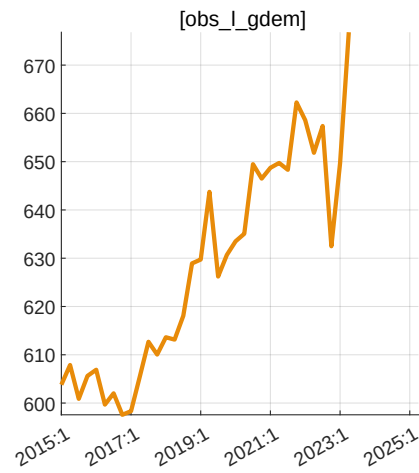
## 1 Observed data



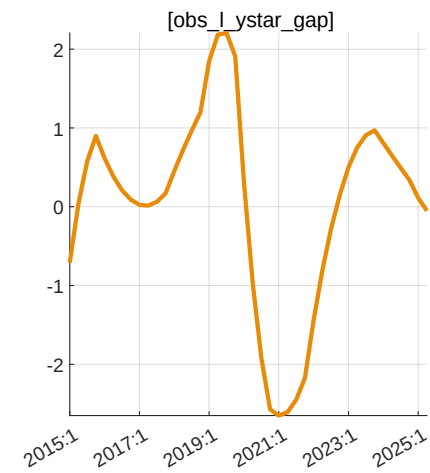
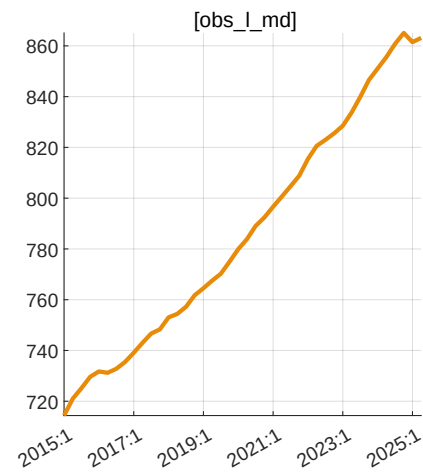
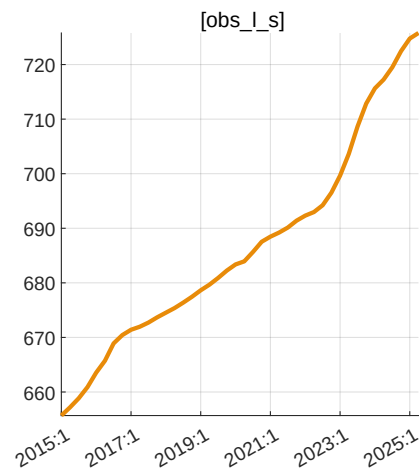
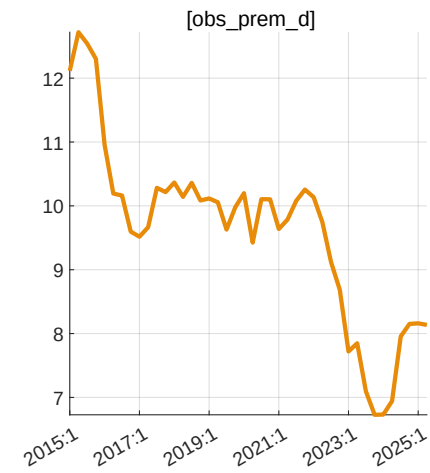
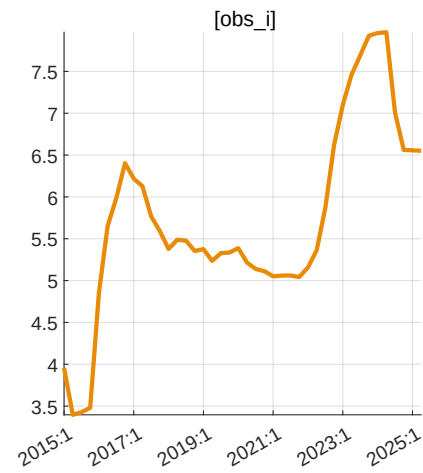
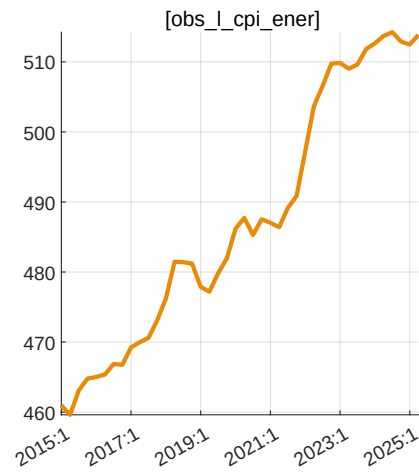
## Observed data (page 1)



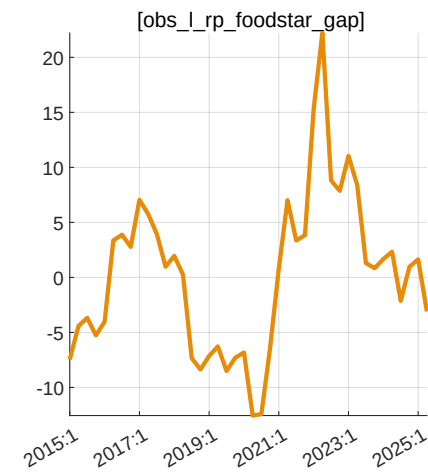
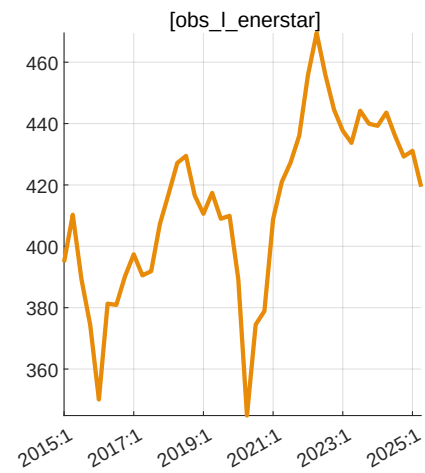
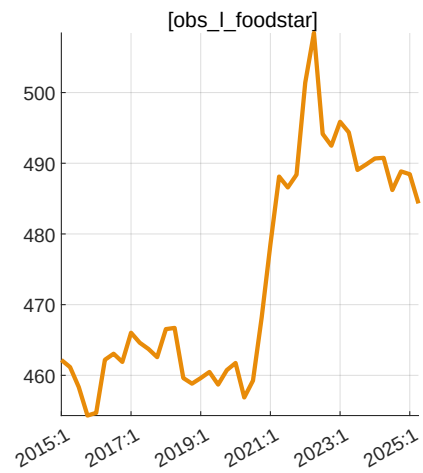
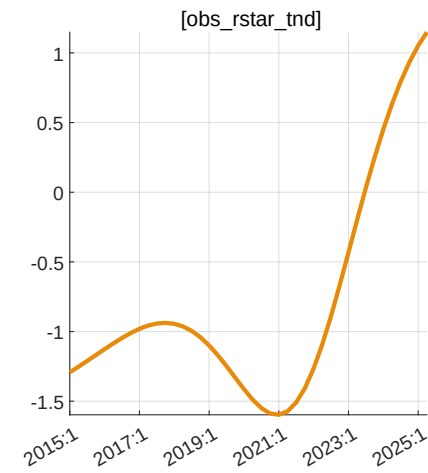
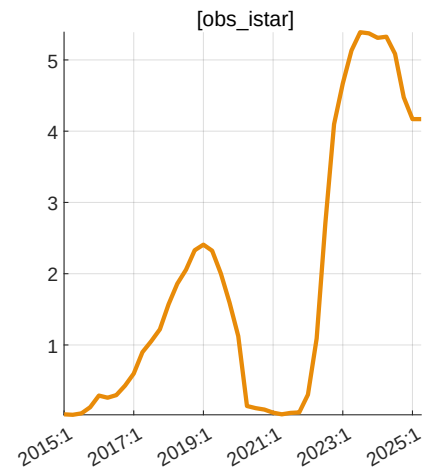
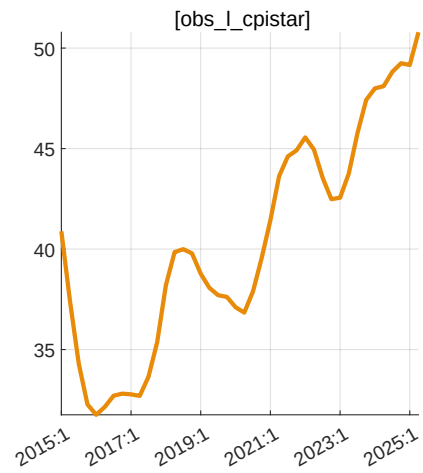
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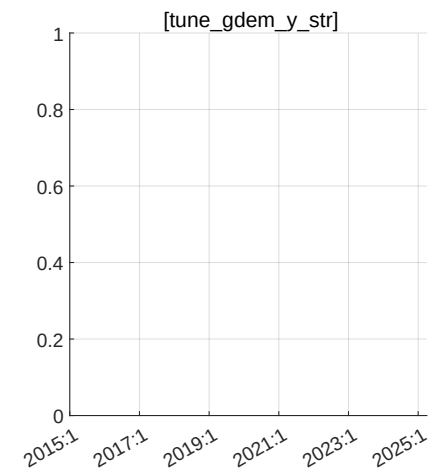
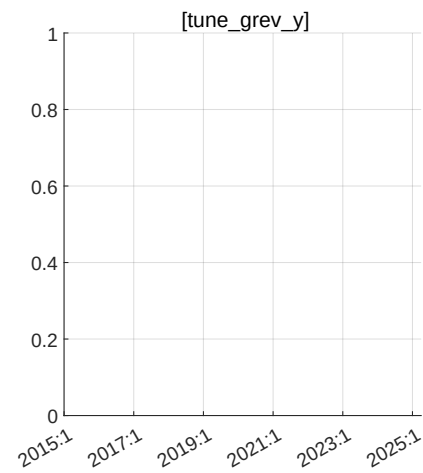
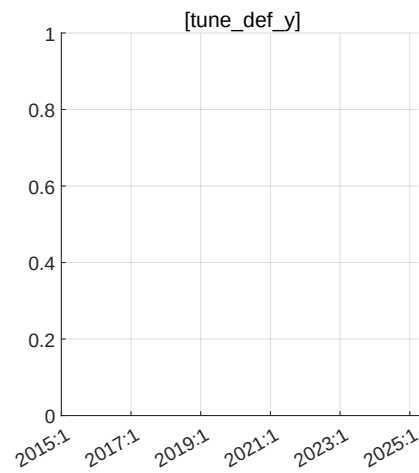
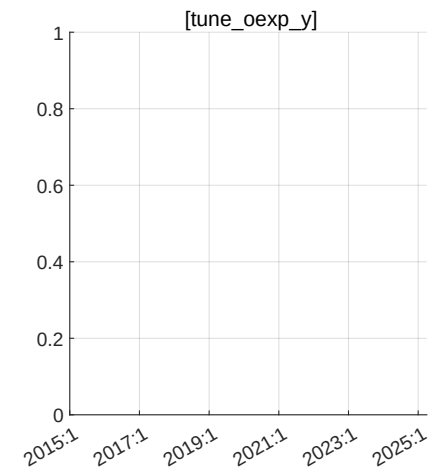
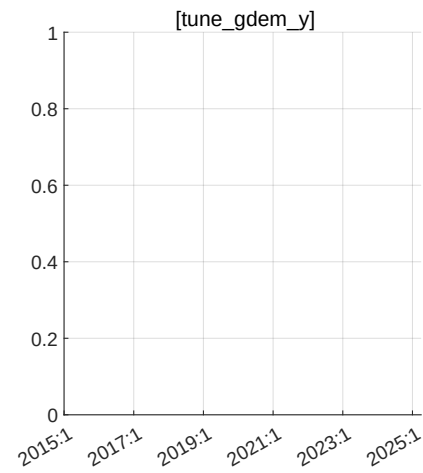
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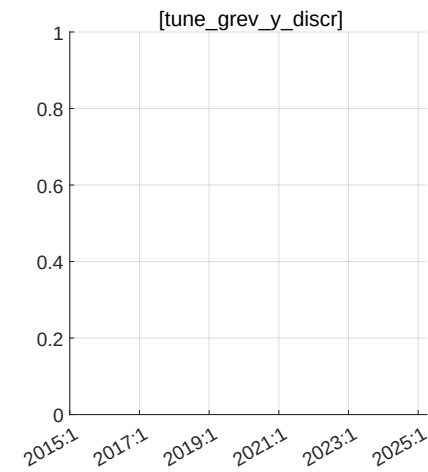
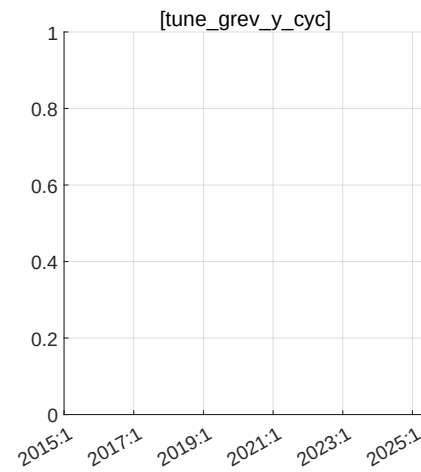
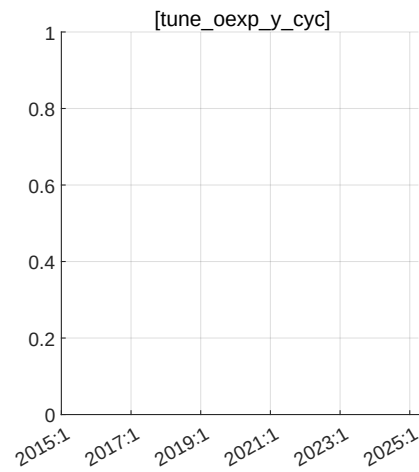
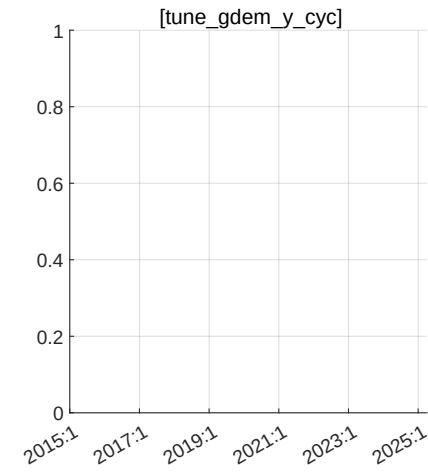
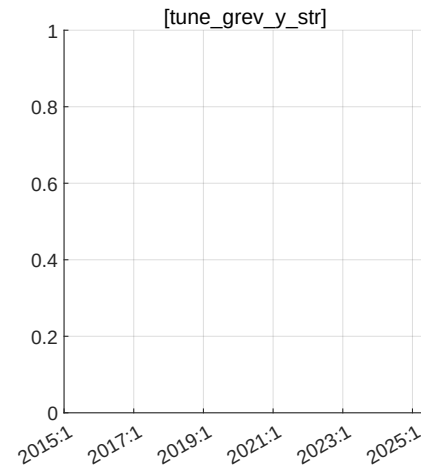
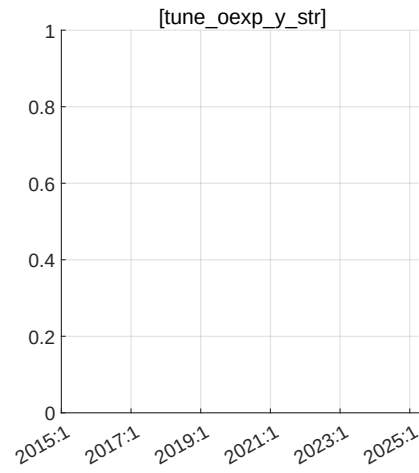
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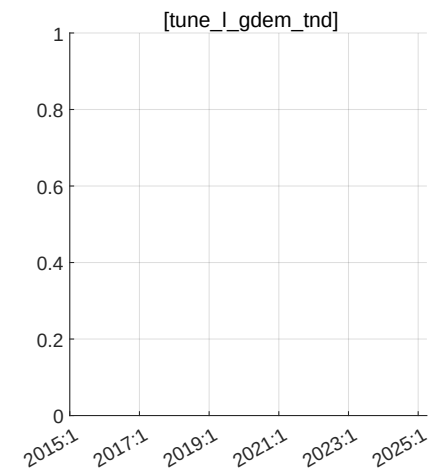
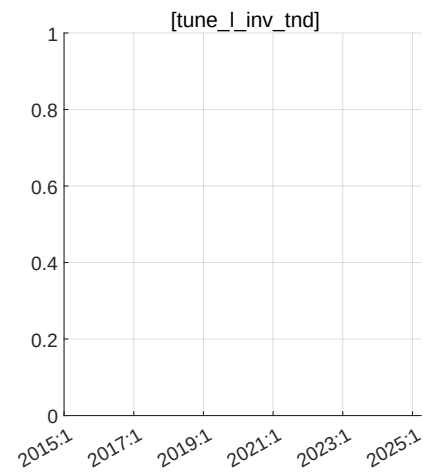
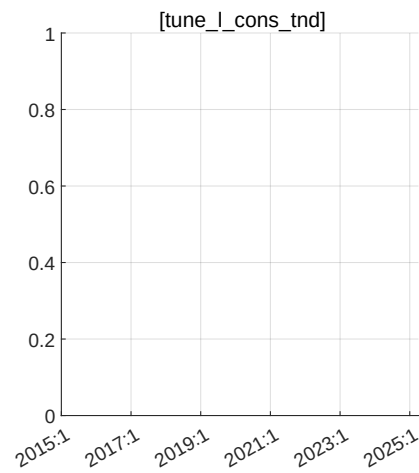
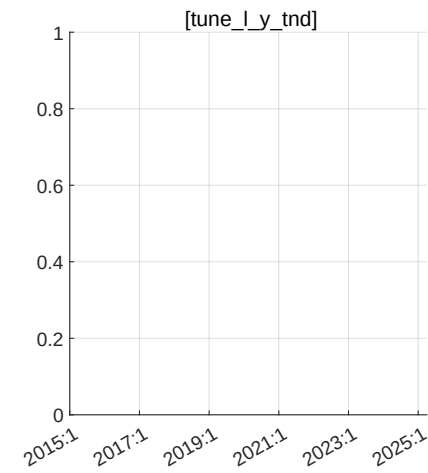
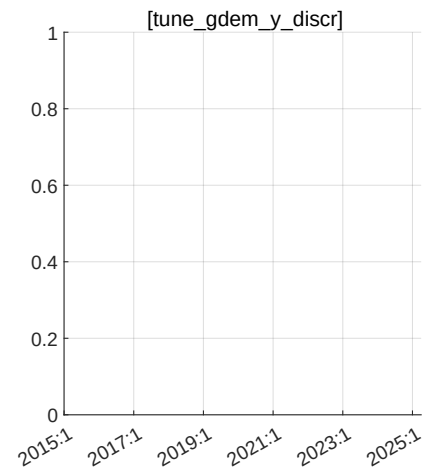
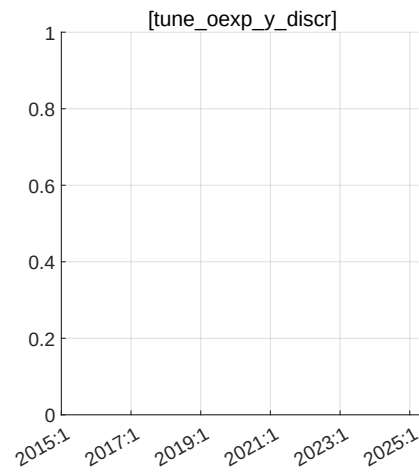


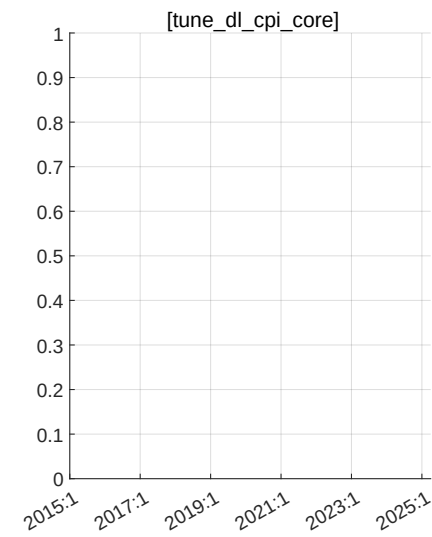
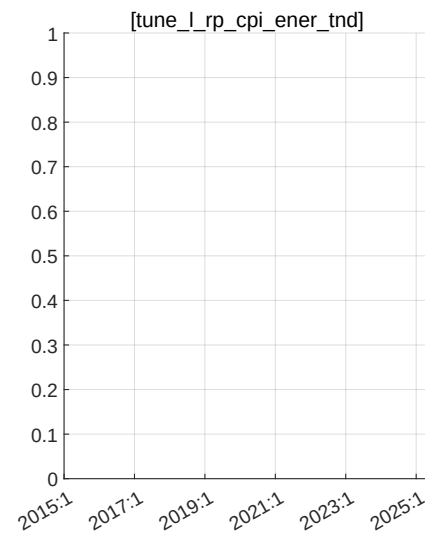
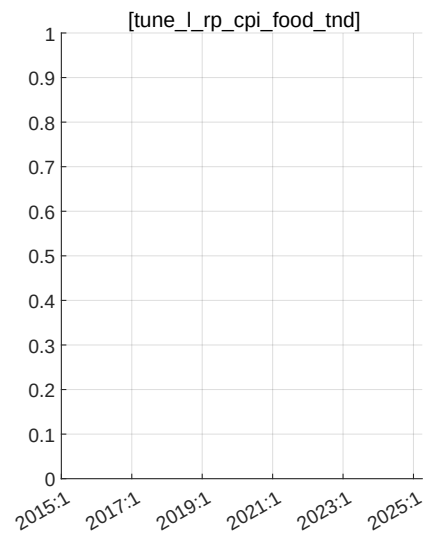
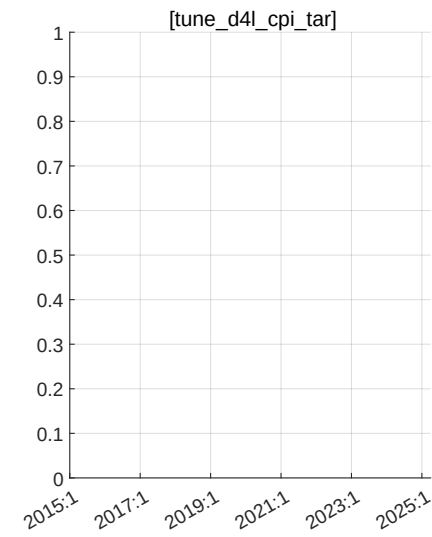
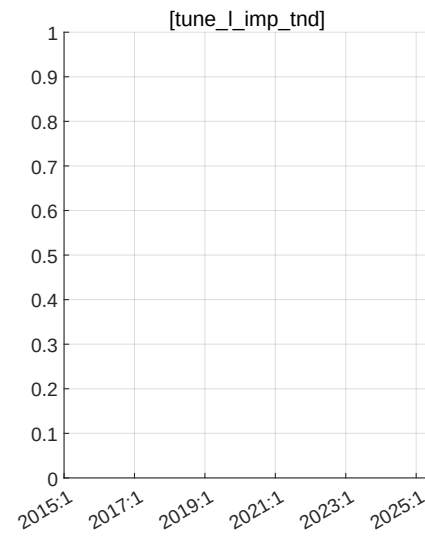
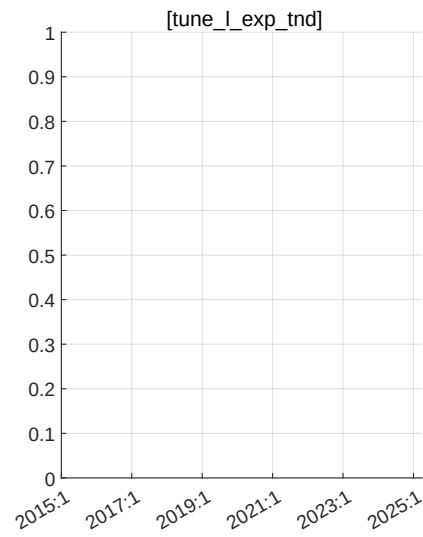
## Observed data (page 5)

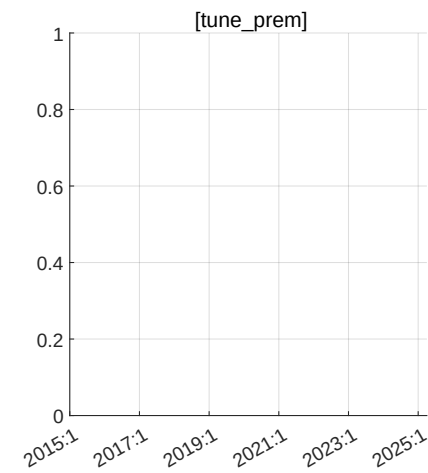
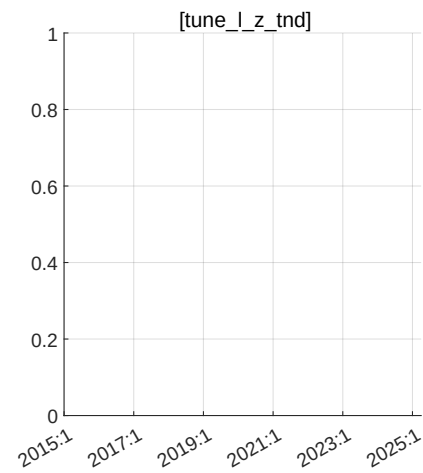
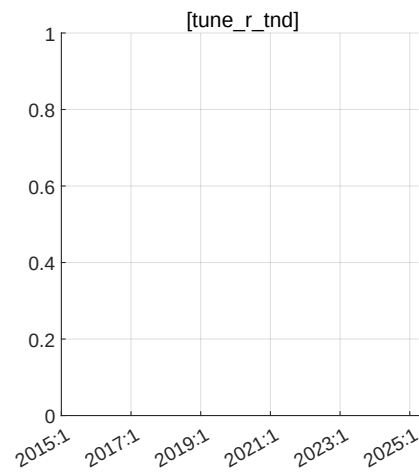
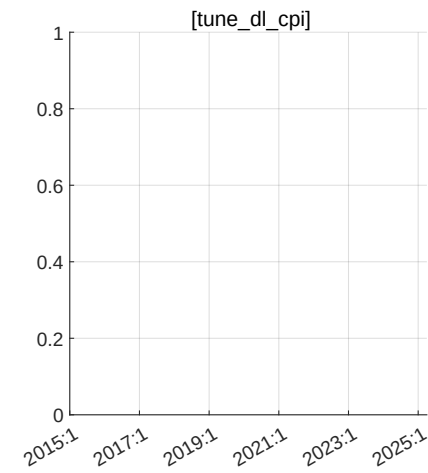
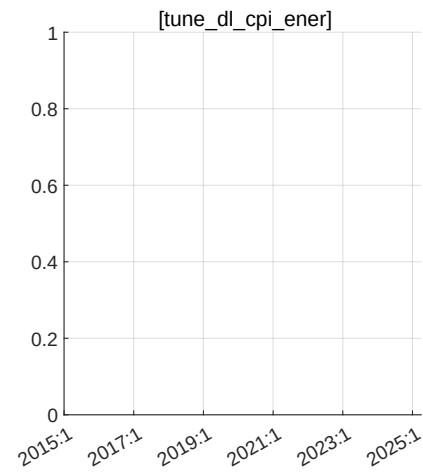
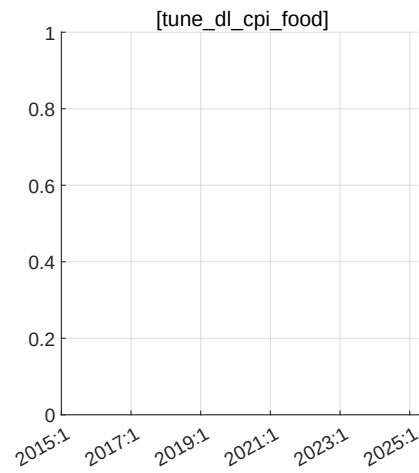


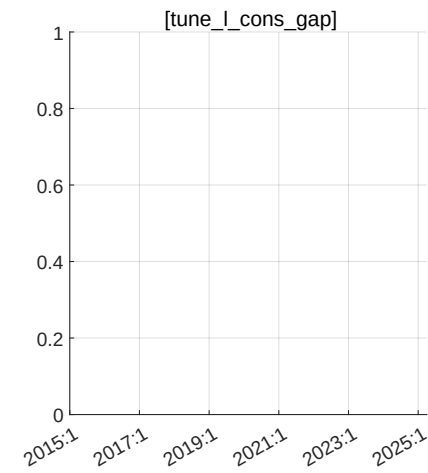
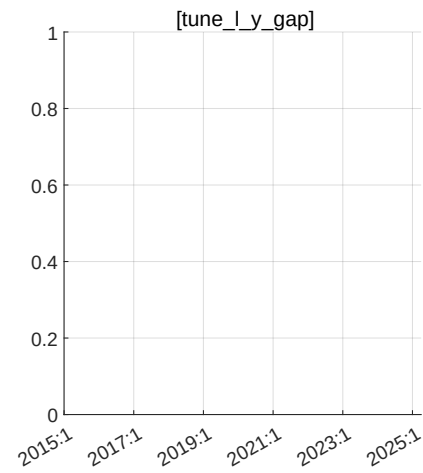
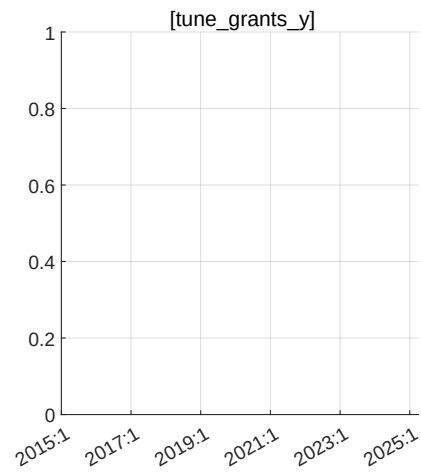
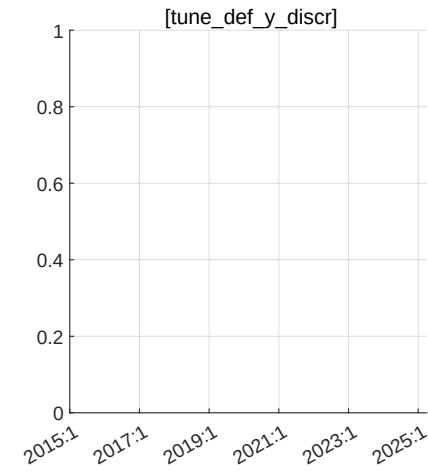
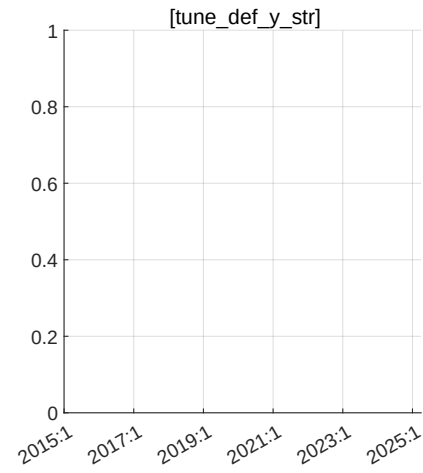
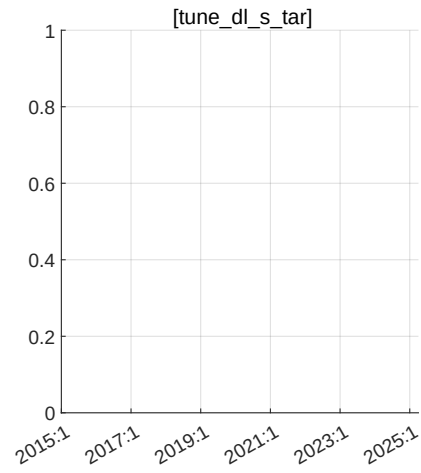


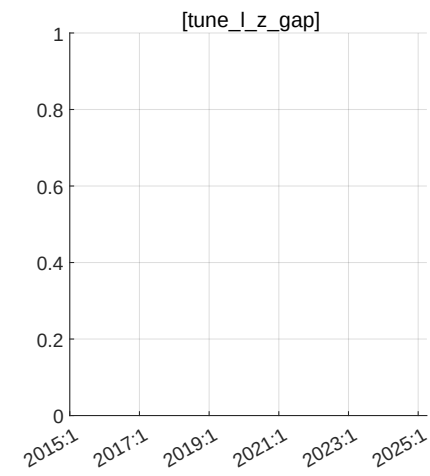
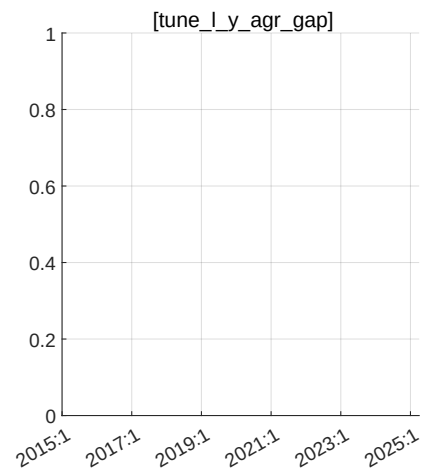
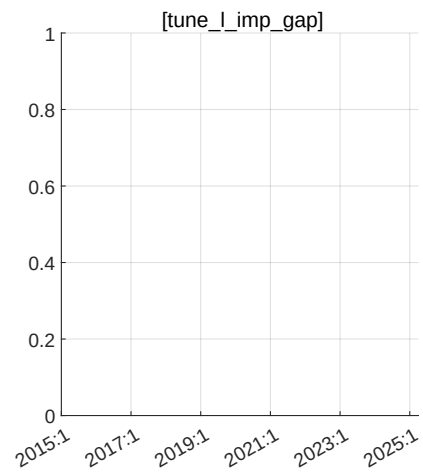
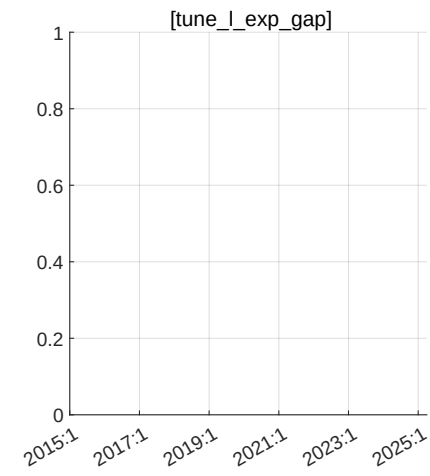
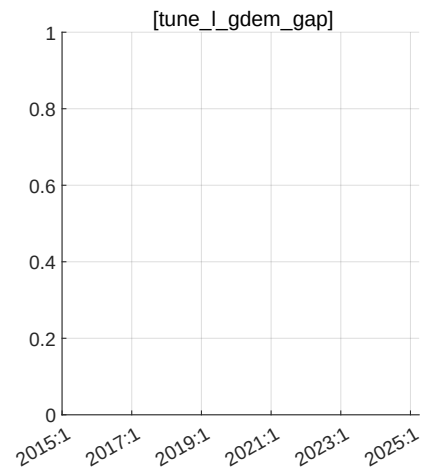
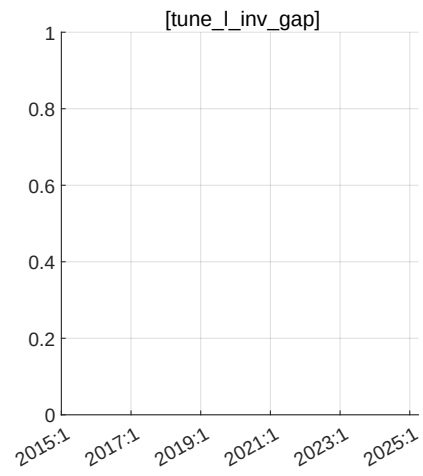
**Observed data (page 6)**

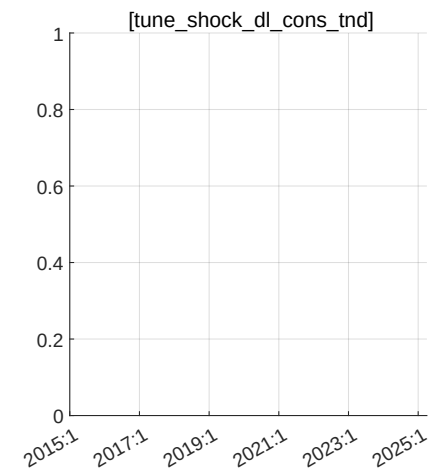
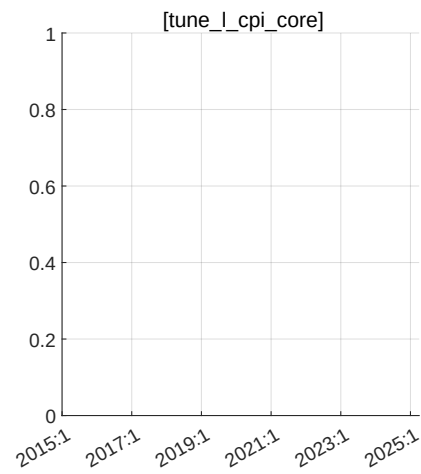
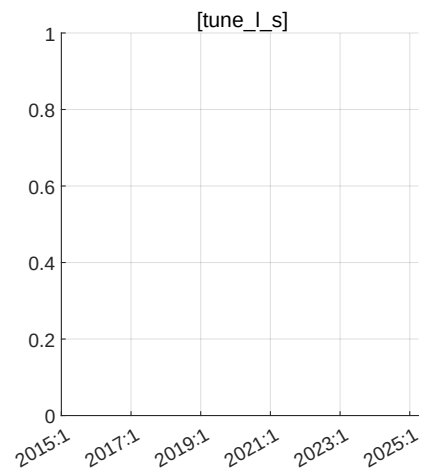
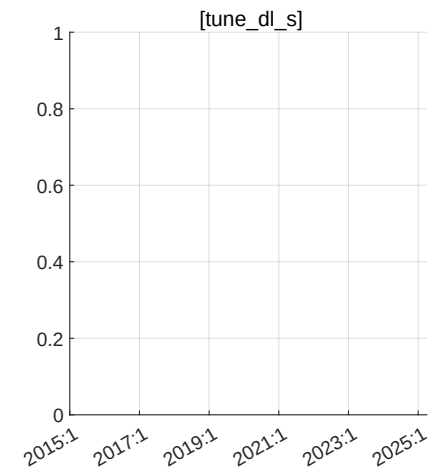
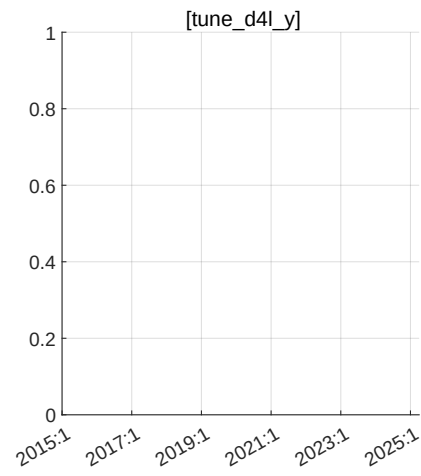
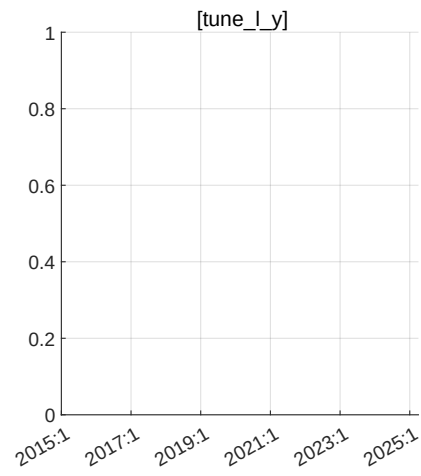
**Observed data (page 7)**

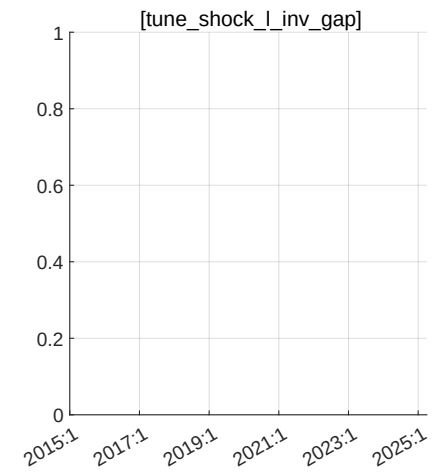
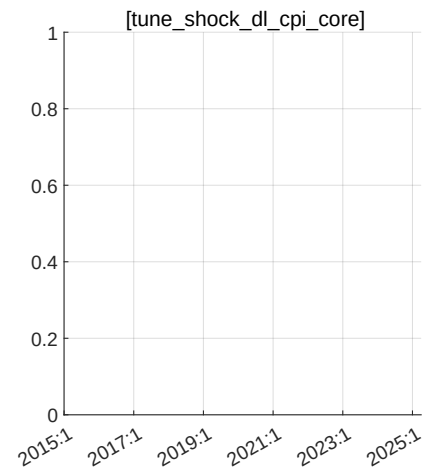
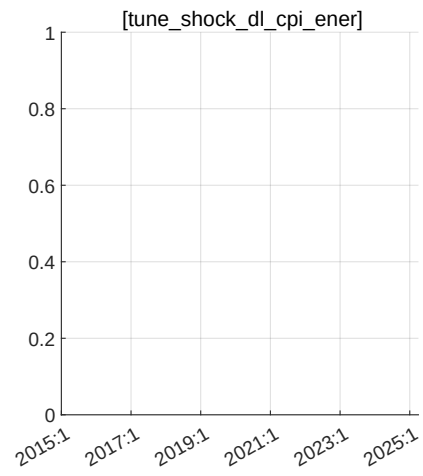
**Observed data (page 8)**

**Observed data (page 9)**

**Observed data (page 10)**

**Observed data (page 11)**

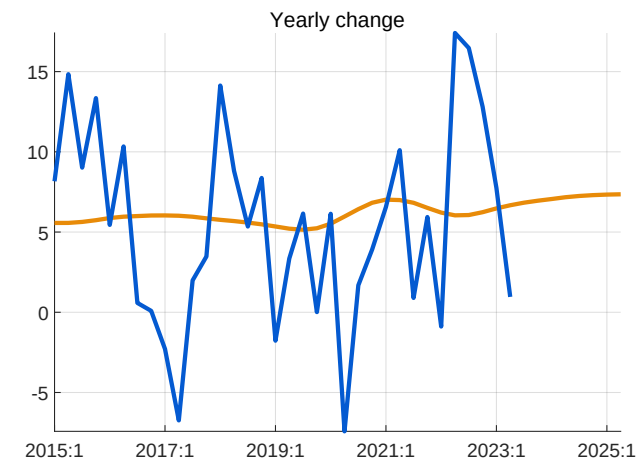
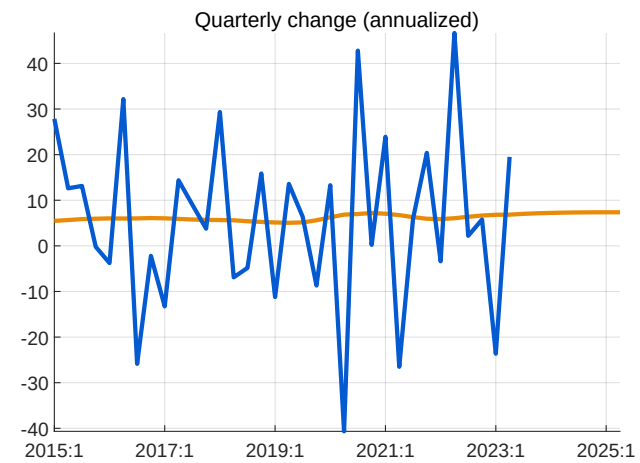
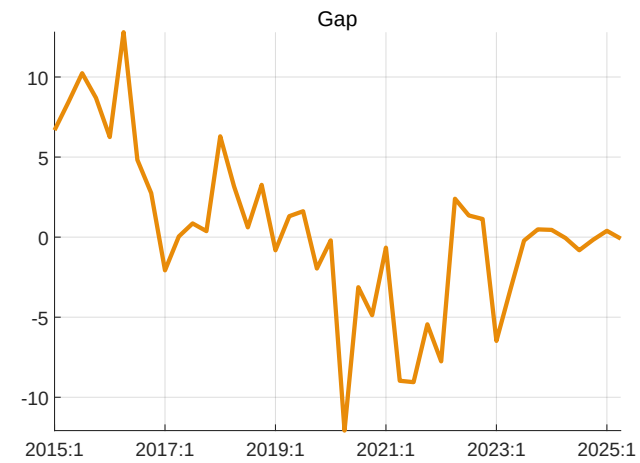
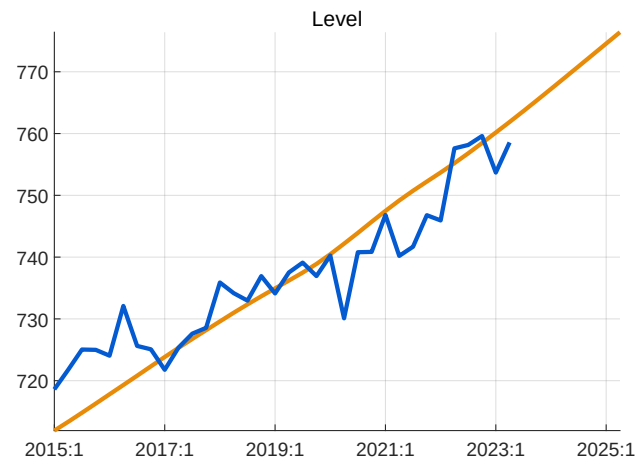
**Observed data (page 12)**

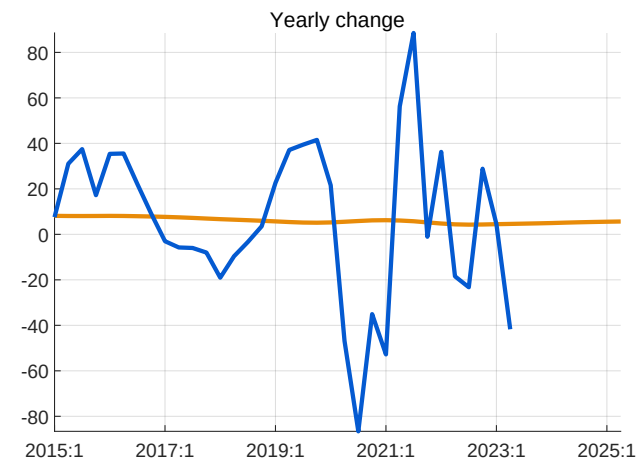
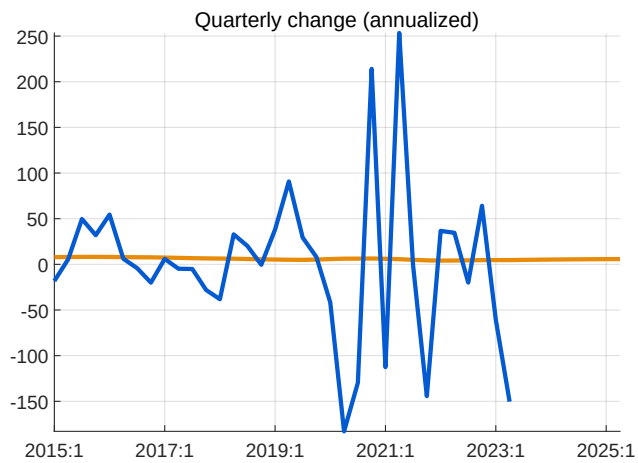
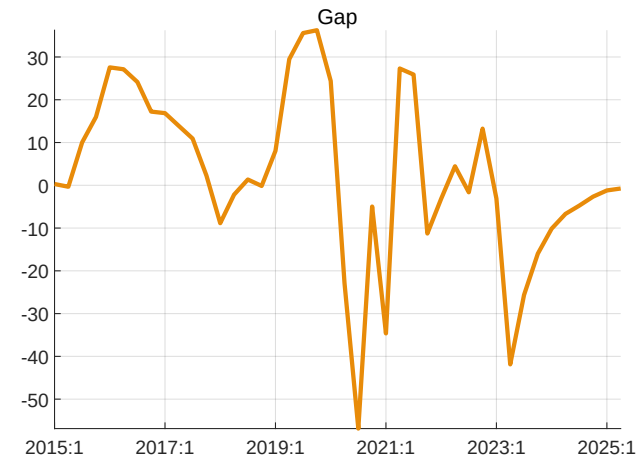
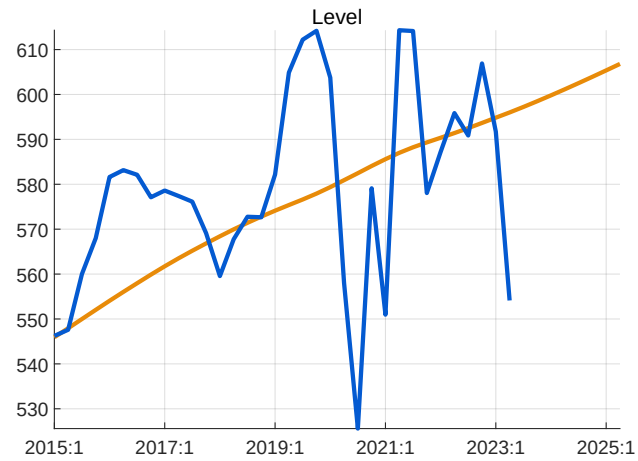
**Observed data (page 13)**



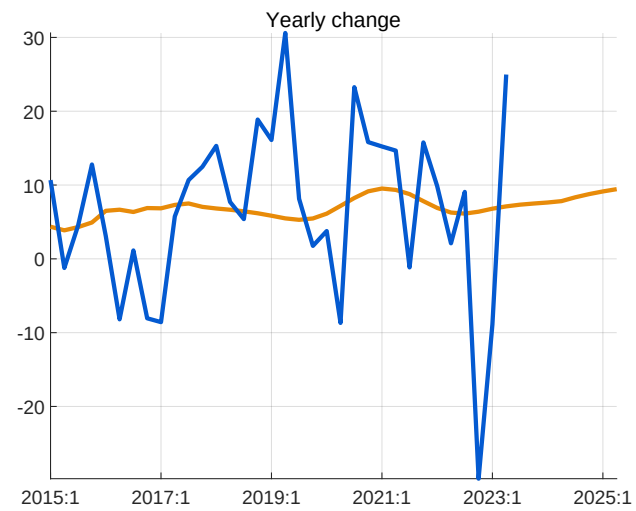
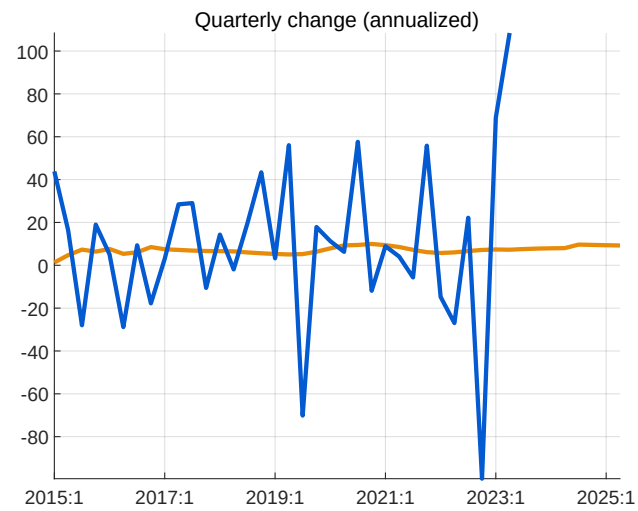
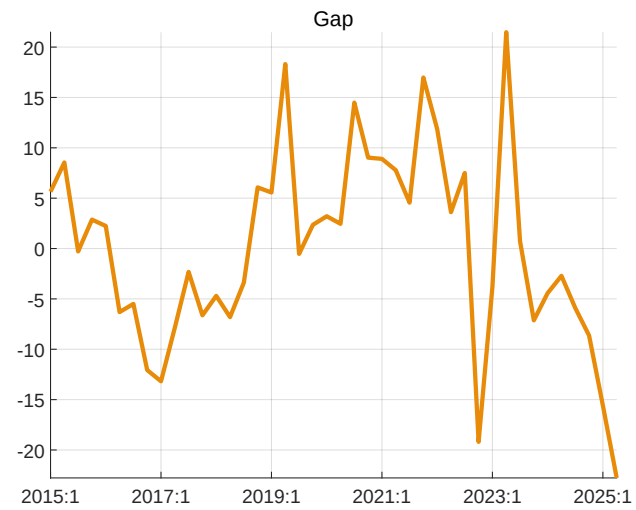
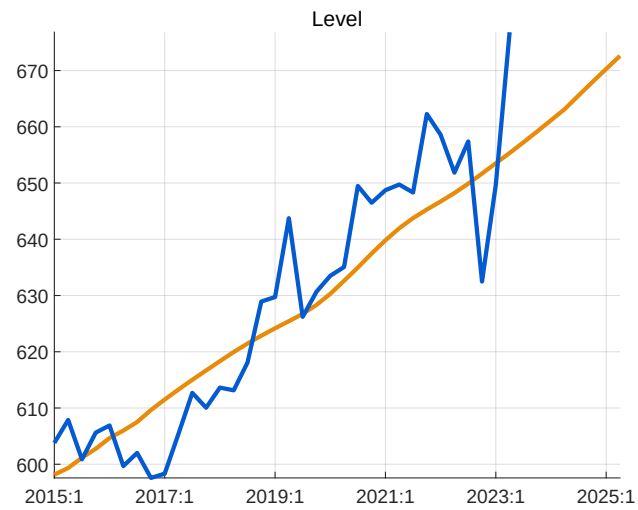
## 2 Trends and gaps

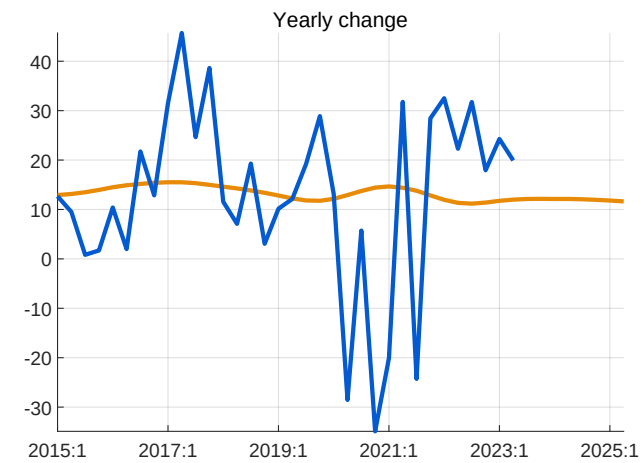
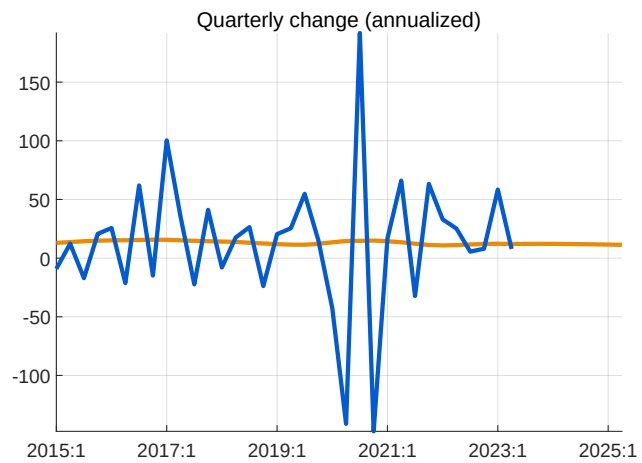
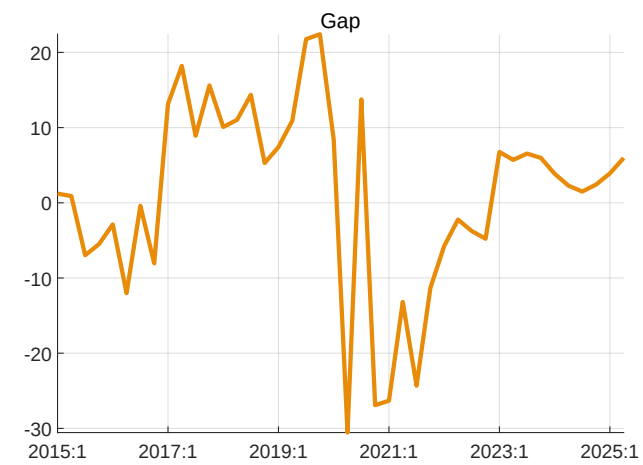
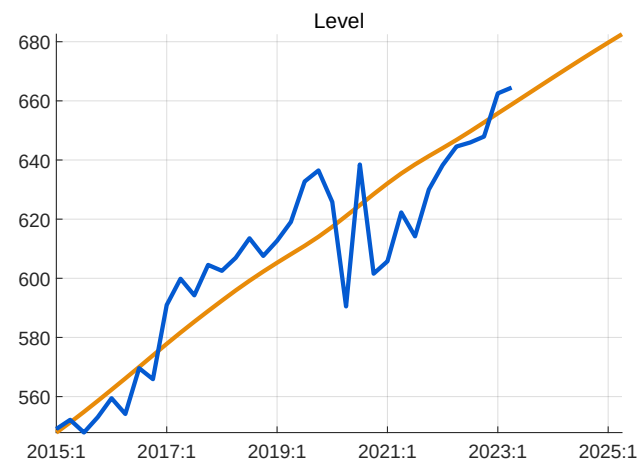


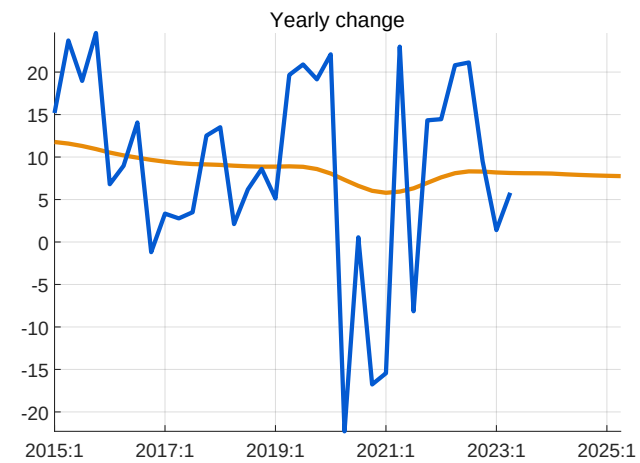
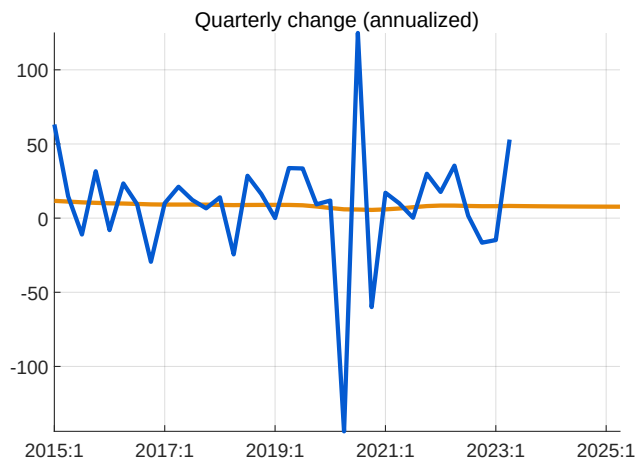
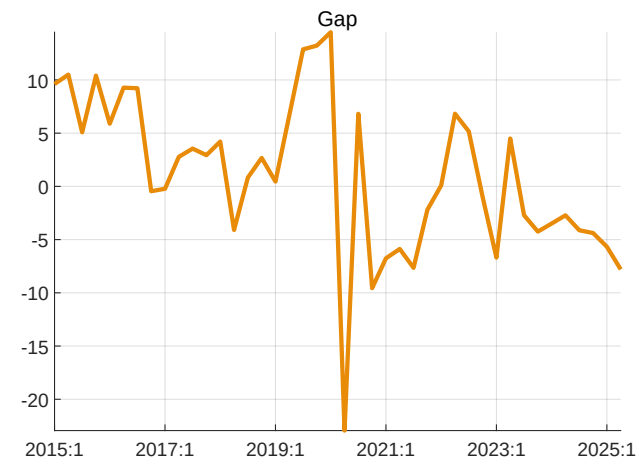
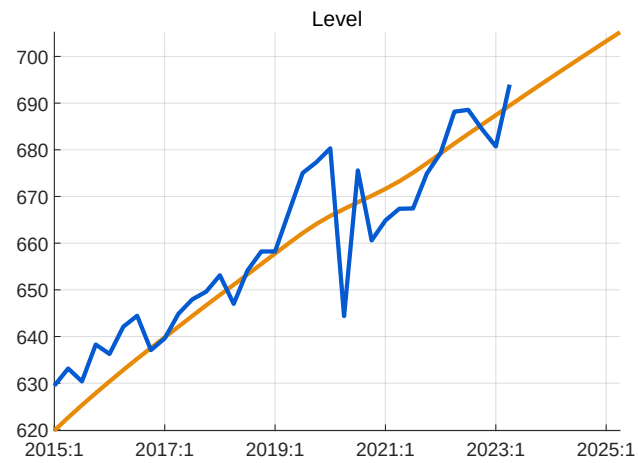
Consumption,  $100 \cdot \log [l\_cons]$ 

Investment,  $100 \cdot \log [I_{inv}]$ 

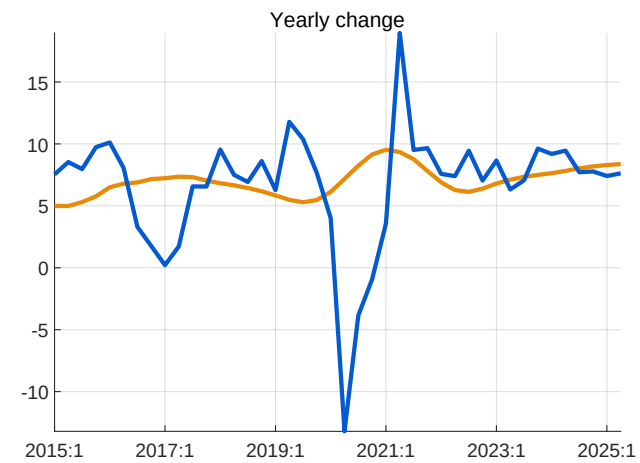
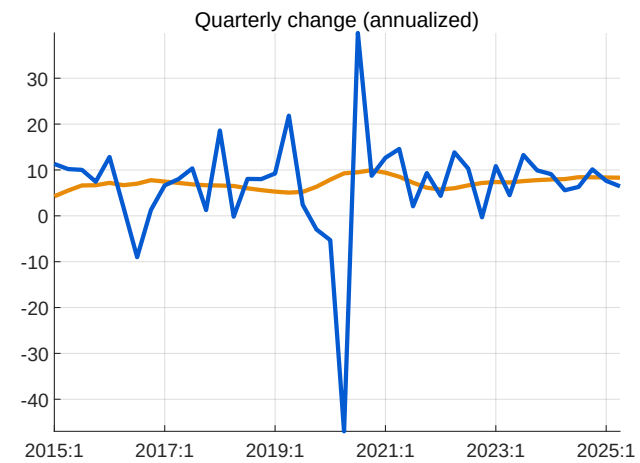
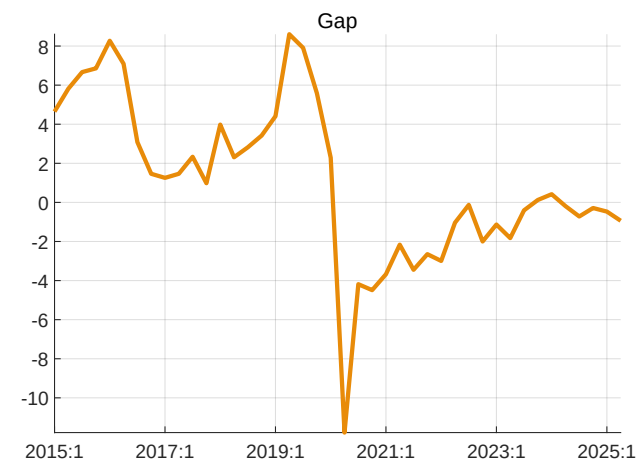
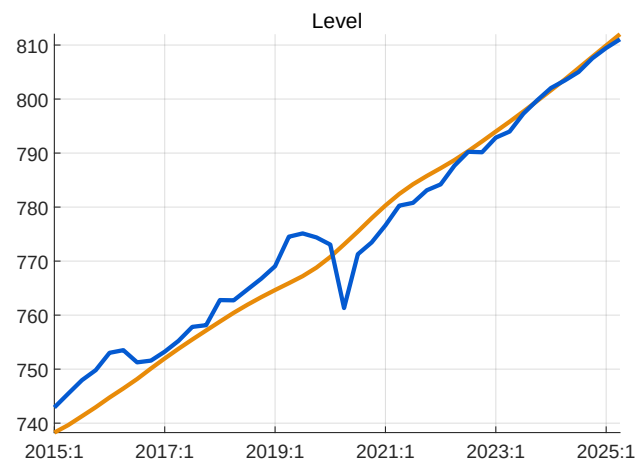
## Gov. demand, 100\*log [l\_gdem]



**Export, 100\*log [l\_exp]**

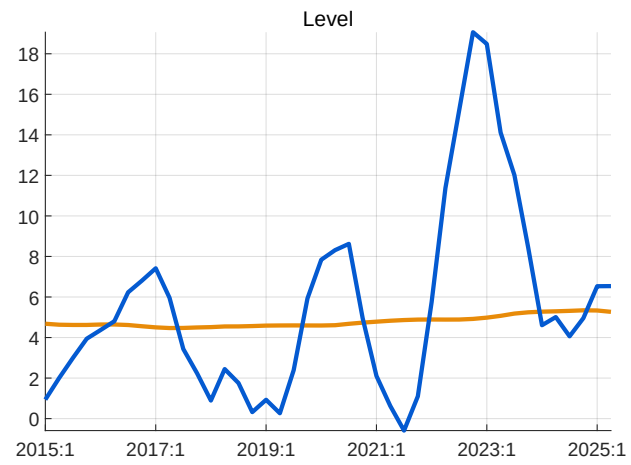
Import,  $100 \cdot \log [l\_imp]$ 

## GDP, 100\*log [l\_y]

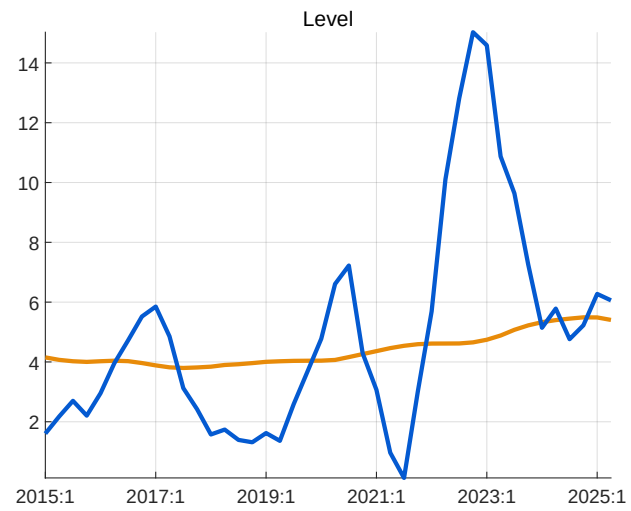




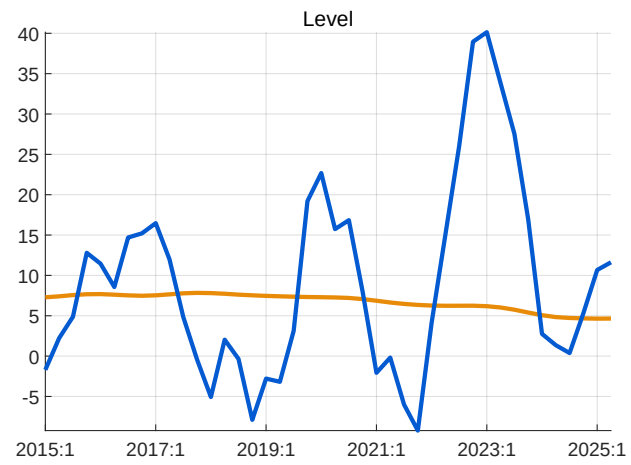
### Headline CPI, YY % [d4l\_cpi]



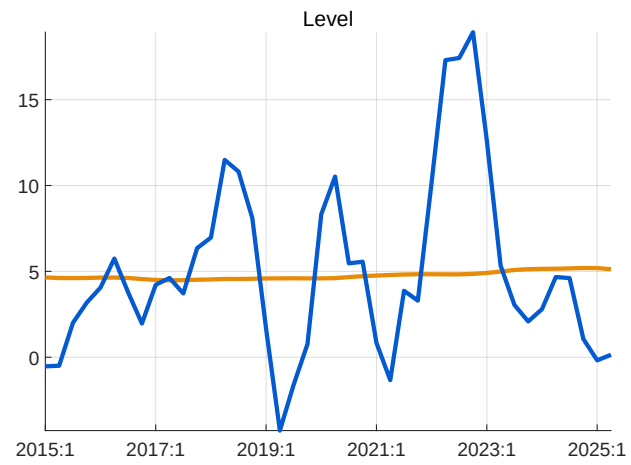
### Core CPI, YY % [d4l\_cpi\_core]

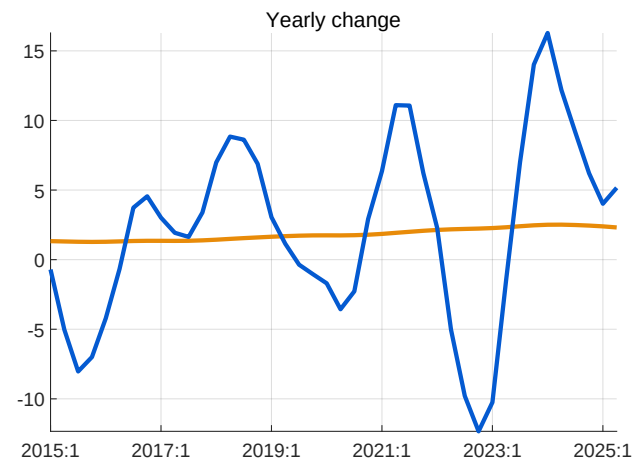
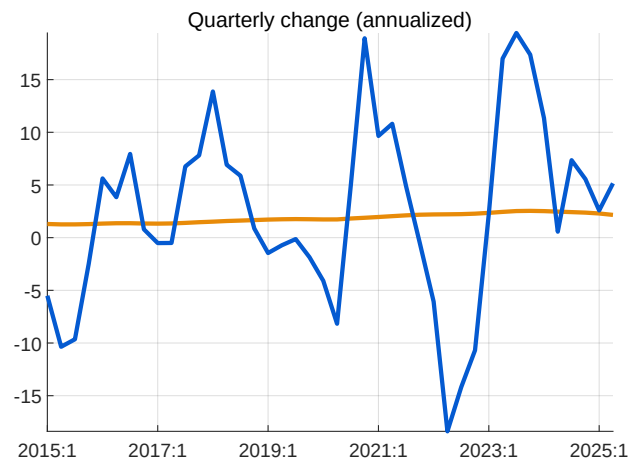
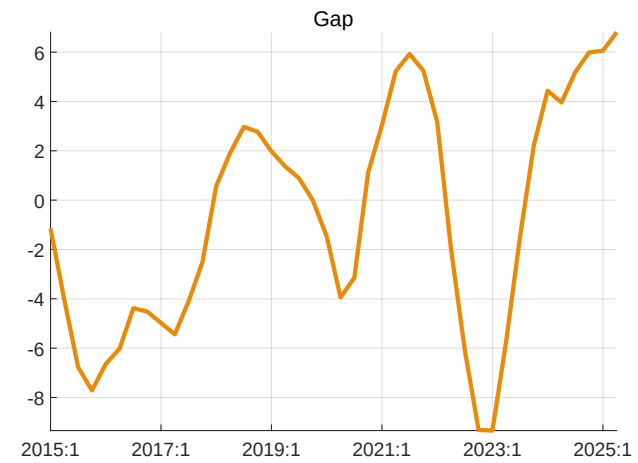
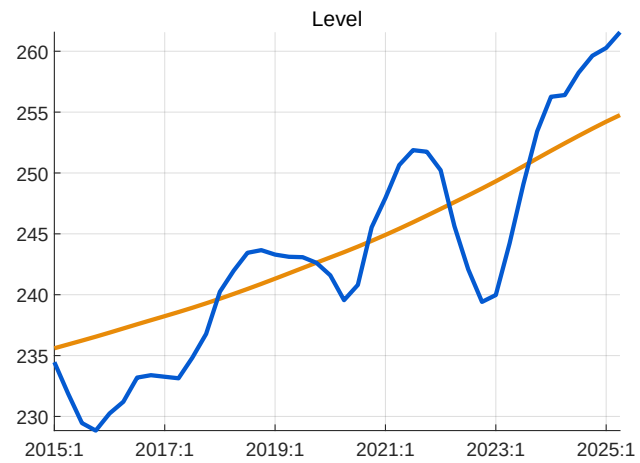


### Food CPI, YY % [d4l\_cpi\_food]

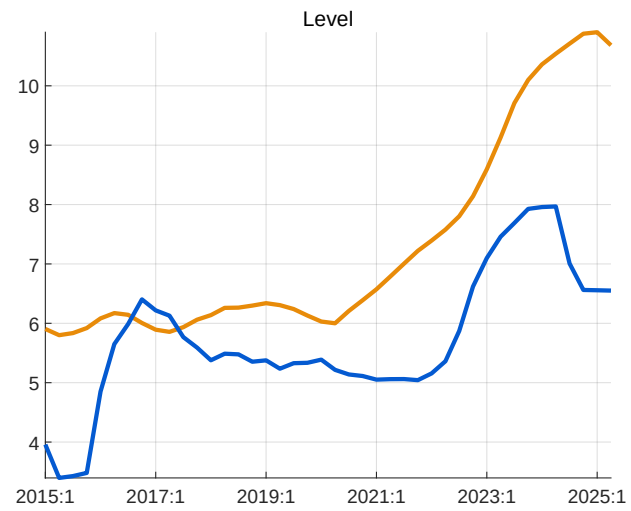


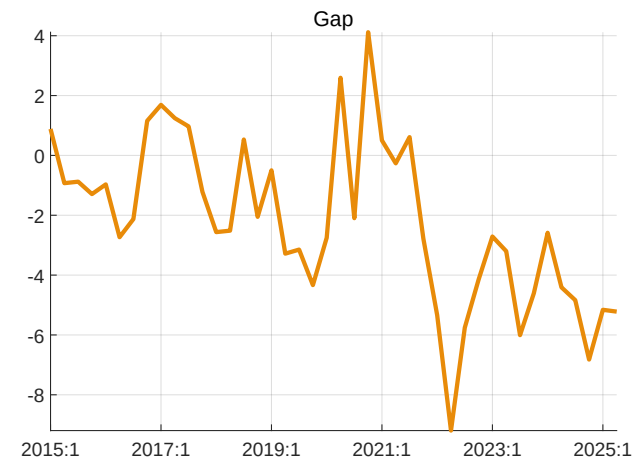
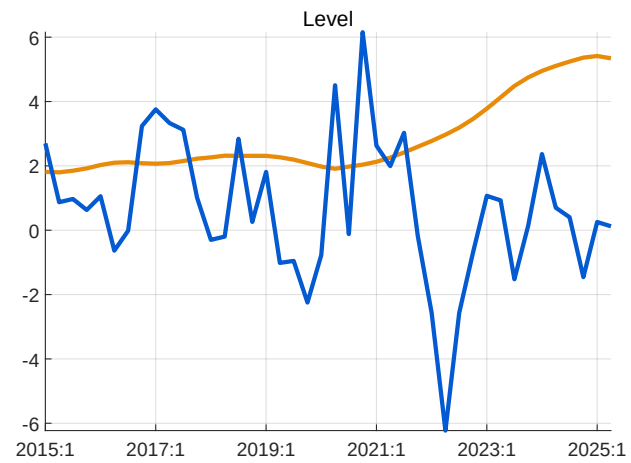
### Energy CPI, YY % [d4l\_cpi\_ener]

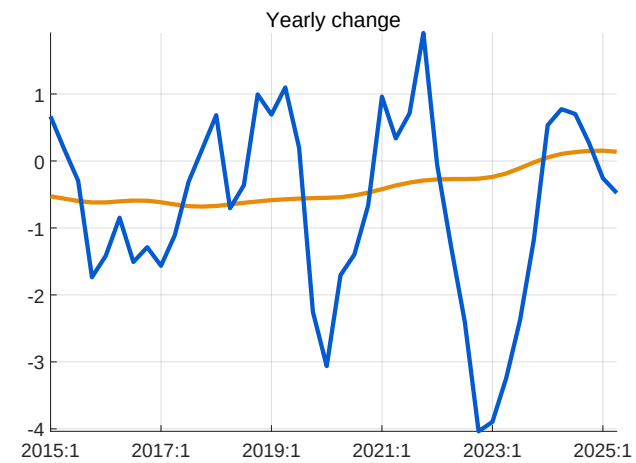
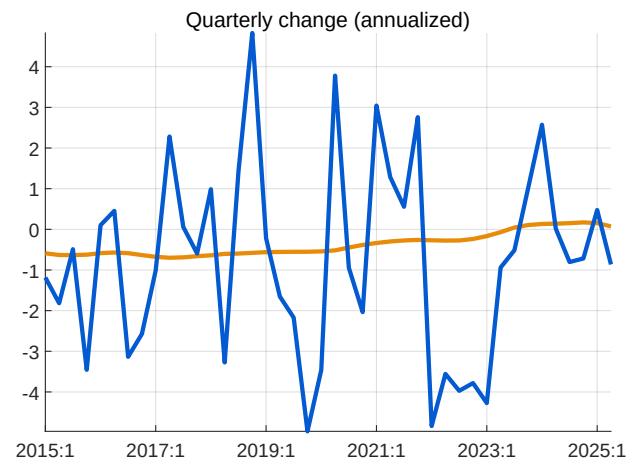
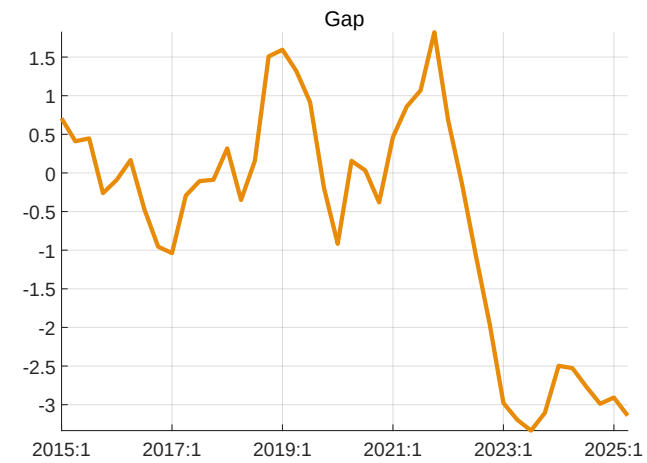
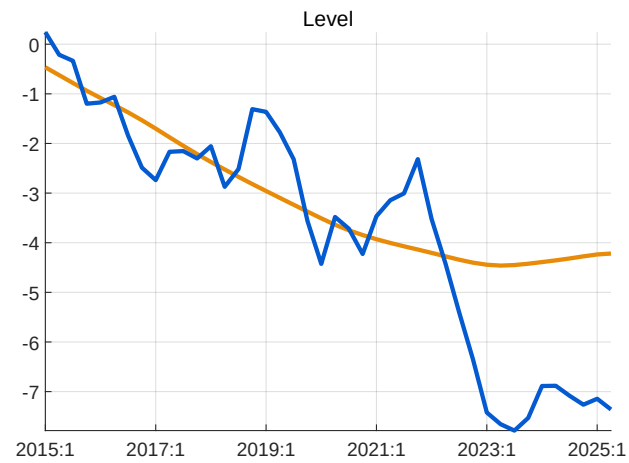


**Real exchn. rate, 100\*log [l\_z]**

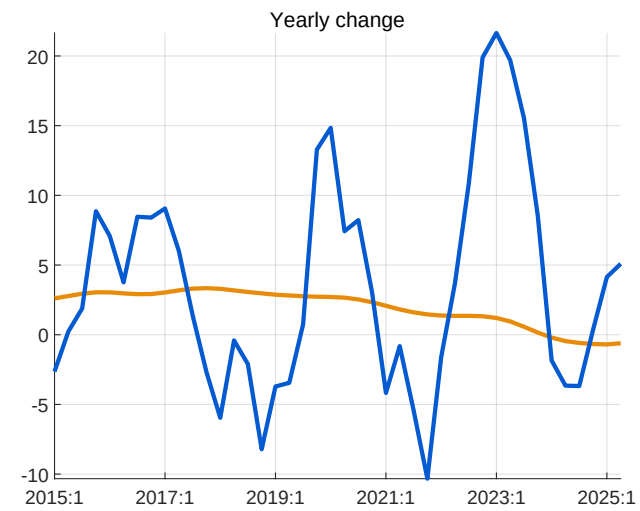
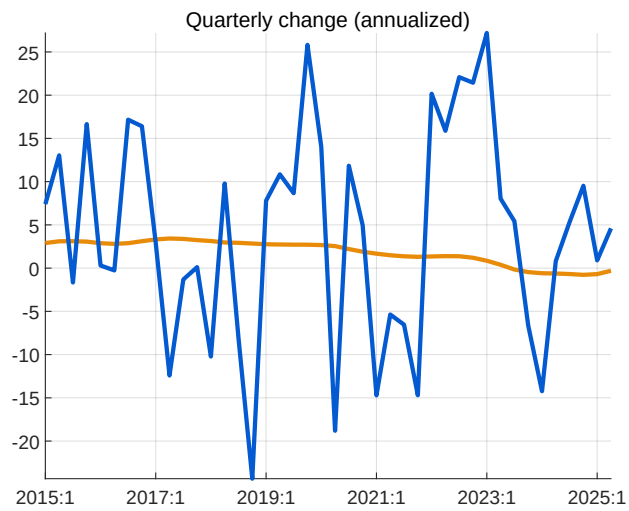
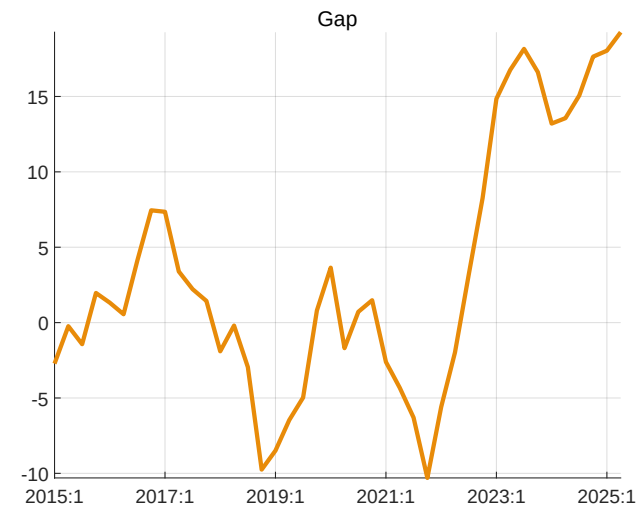
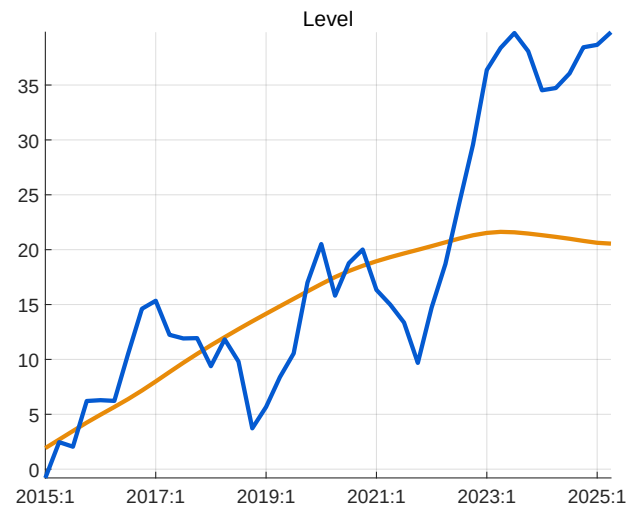
### Policy rate (IB rate used), % [i]

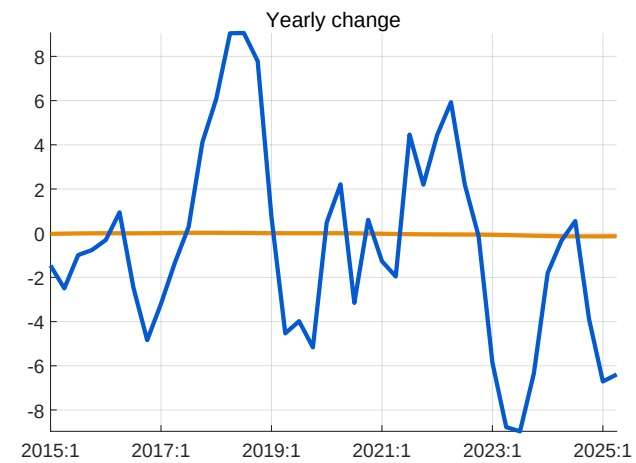
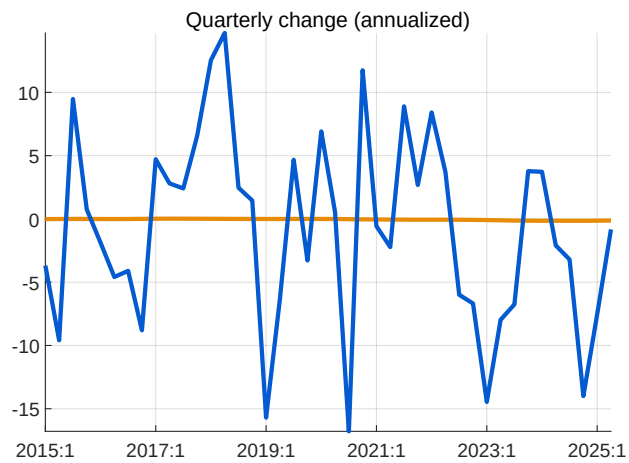
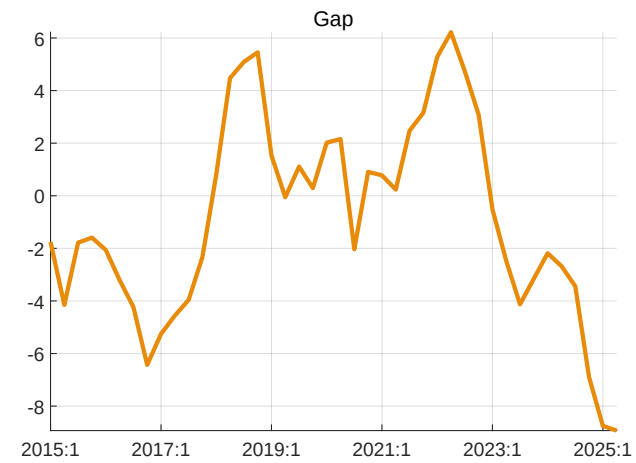
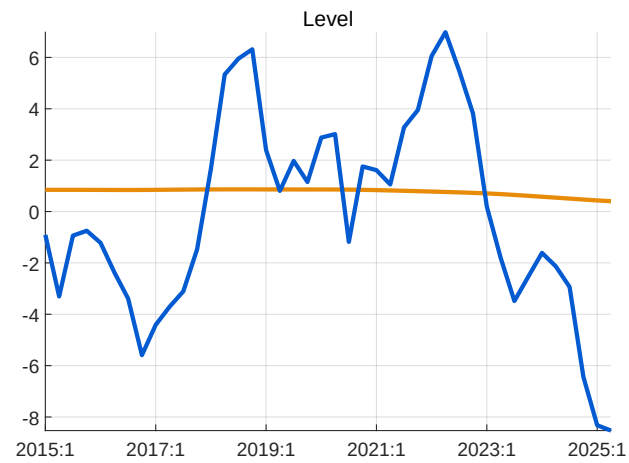


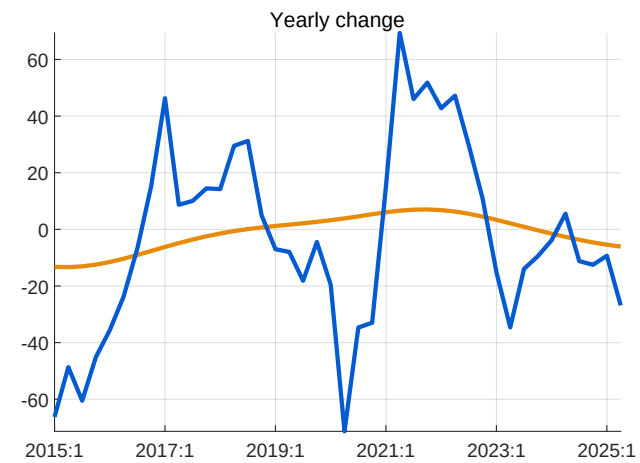
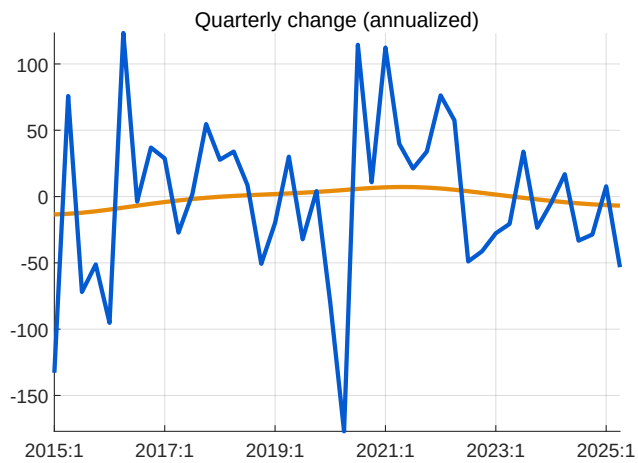
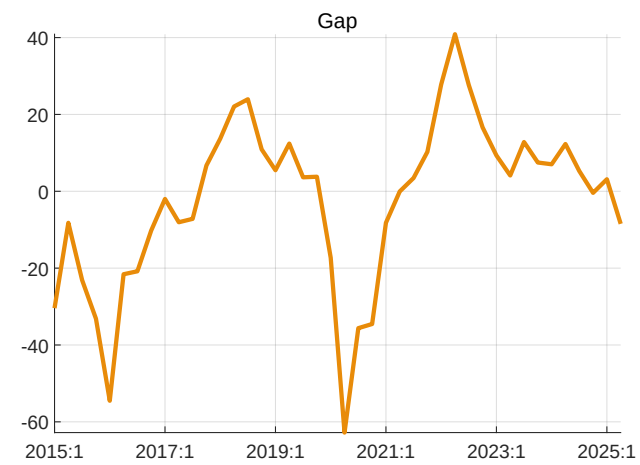
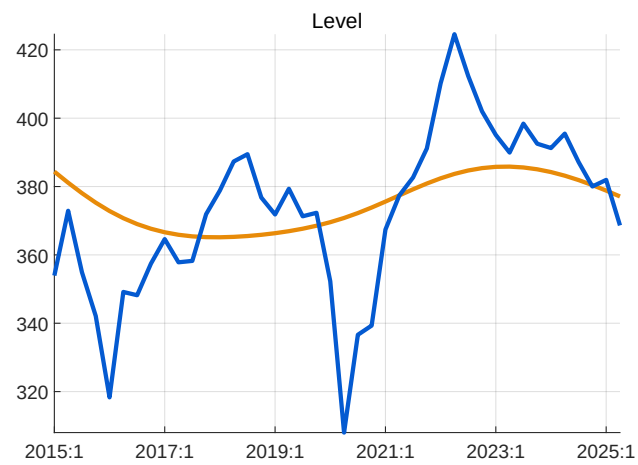
**Real interest (policy) rate, % [r]**

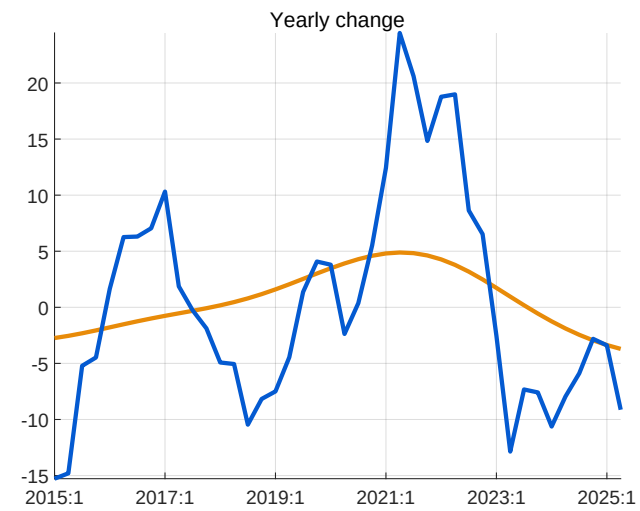
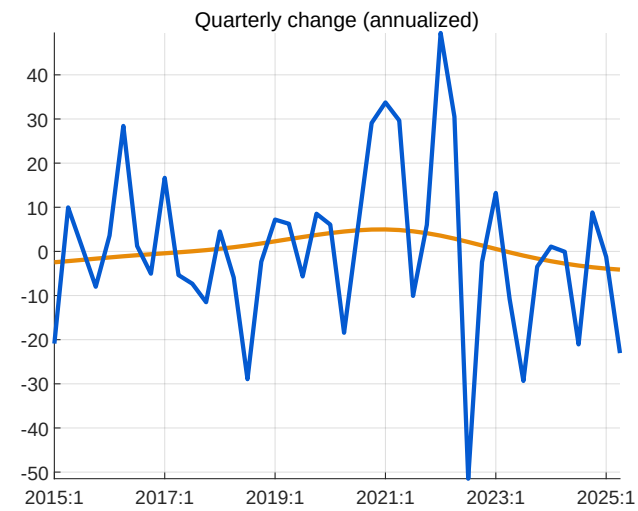
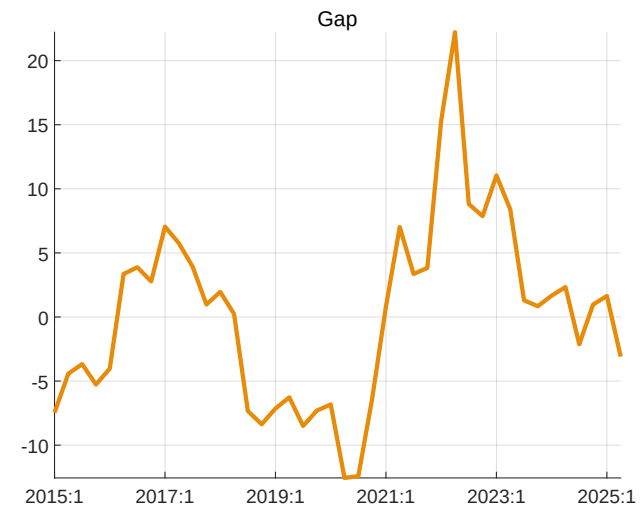
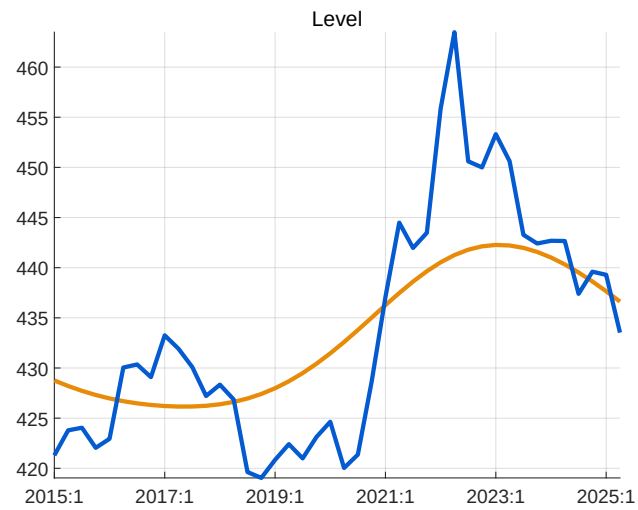
Core rel. price,  $100 \cdot \log [l\_rp\_cpi\_core]$ 



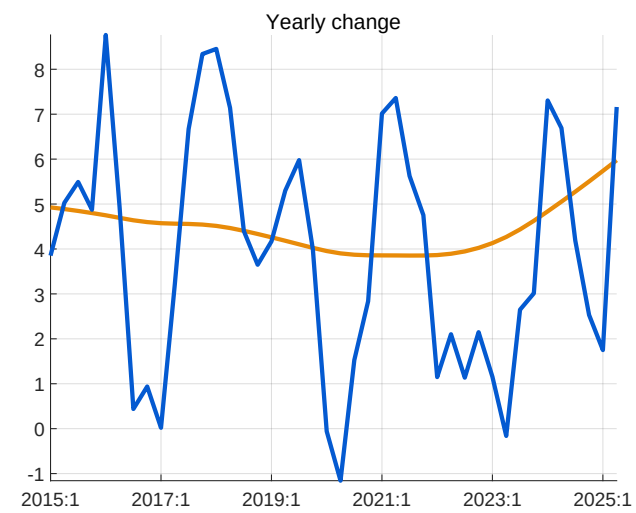
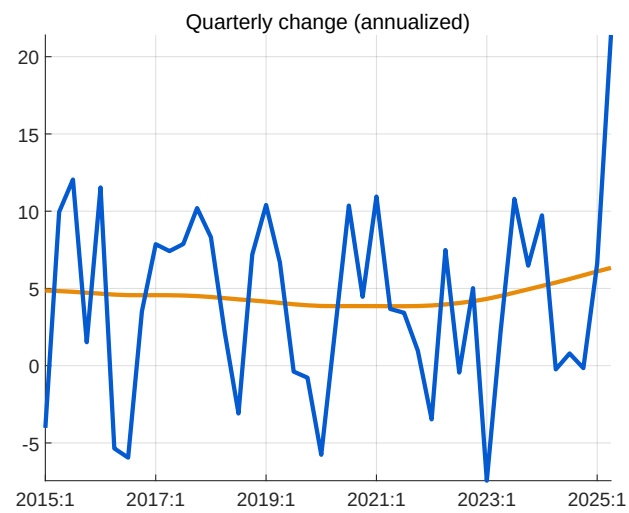
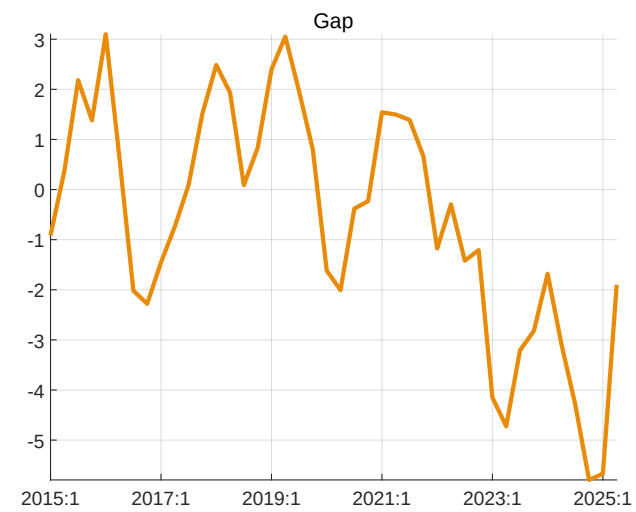
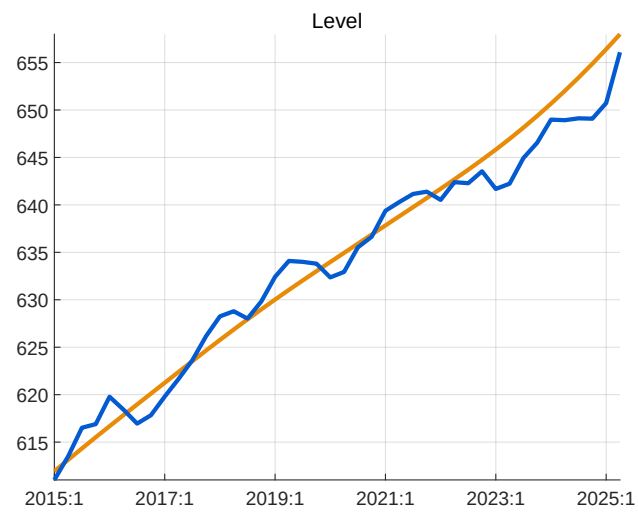
**Food rel. price,  $100 \cdot \log [l\_rp\_cpi\_food]$** 

**Energy rel. price, 100\*log [l\_rp\_cpi\_ener]**

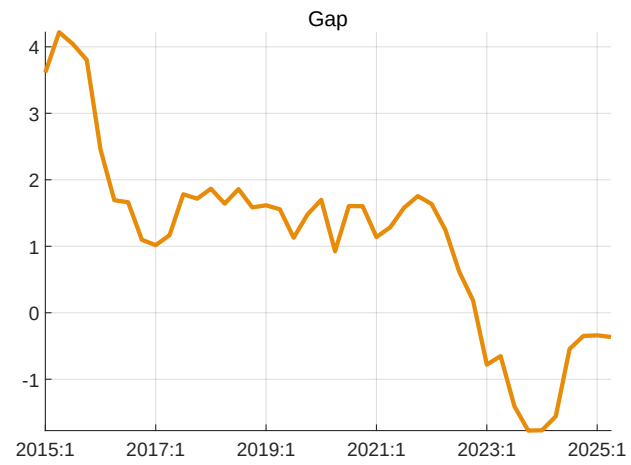
Foreign rel. ener. price,  $100 \cdot \log [l\_rp\_enerstar]$ 

**Foreign rel. food price,  $100 \cdot \log [l\_rp\_foodstar]$** 

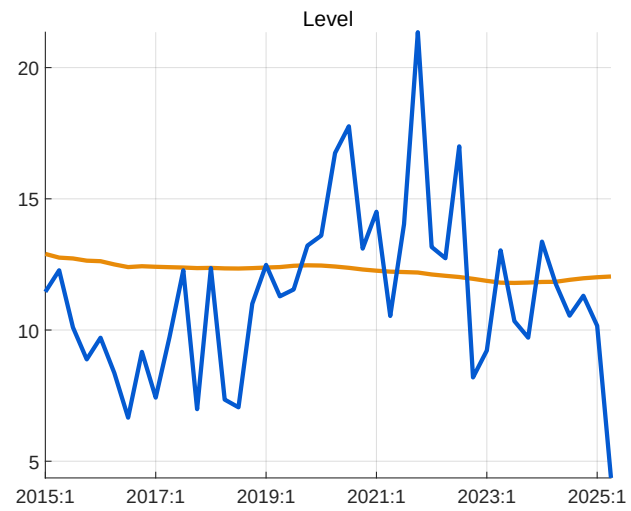
## Agric. GDP, 100\*log [l\_y\_agr]

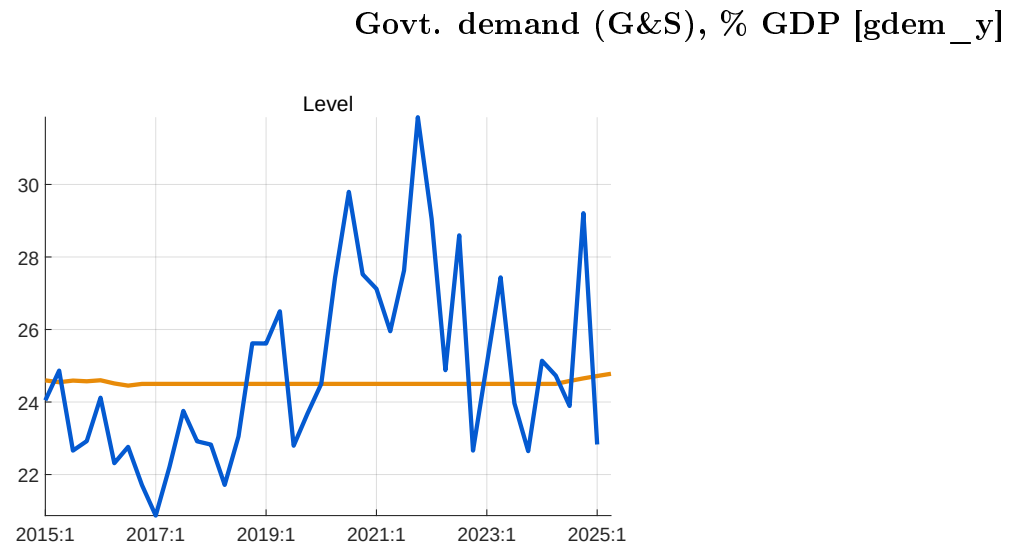


### Lending premium, % [prem\_d]



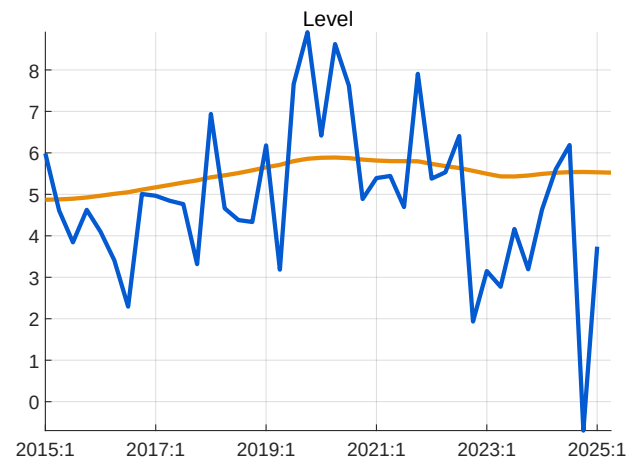
### Budget deficit, % of GDP [def\_y]



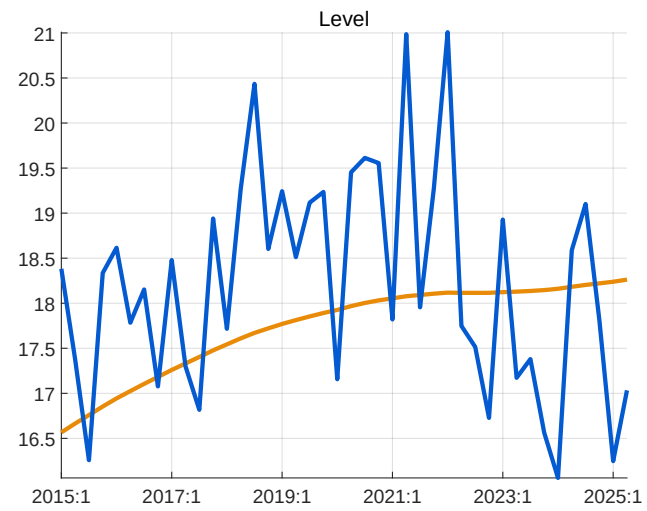




### Other govt. exp., % to GDP [oexp\_y]



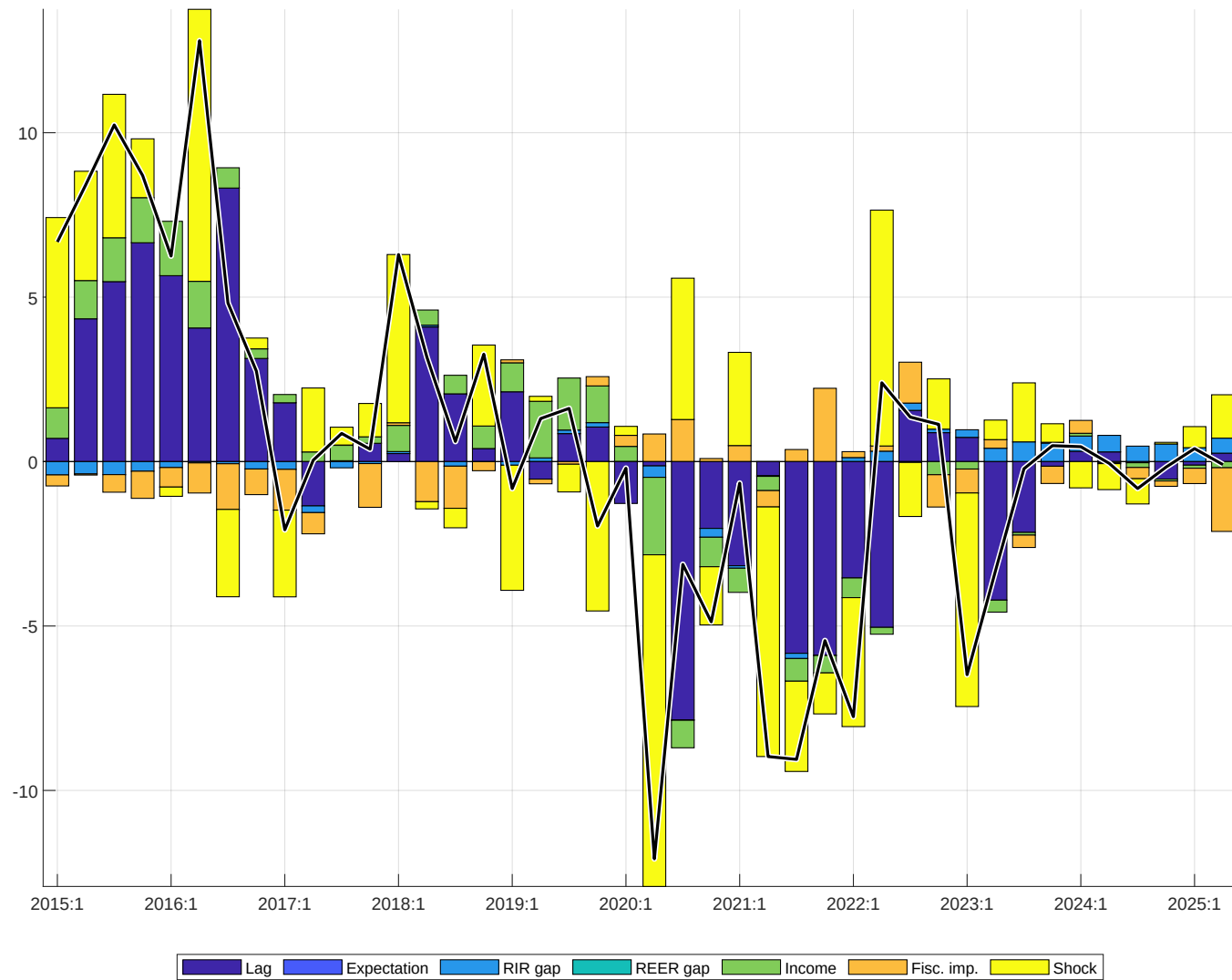
### Govt. rev., % of GDP [grev\_y]

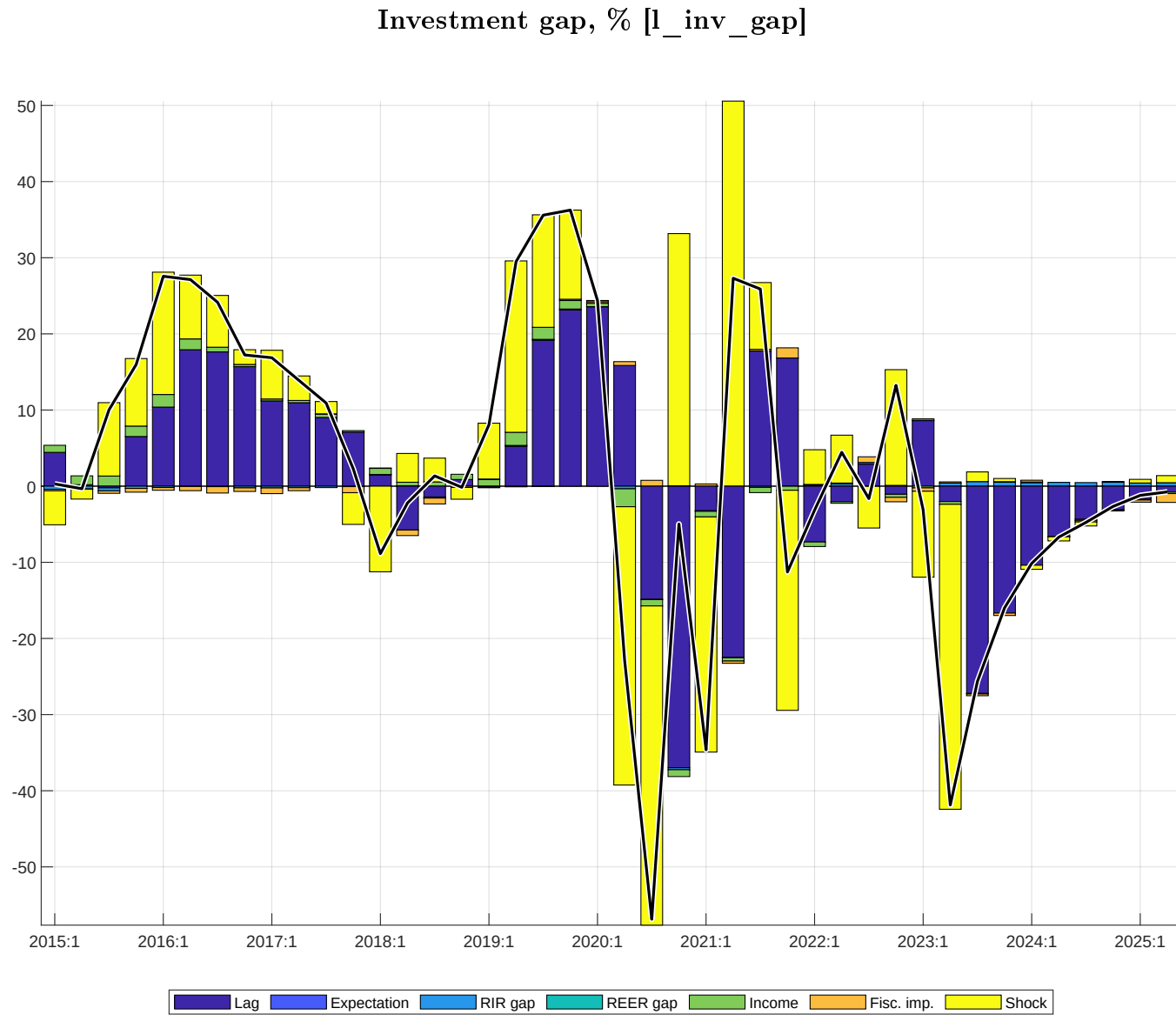


### 3 Decomposition of equations: charts

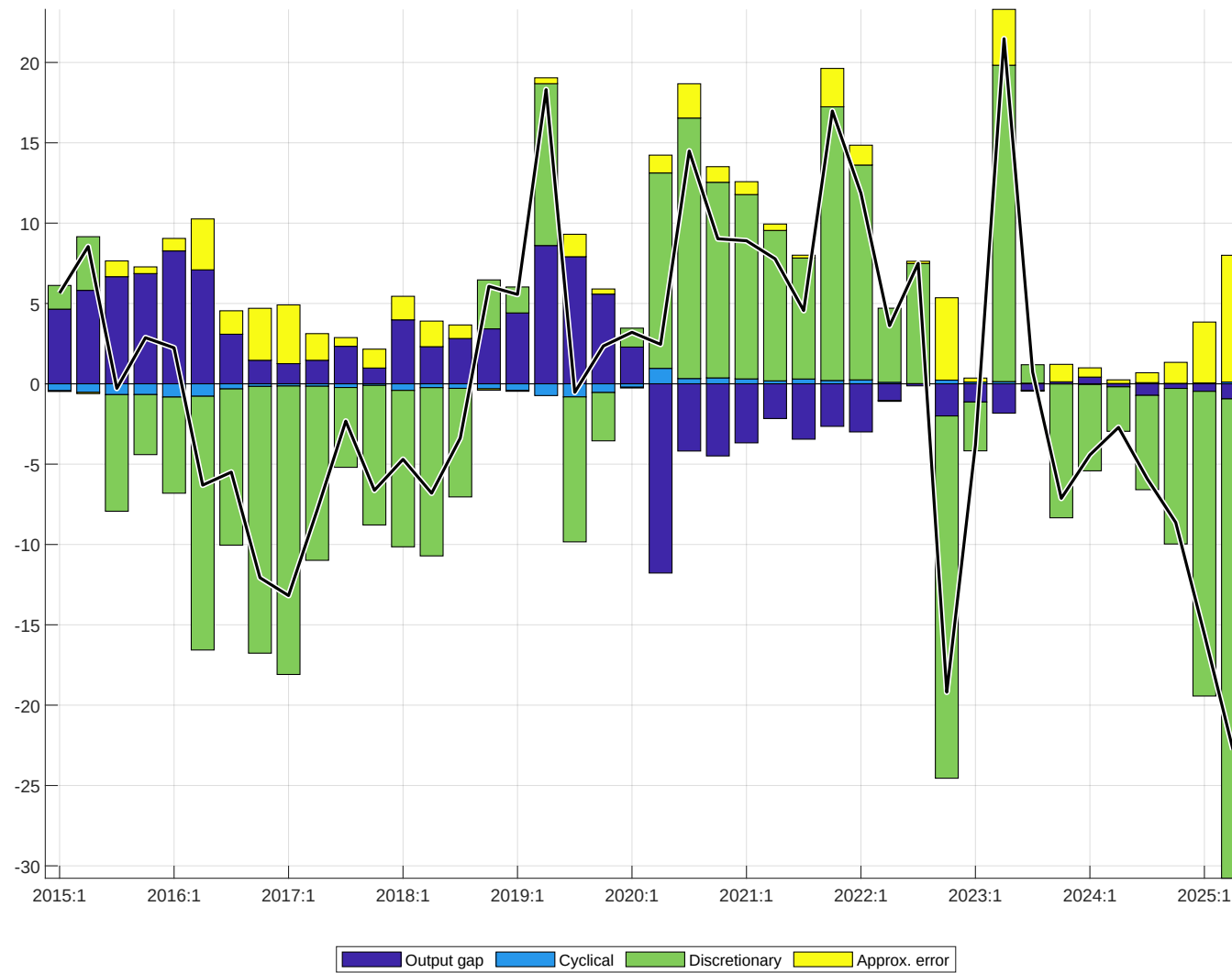


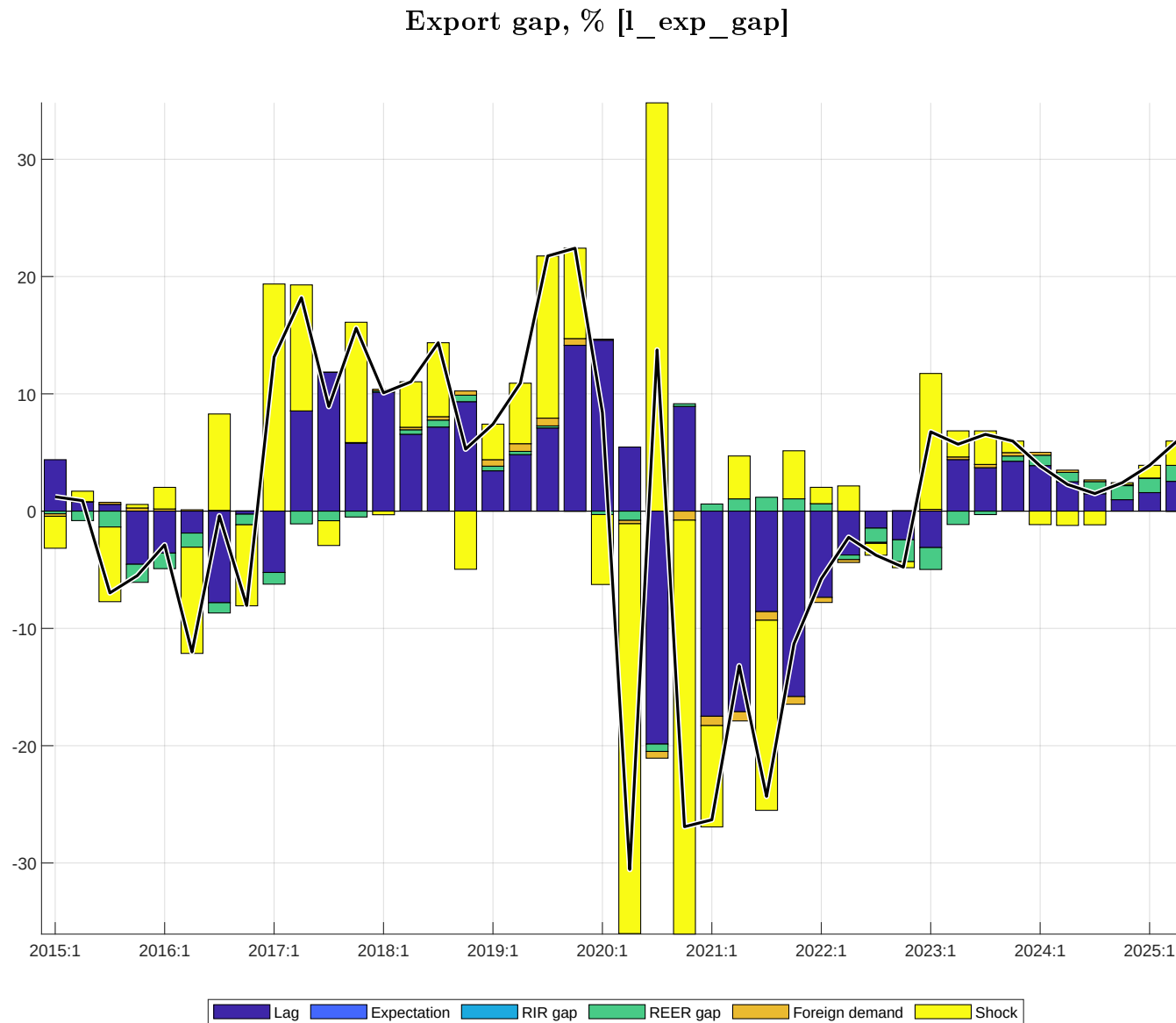
## Consumption gap, % [l\_cons\_gap]



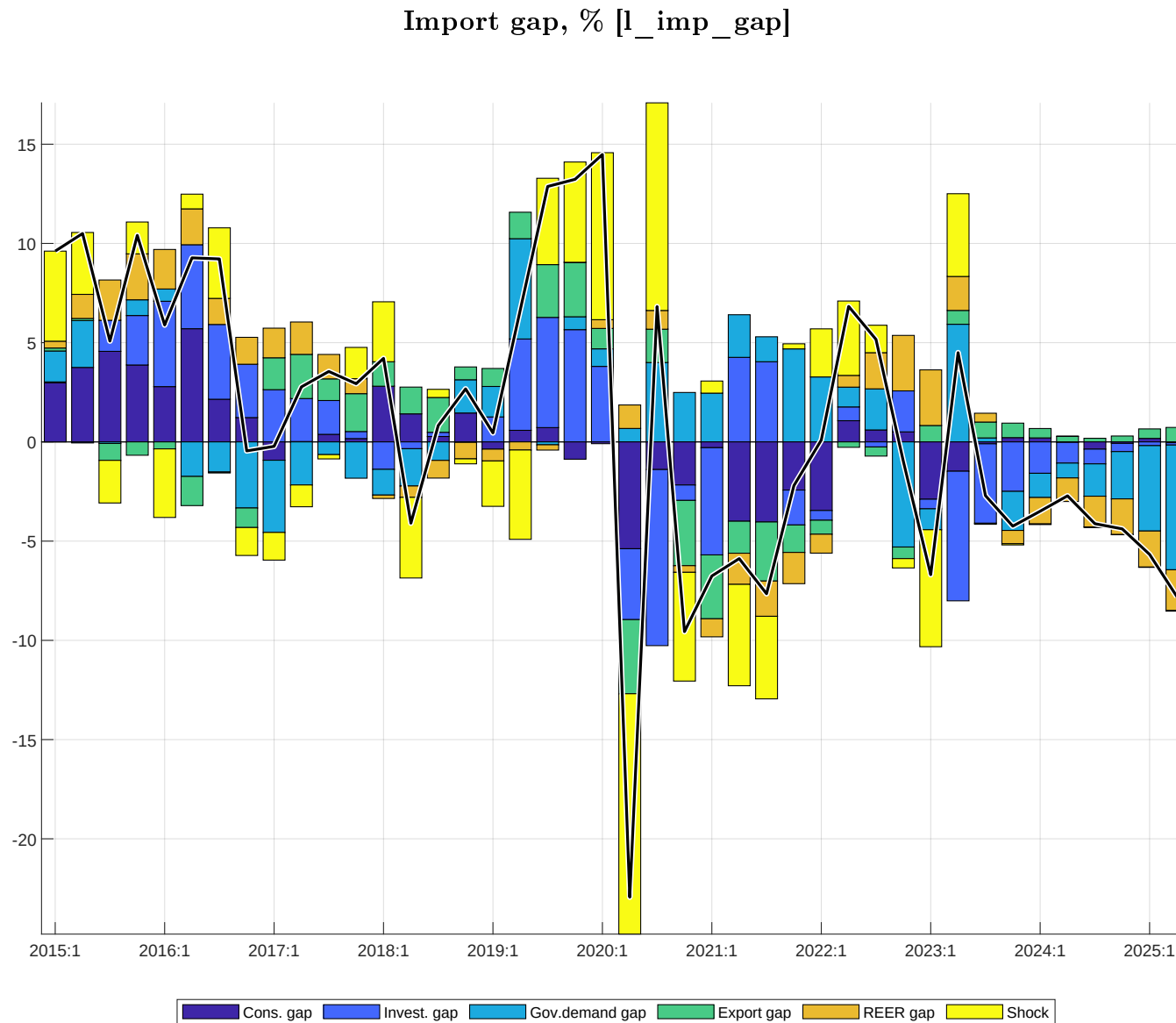


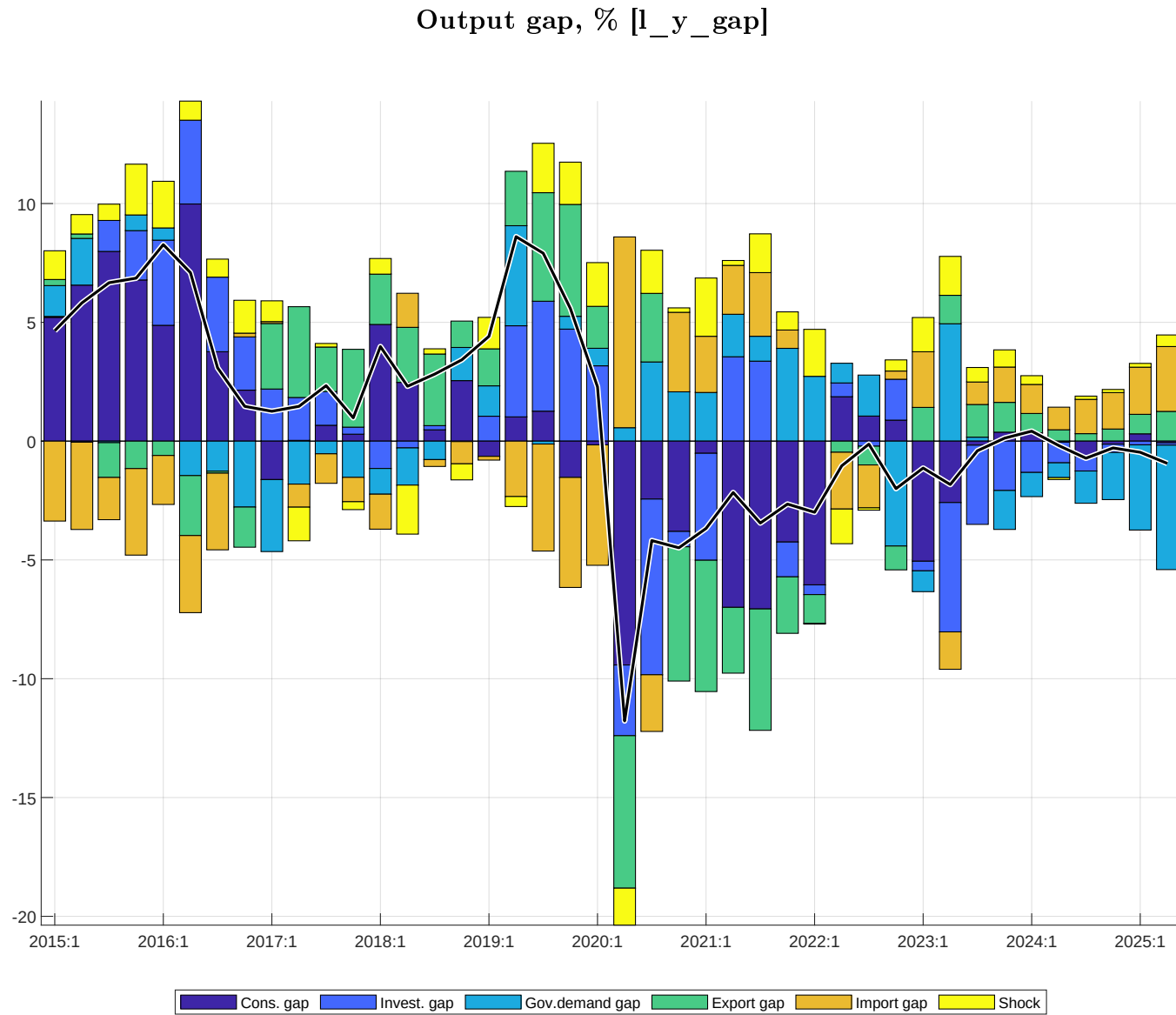
## Gov. demand gap, % [l\_gdem\_gap]



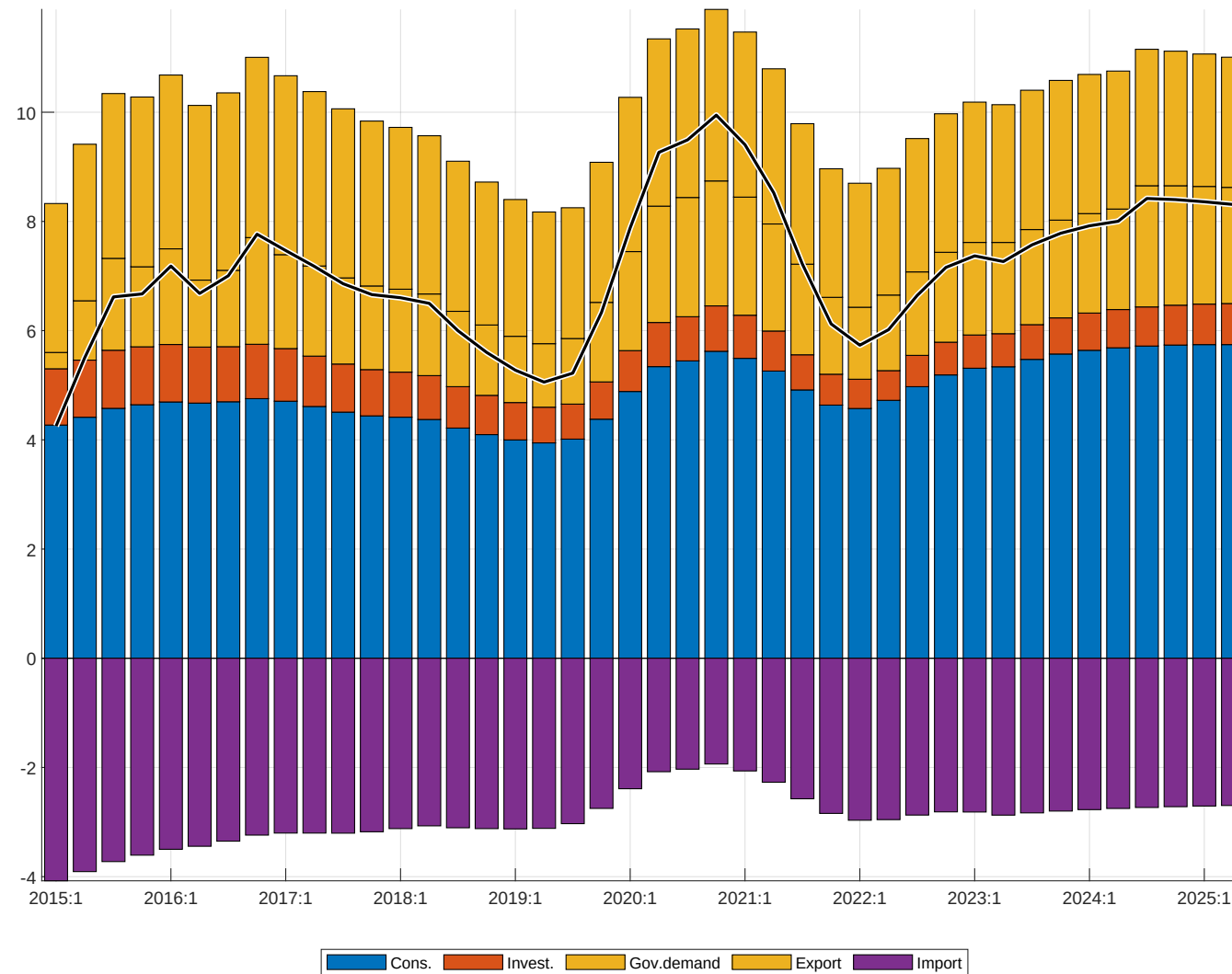




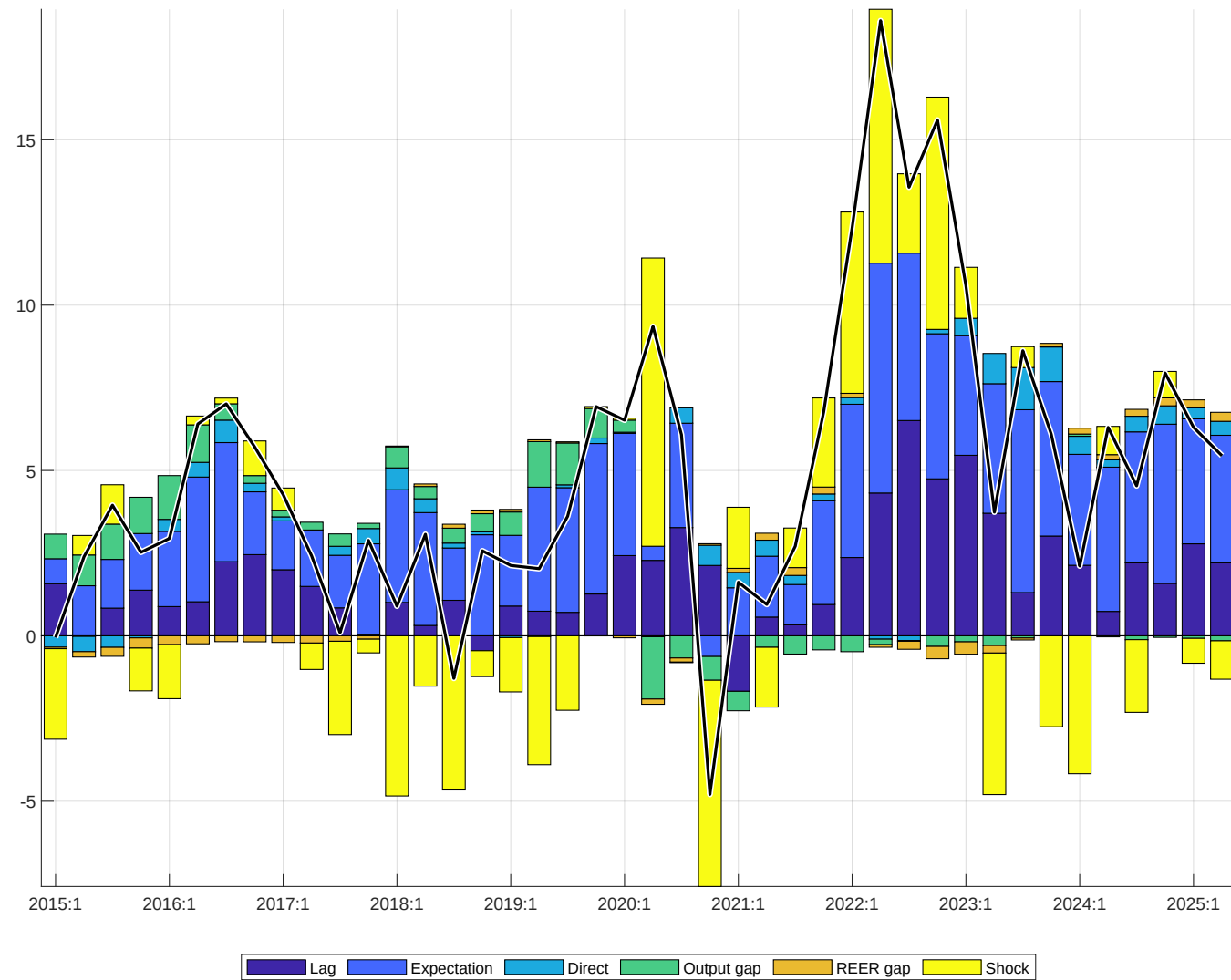




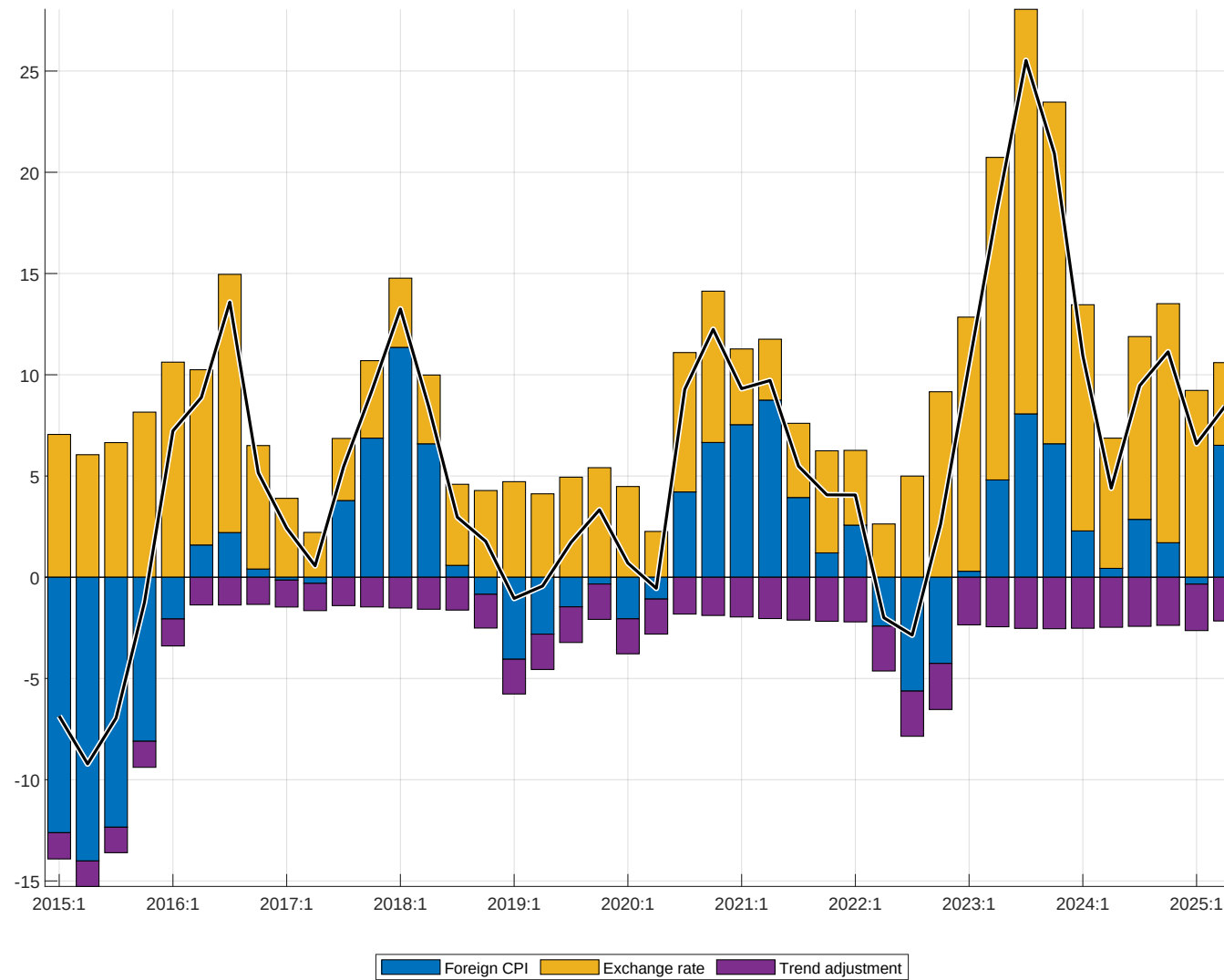
## GDP tnd., ann. QQ % [dl\_y\_tnd]



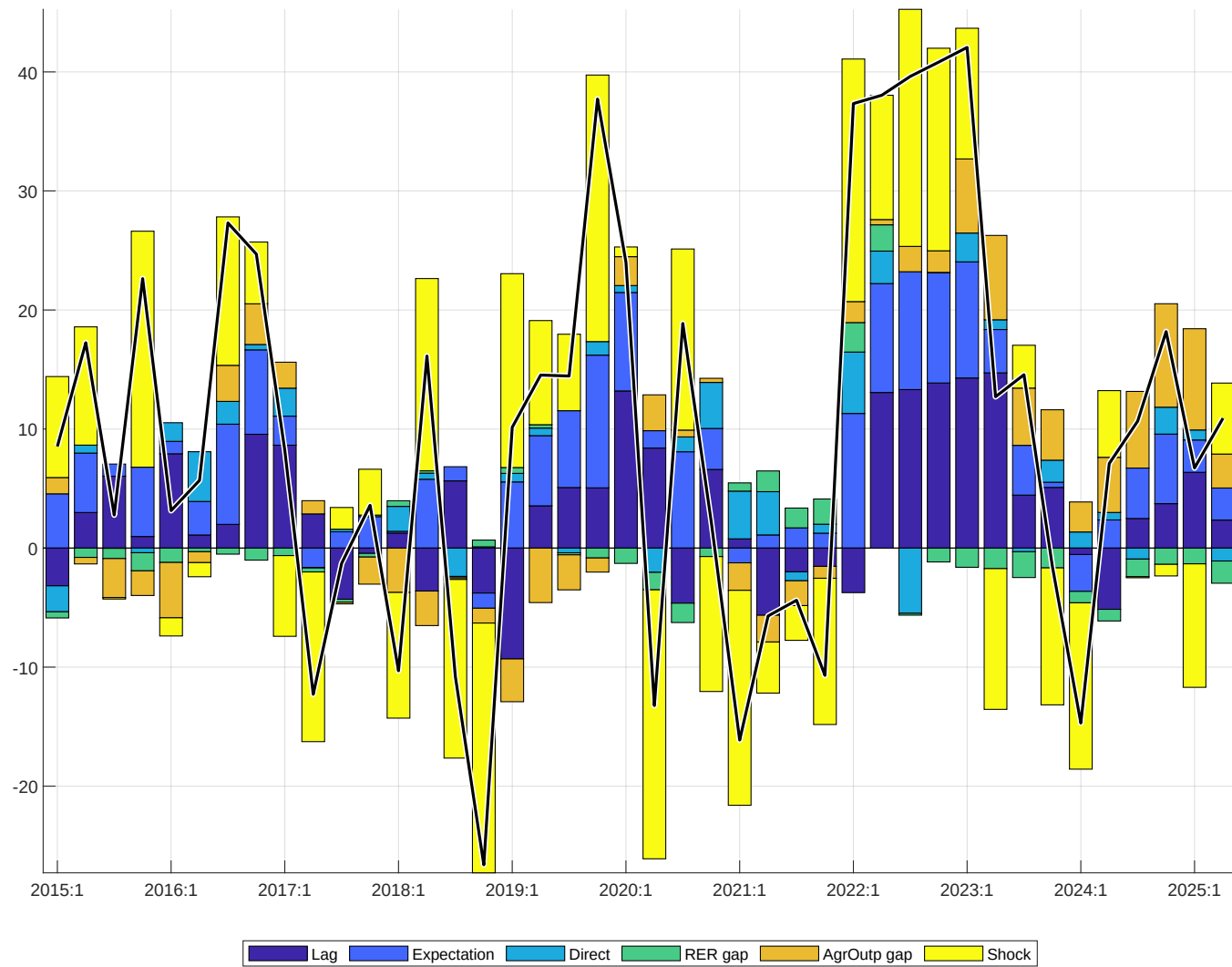
## Core CPI, ann. QQ % [dl\_cpi\_core]



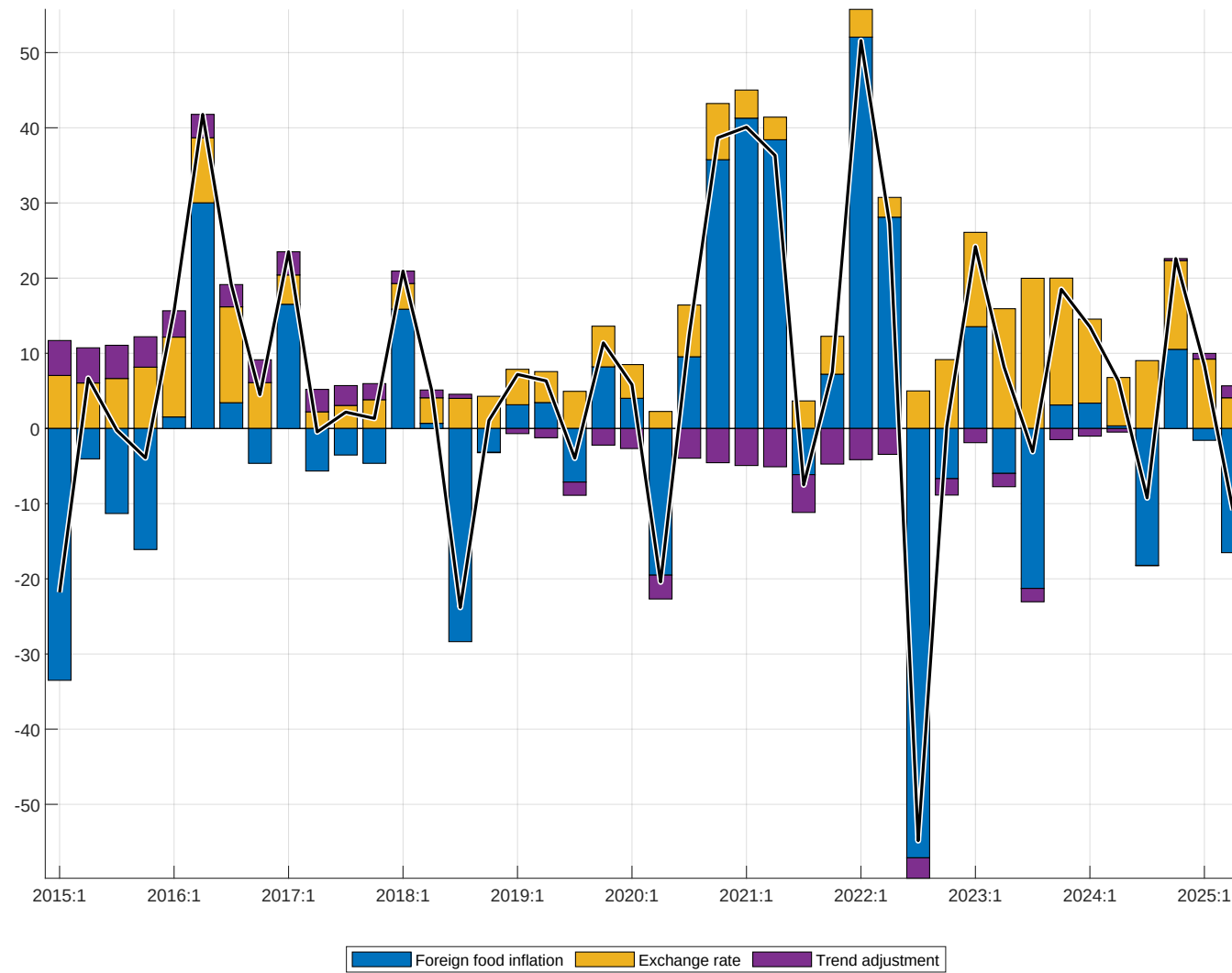
## Dir. ext. eff. in core, ann. QQ % [dl\_cpi\_core\_direct]



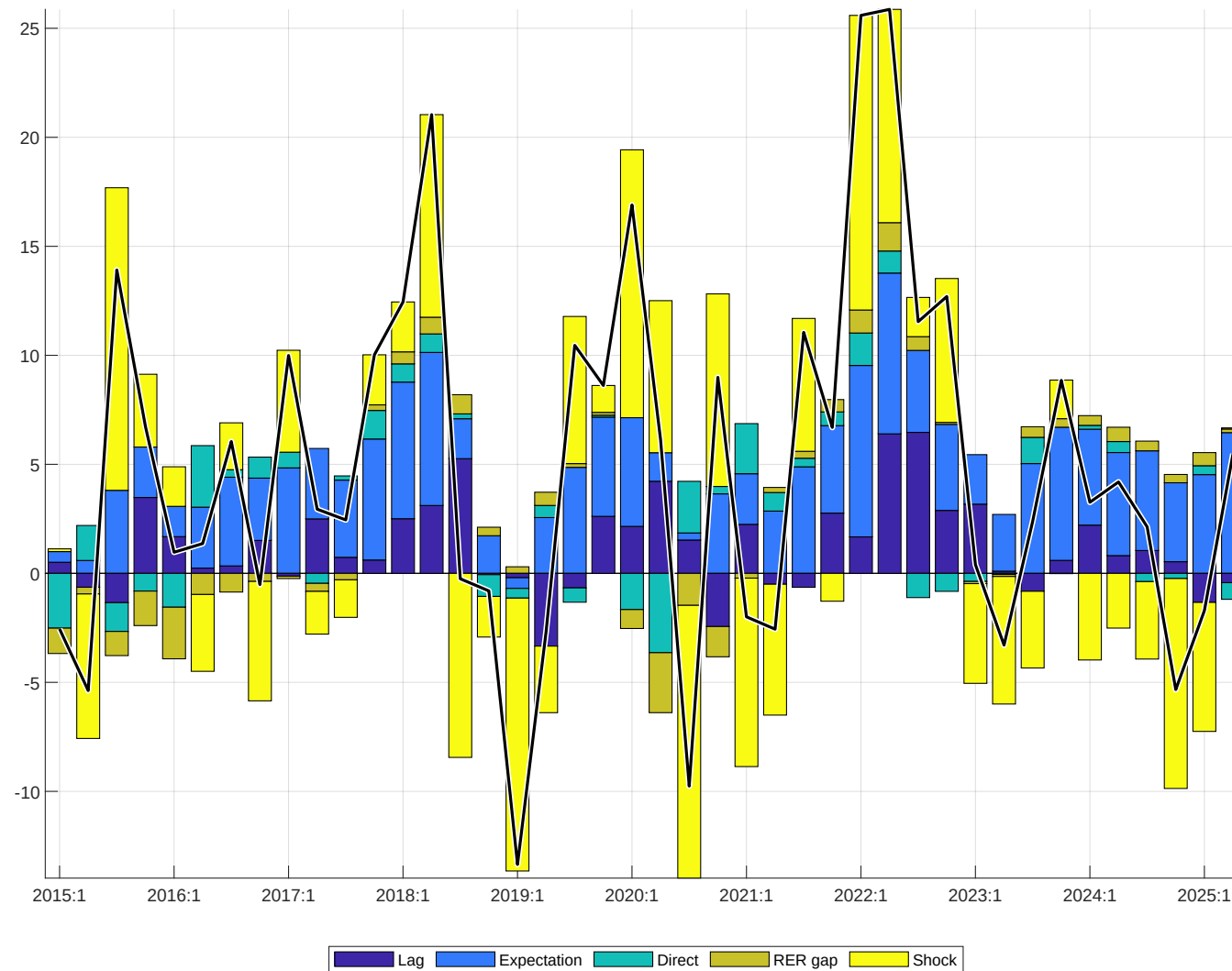
## Food CPI, ann. QQ % [dl\_cpi\_food]



## Dir. ext. eff. in food, ann. QQ % [dl\_cpi\_food\_direct]

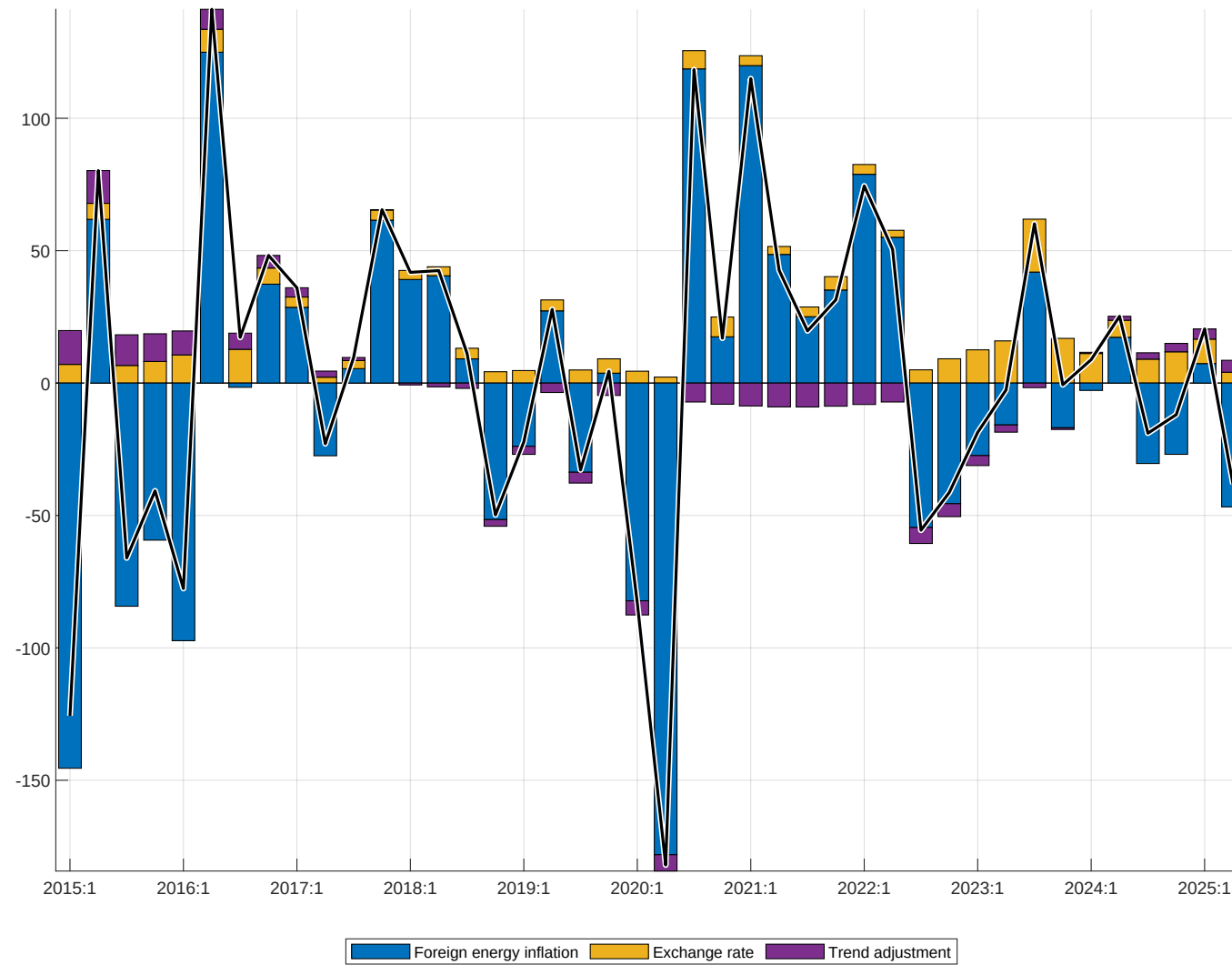


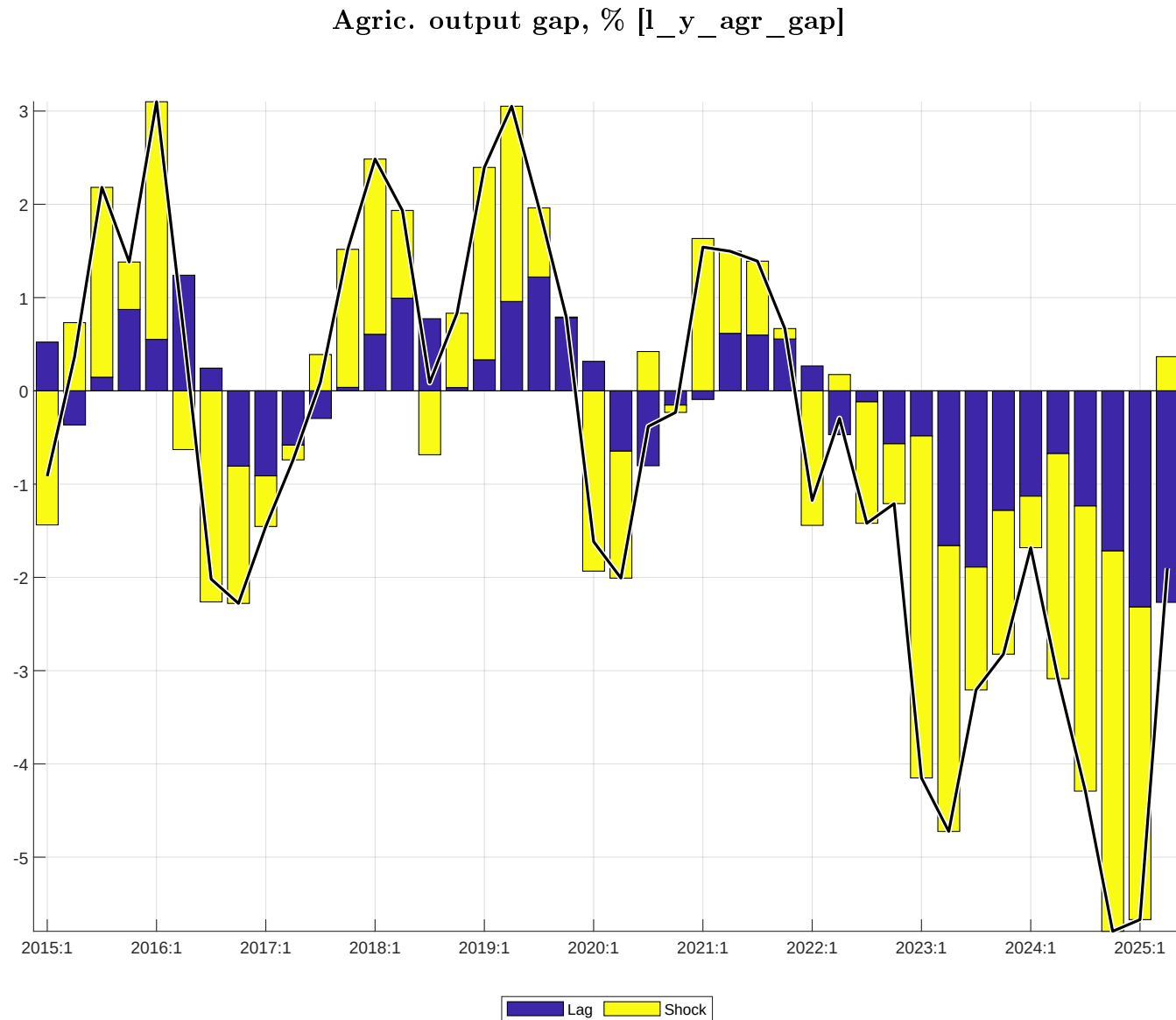
## Energy CPI, ann. QQ % [dl\_cpi\_ener]



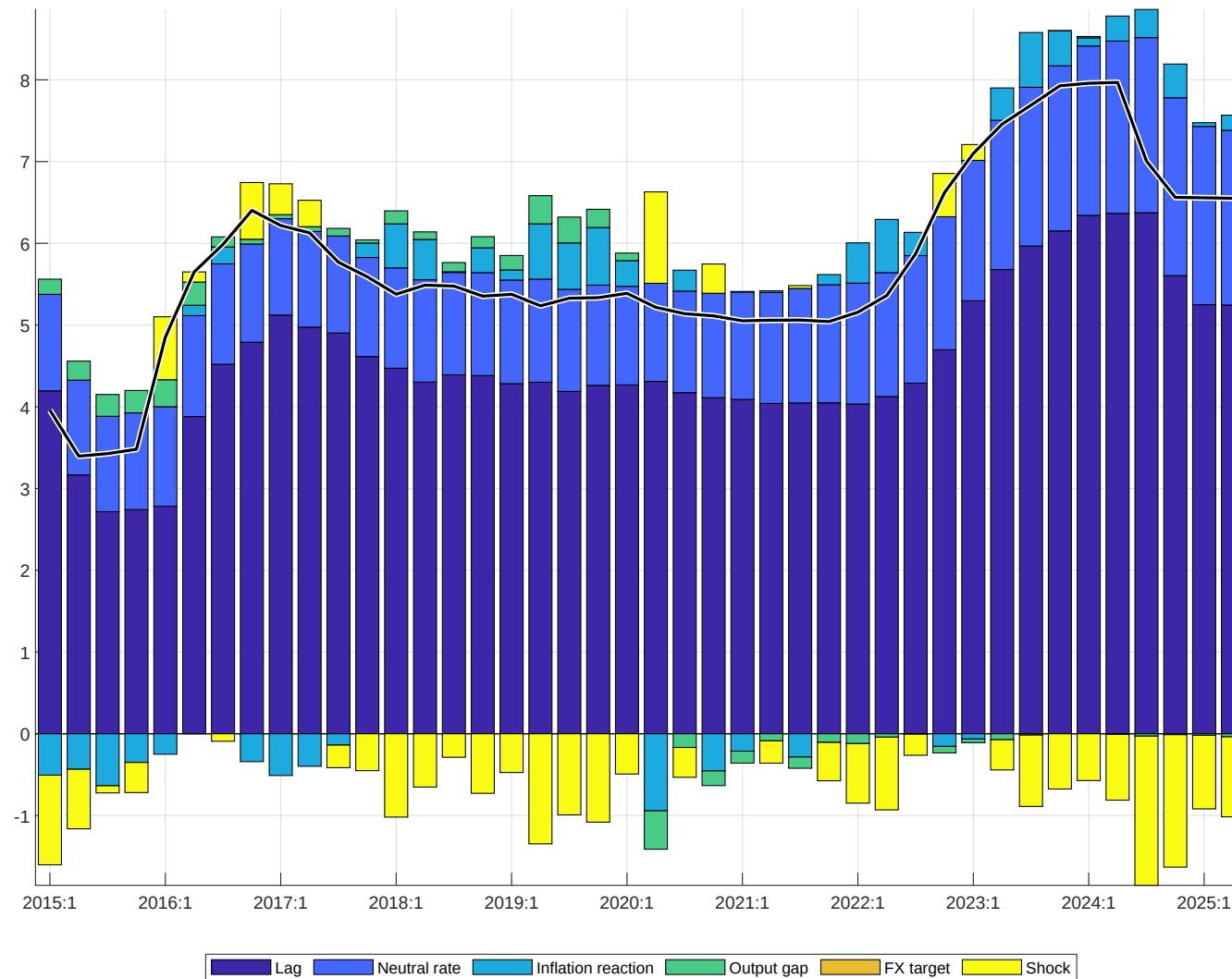


## Dir. ext. eff. in energy, ann. QQ % [dl\_cpi\_ener\_direct]

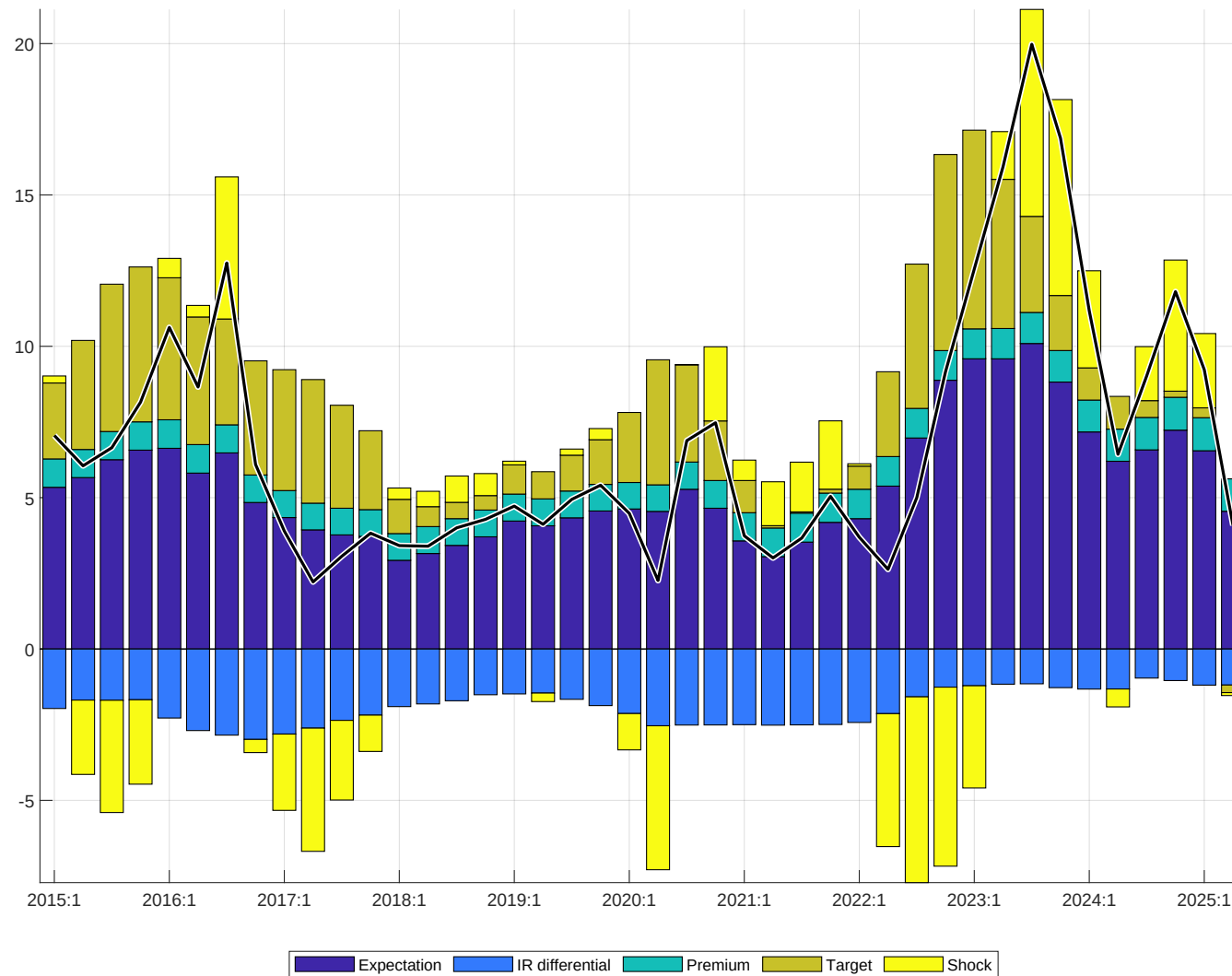




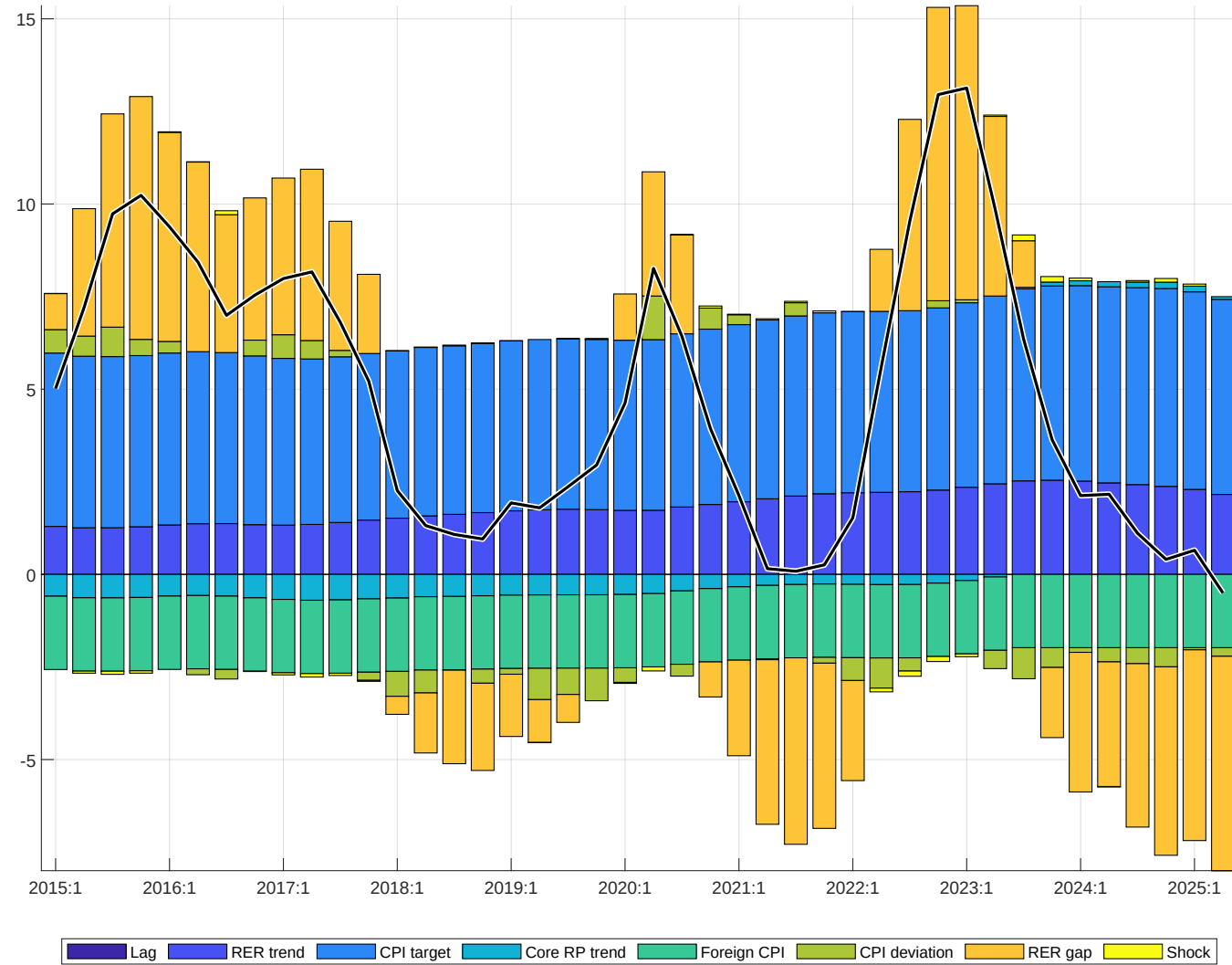
## Policy rate (IB rate used), % [i]



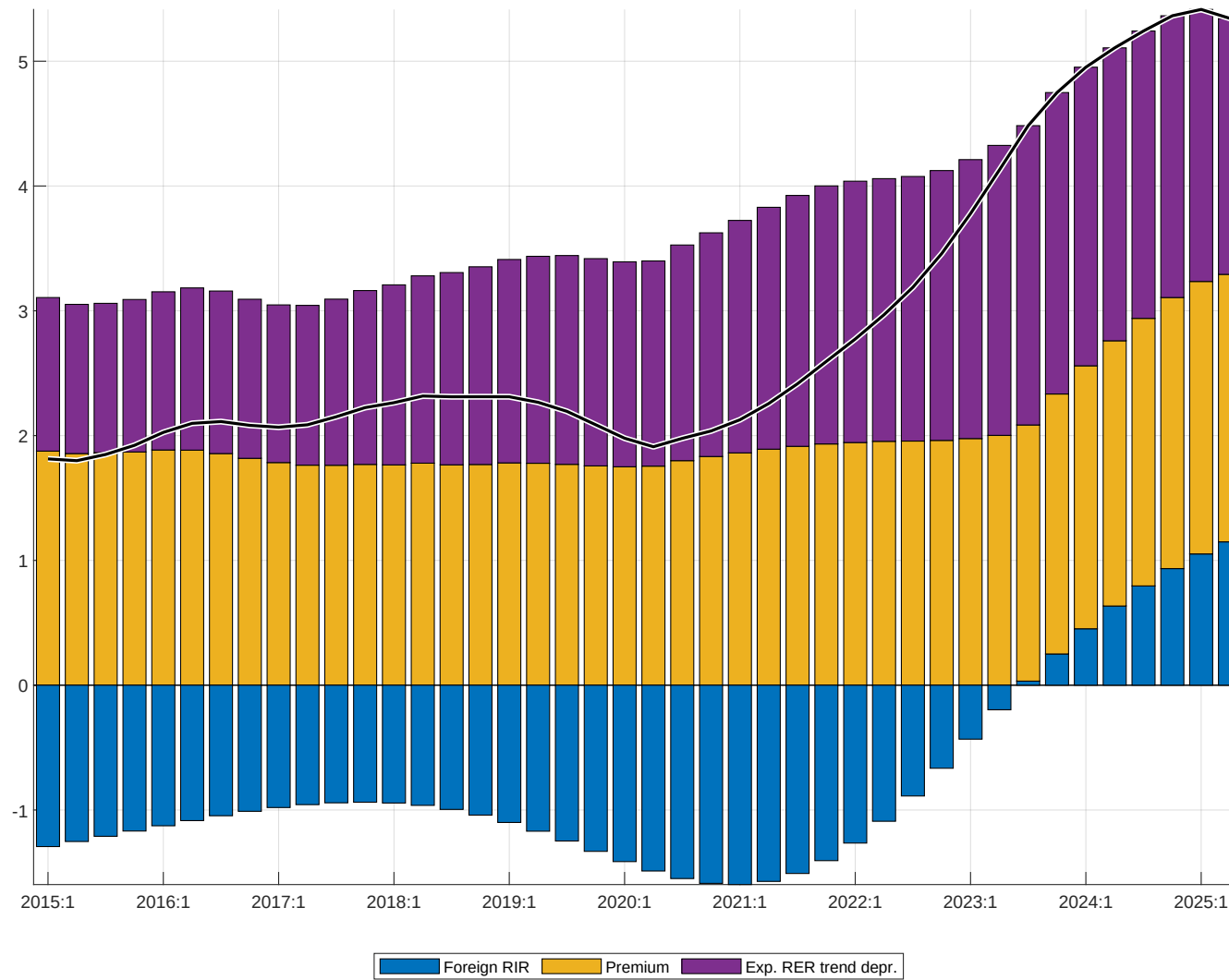
## Exchange rate, QQ annualized % [dl\_s]



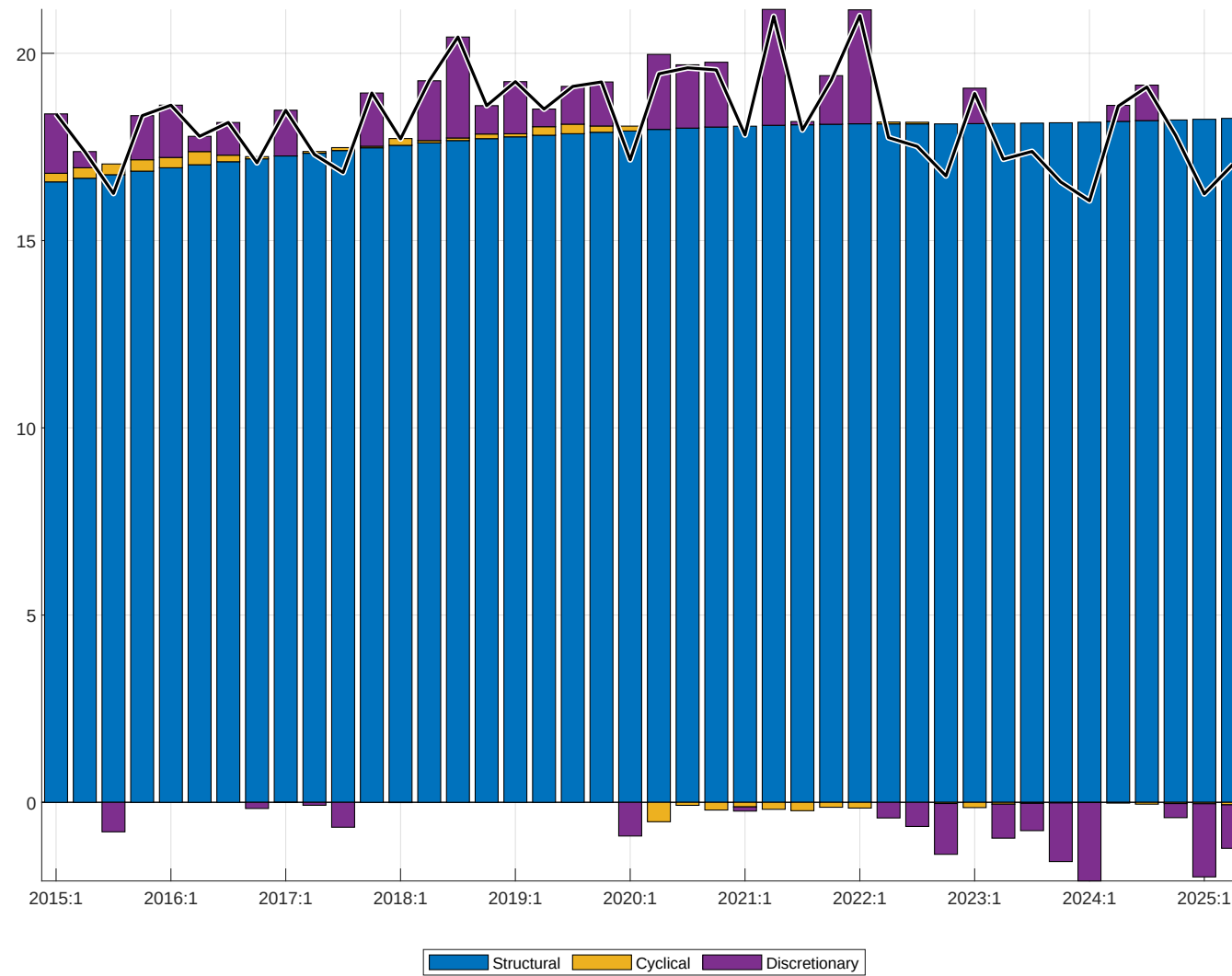
## Exchange rate target, QQ % [dl\_s\_tar]



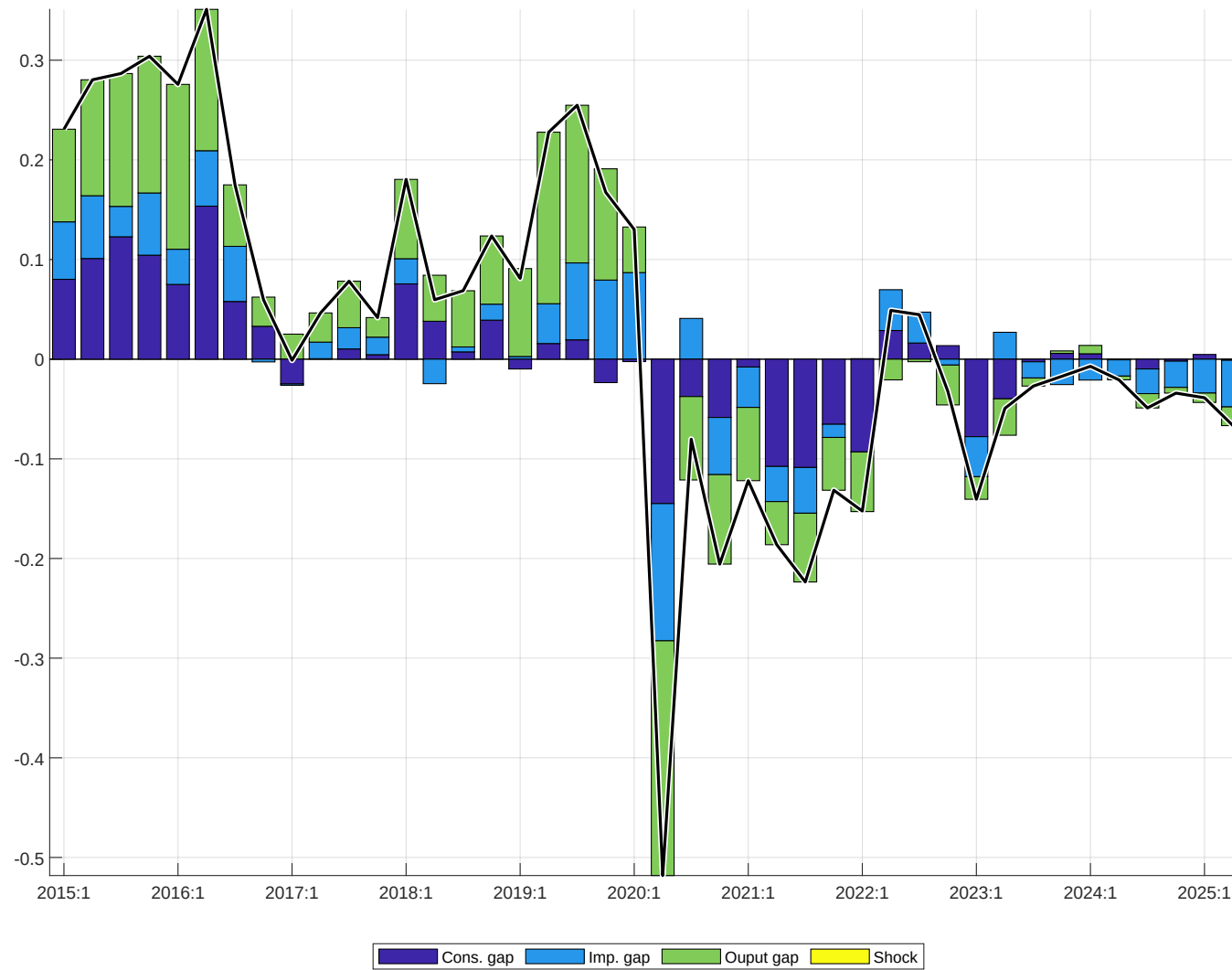
## RIR (policy) trend, % [r\_tnd]



## Govt. rev., % of GDP [grev\_y]

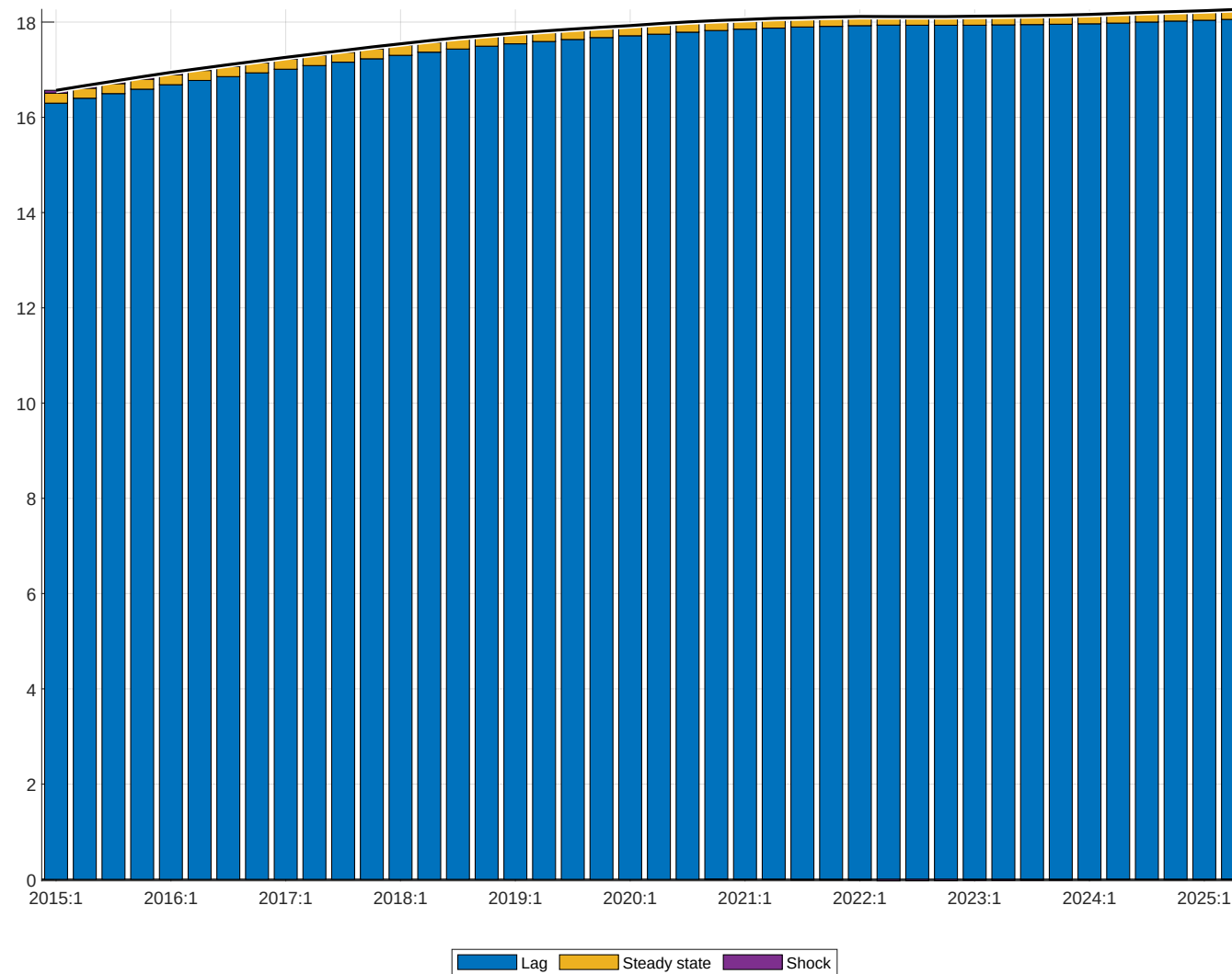


## Cyc. govt. rev., % of GDP [grev\_y\_cyc]

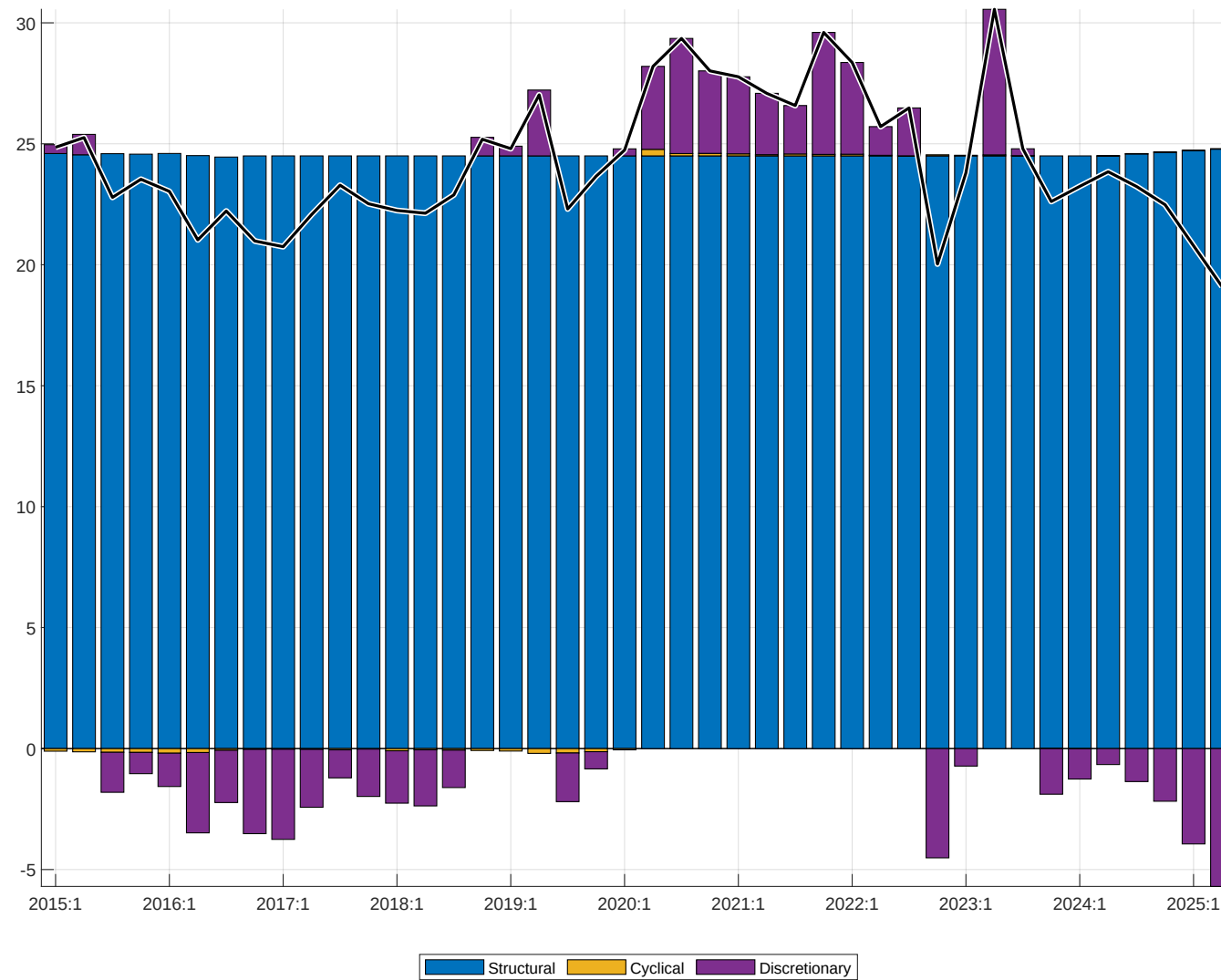


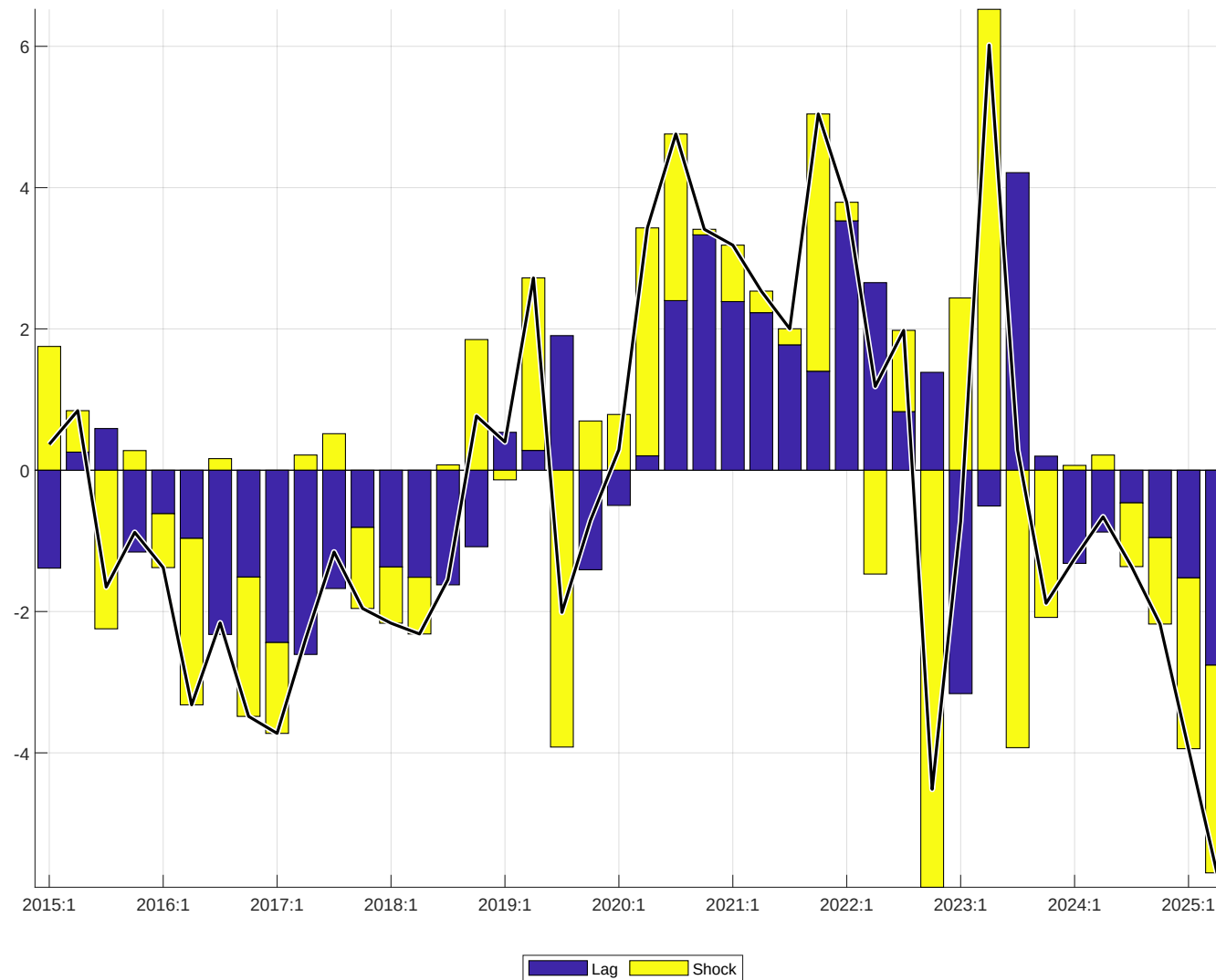


## Struct. govt. rev., % of GDP [grev\_y\_str]

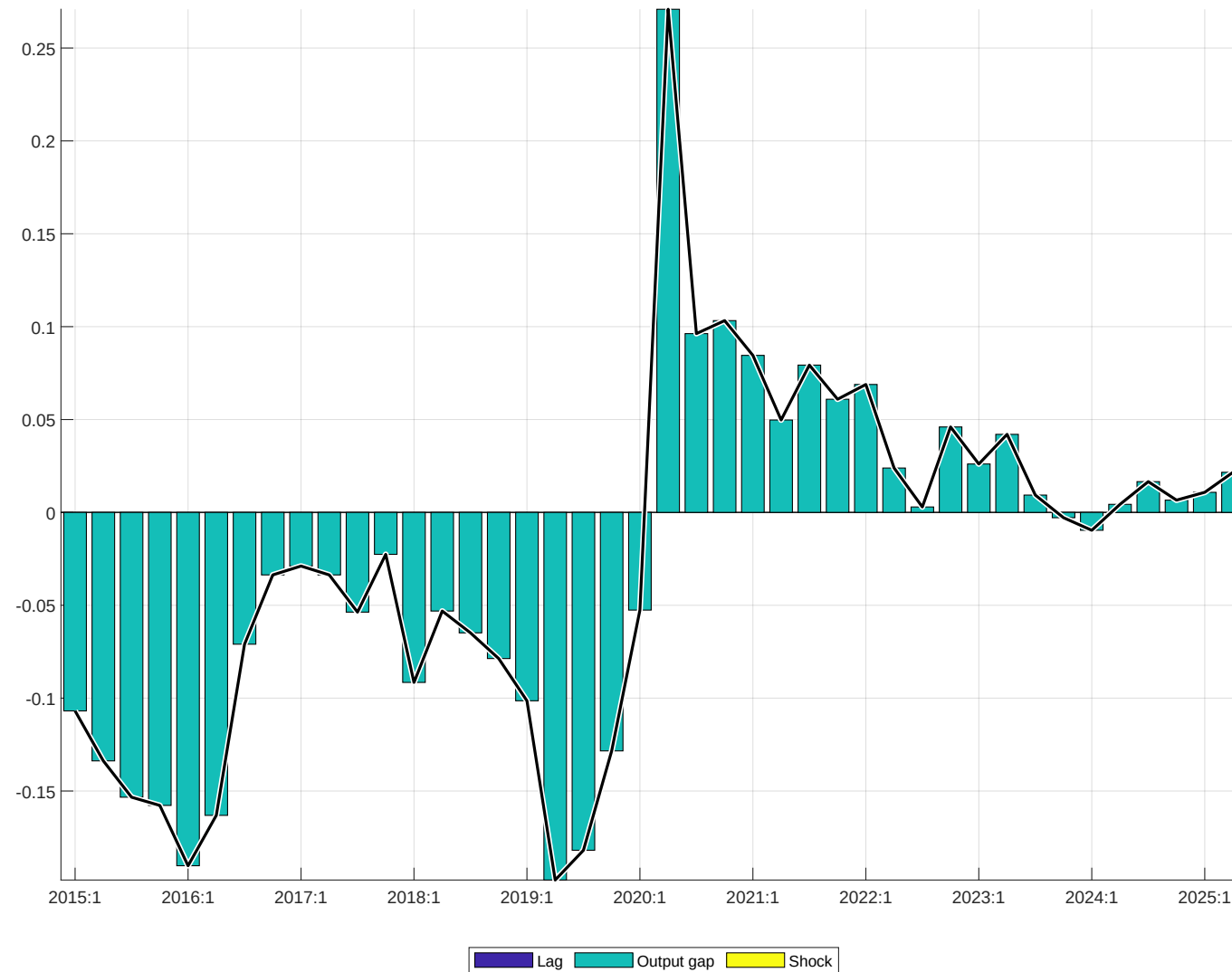


## Govt. demand (G&amp;S), % GDP [gdem\_y]

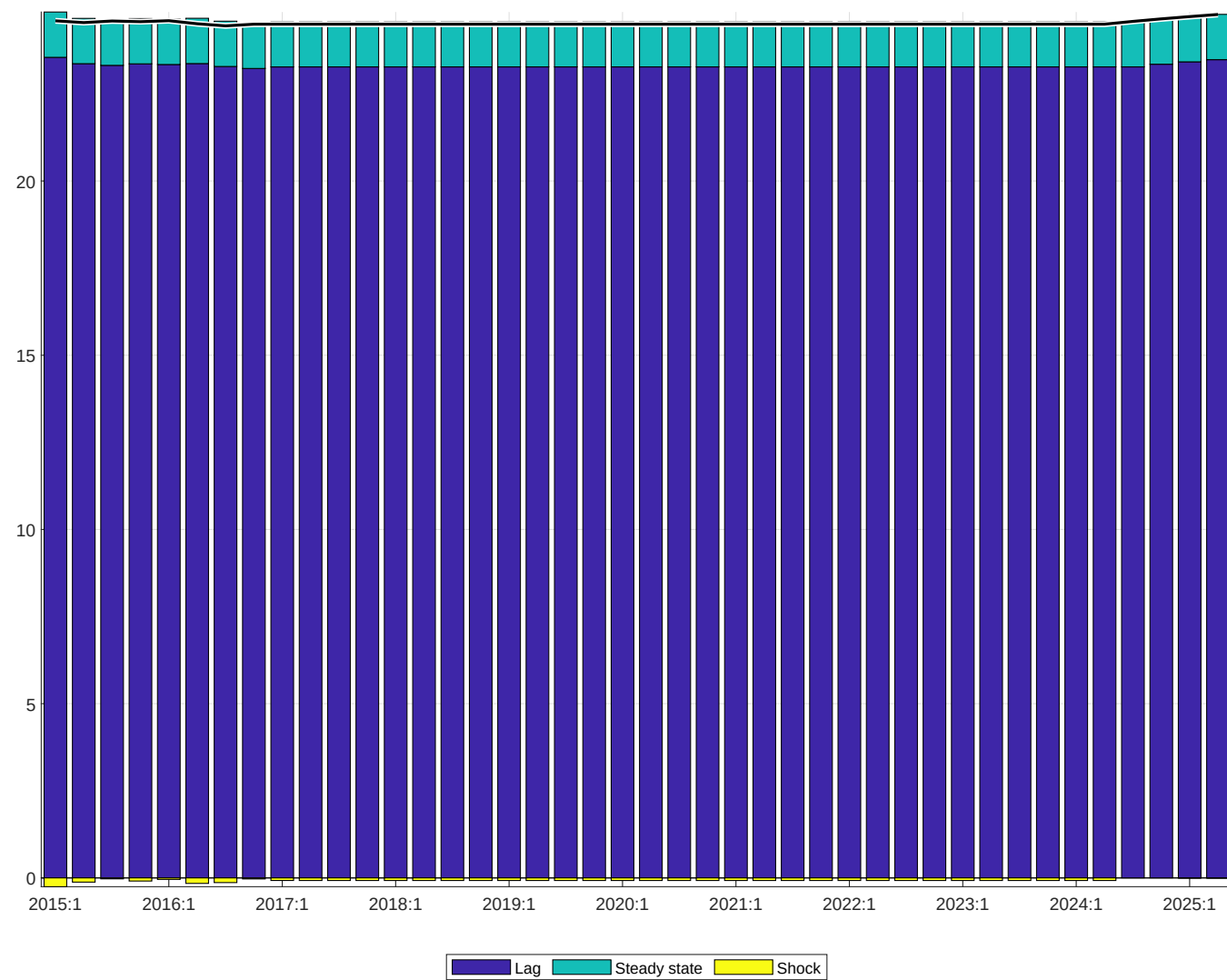


**Discr. govt. demand (G&S), % GDP [gdem\_y\_discr]**

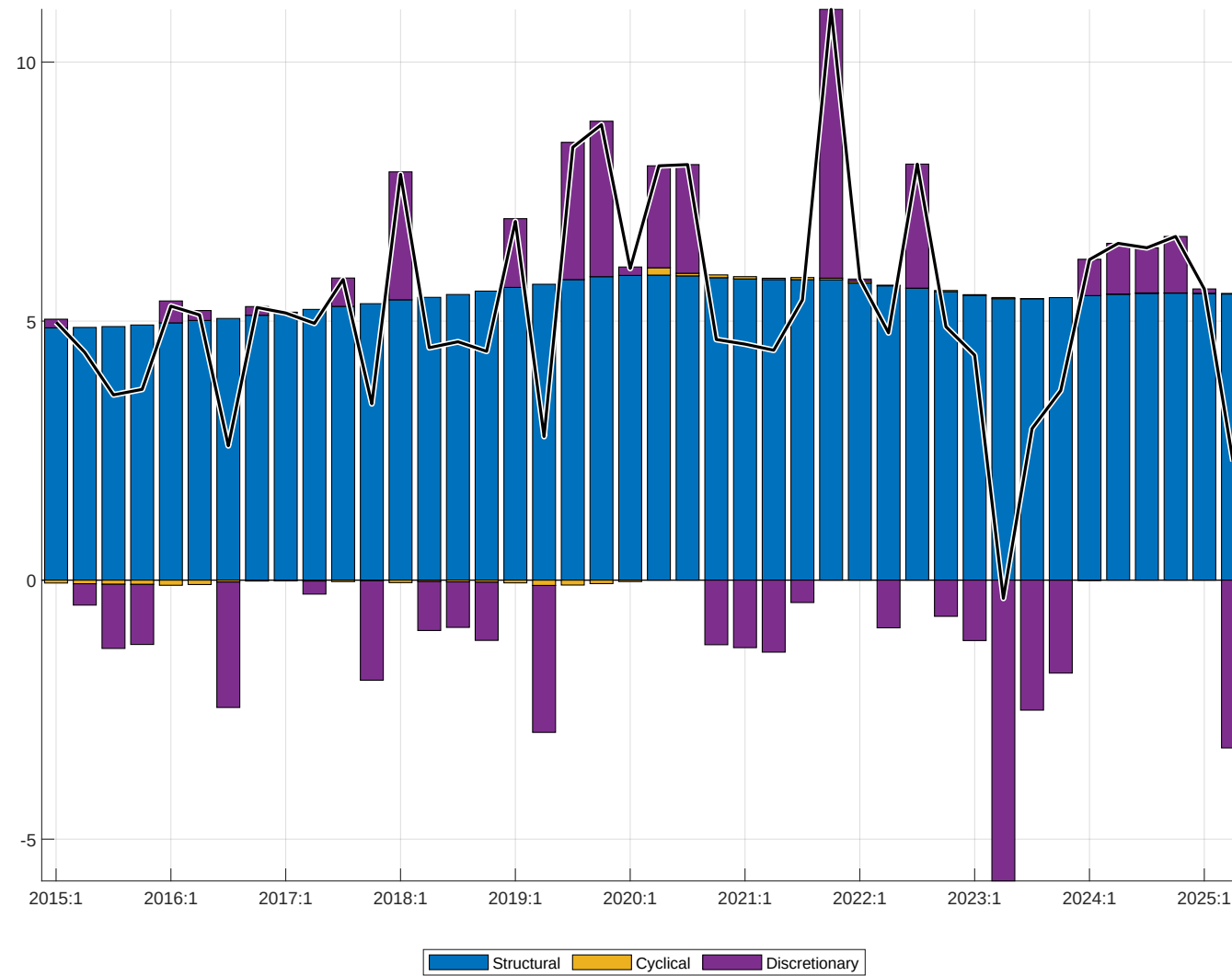
## Cyc. govt. demand (G&amp;S), % GDP [gdem\_y\_cyc]



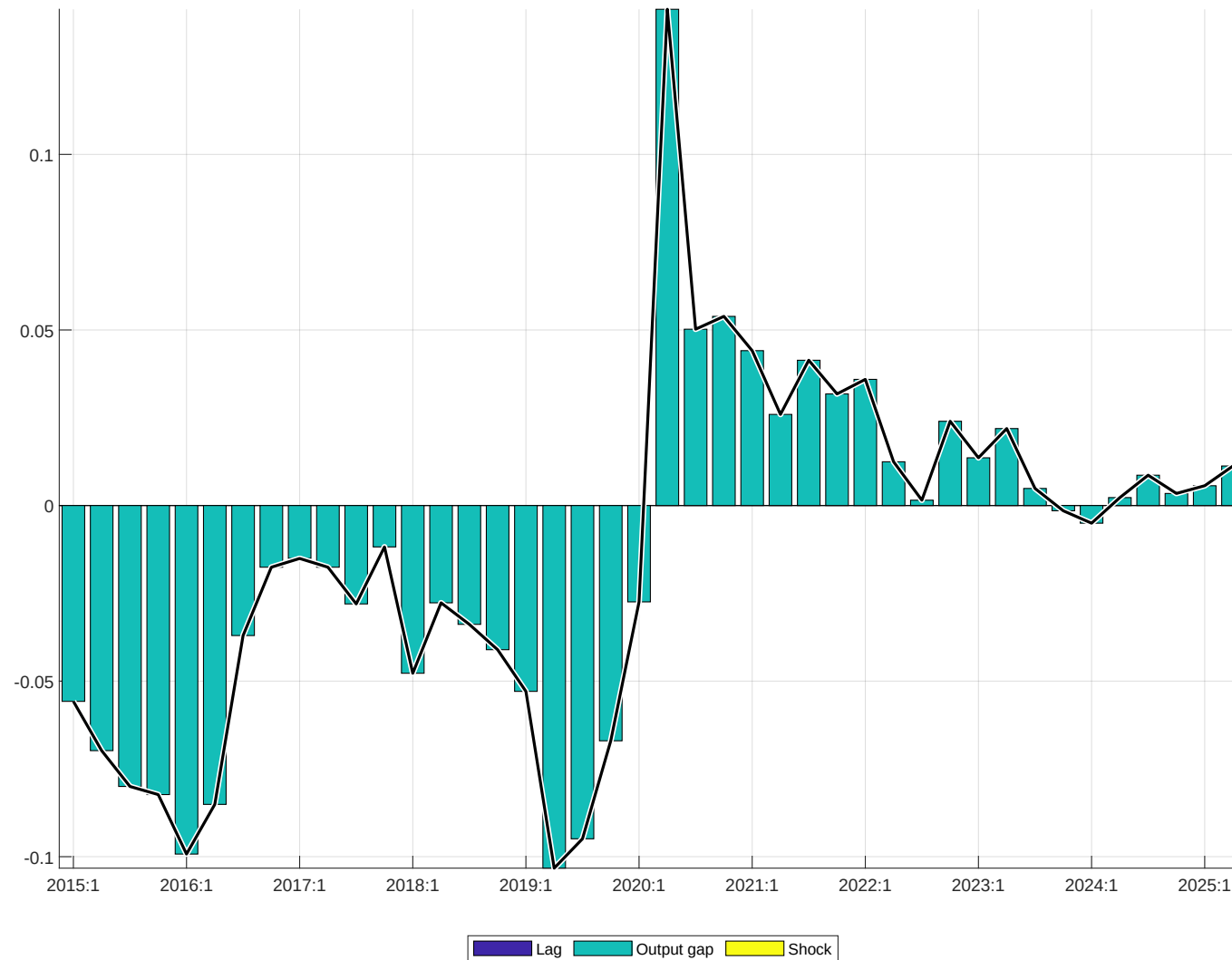
## Struct. govt. demand (G&amp;S), % GDP [gdem\_y\_str]



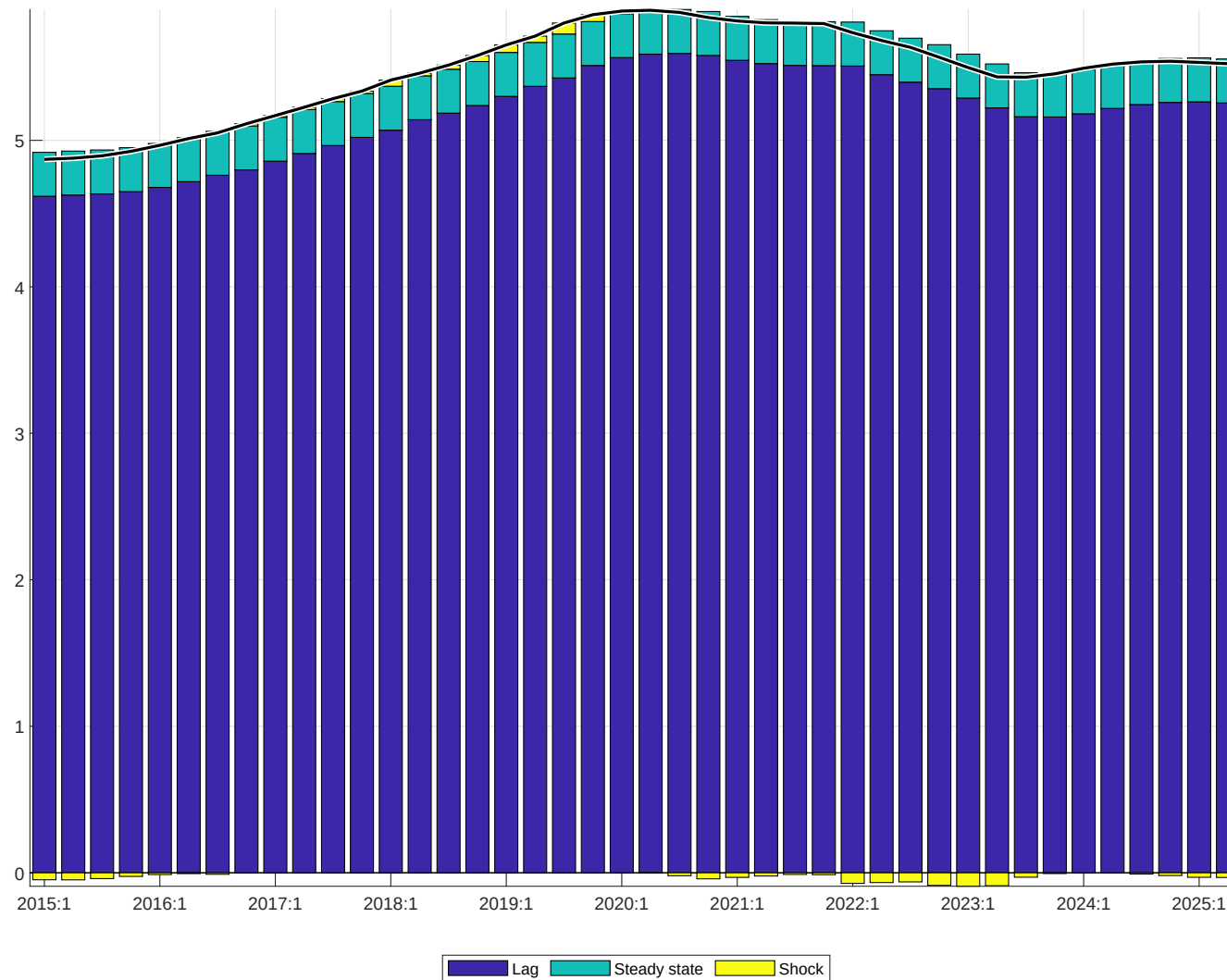
## Other govt. exp., % to GDP [oexp\_y]



## Cyc. other govt. exp., % to GDP [oexp\_y\_cyc]

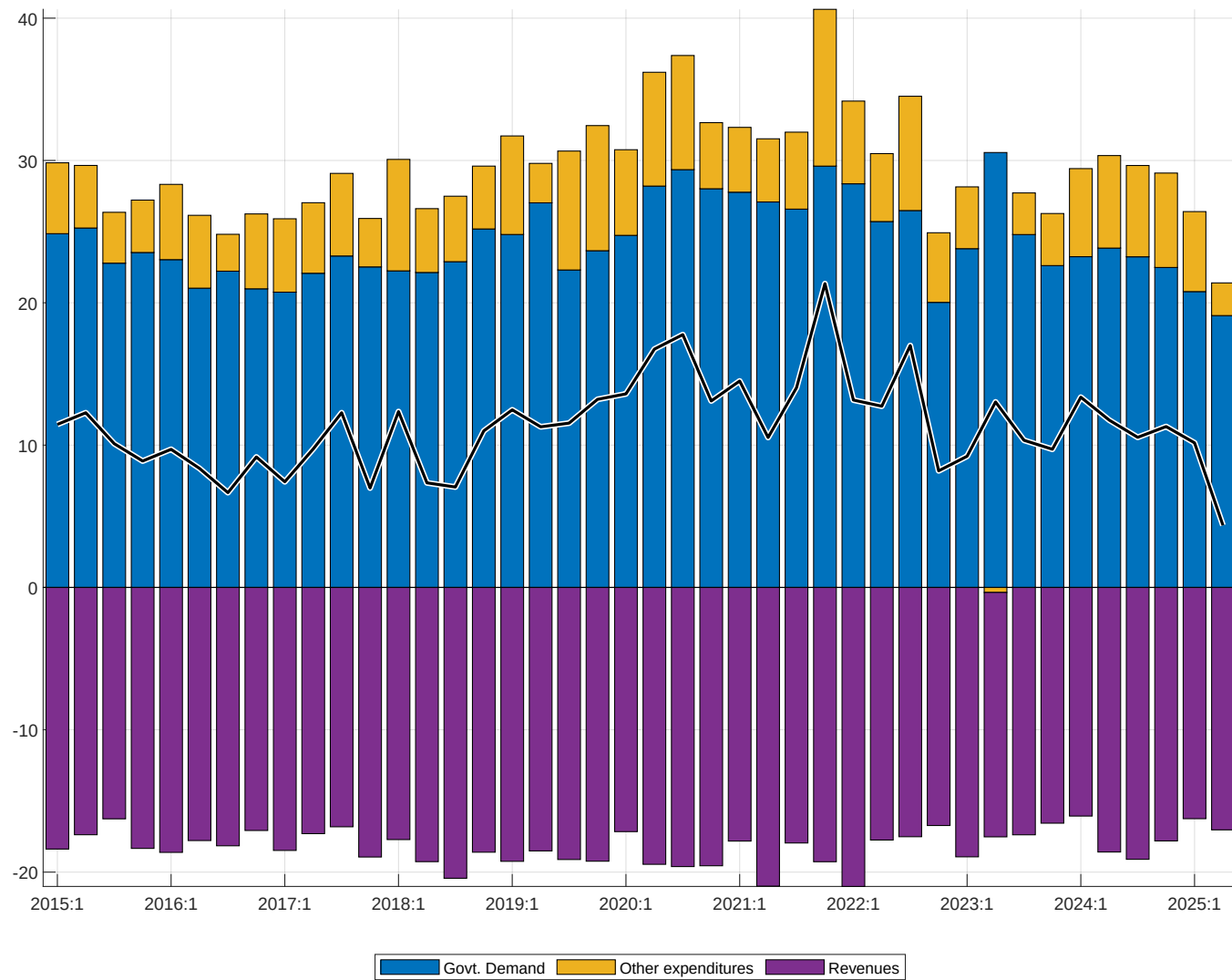


## Struct. other govt. exp., % to GDP [oexp\_y\_str]

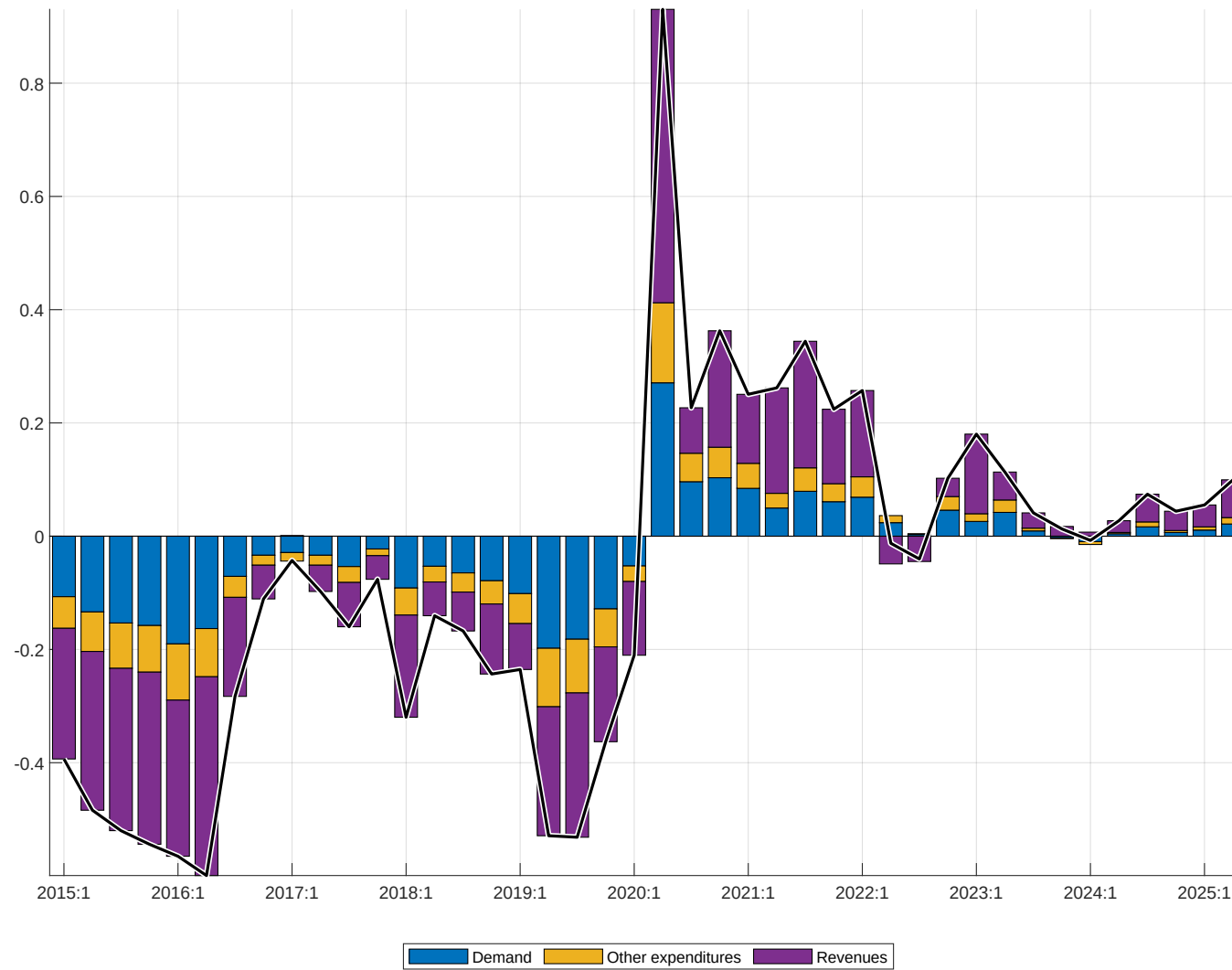




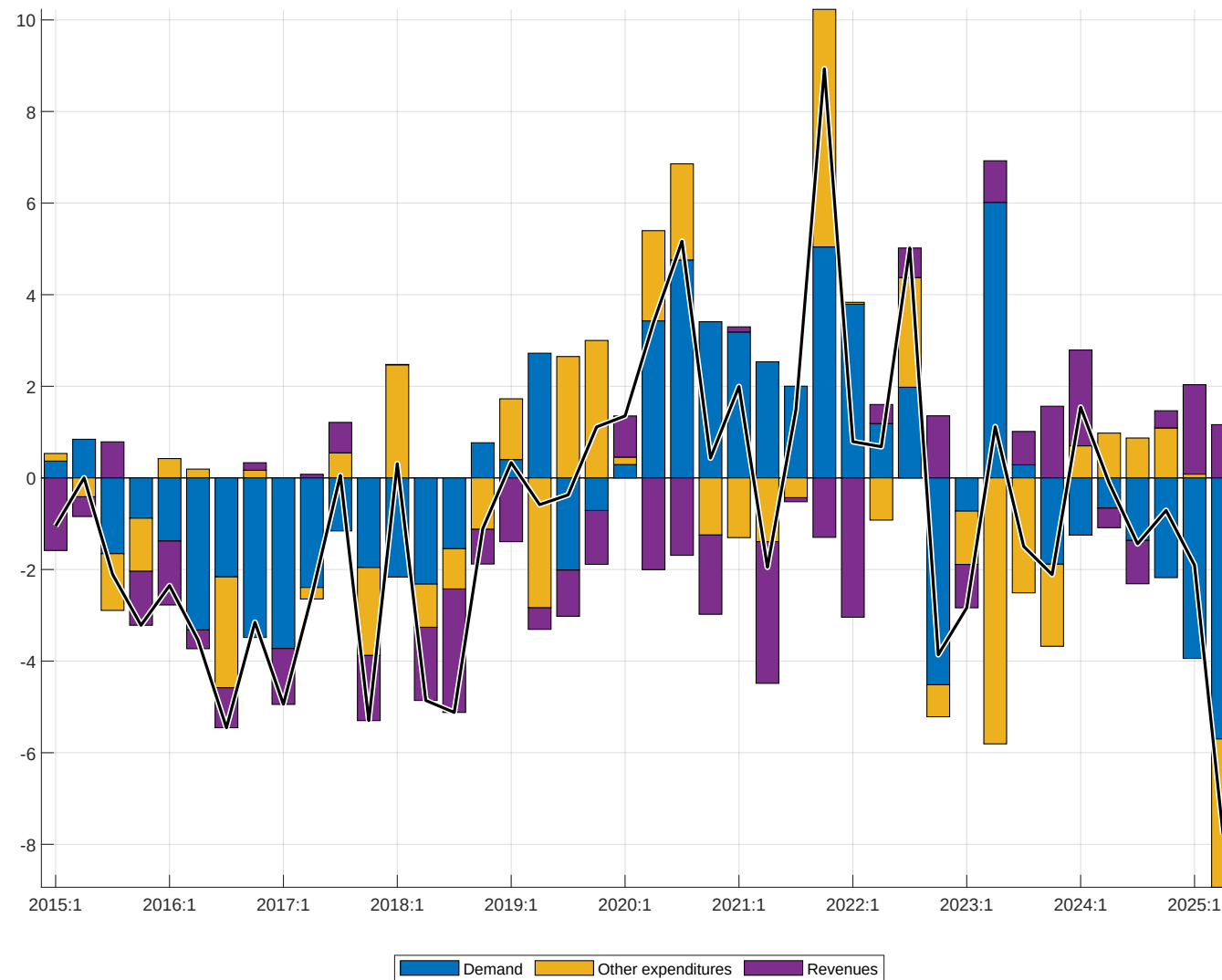
Budget deficit, % of GDP [def\_y]



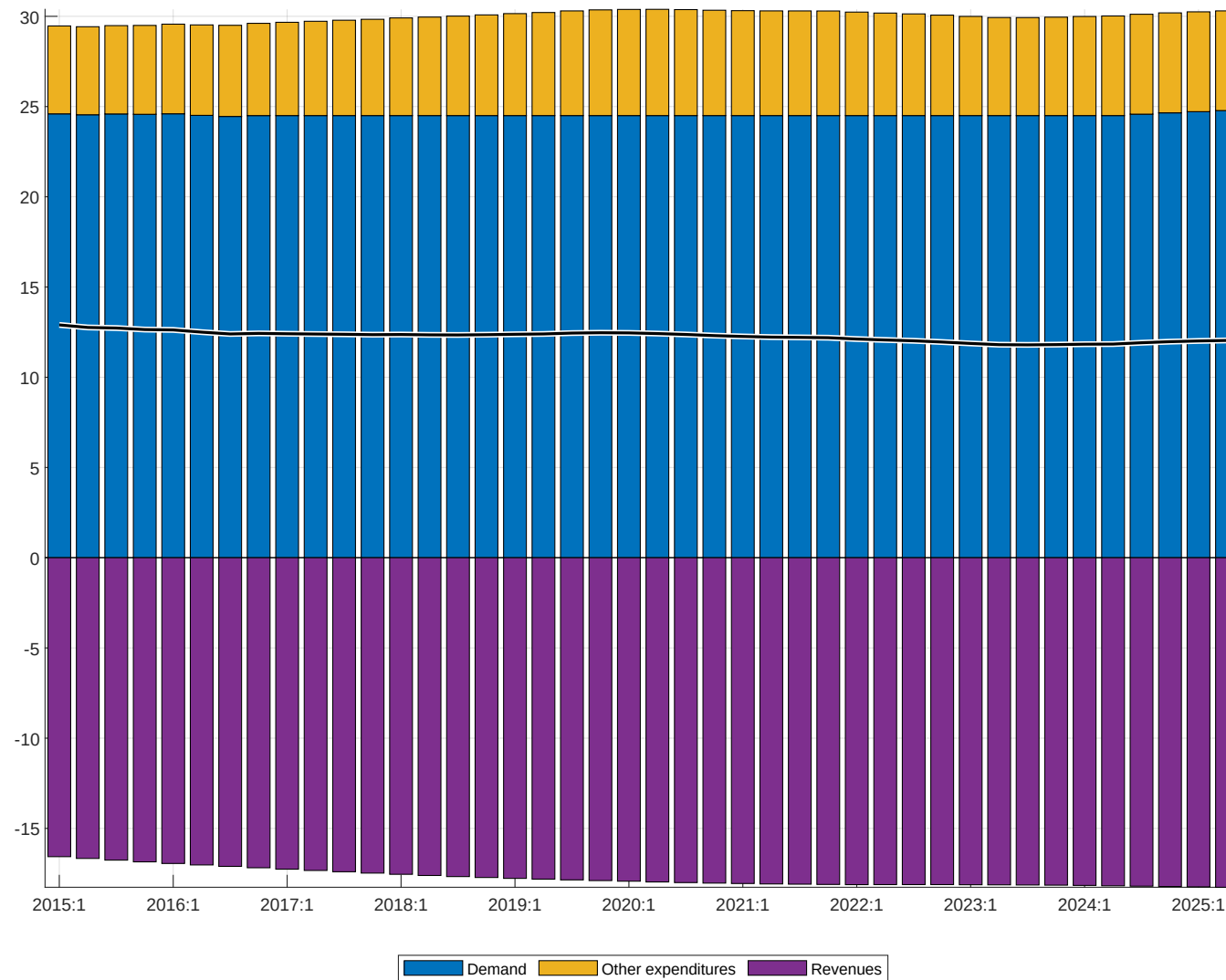
## Cyc. deficit, % of GDP [def\_y\_cyc]



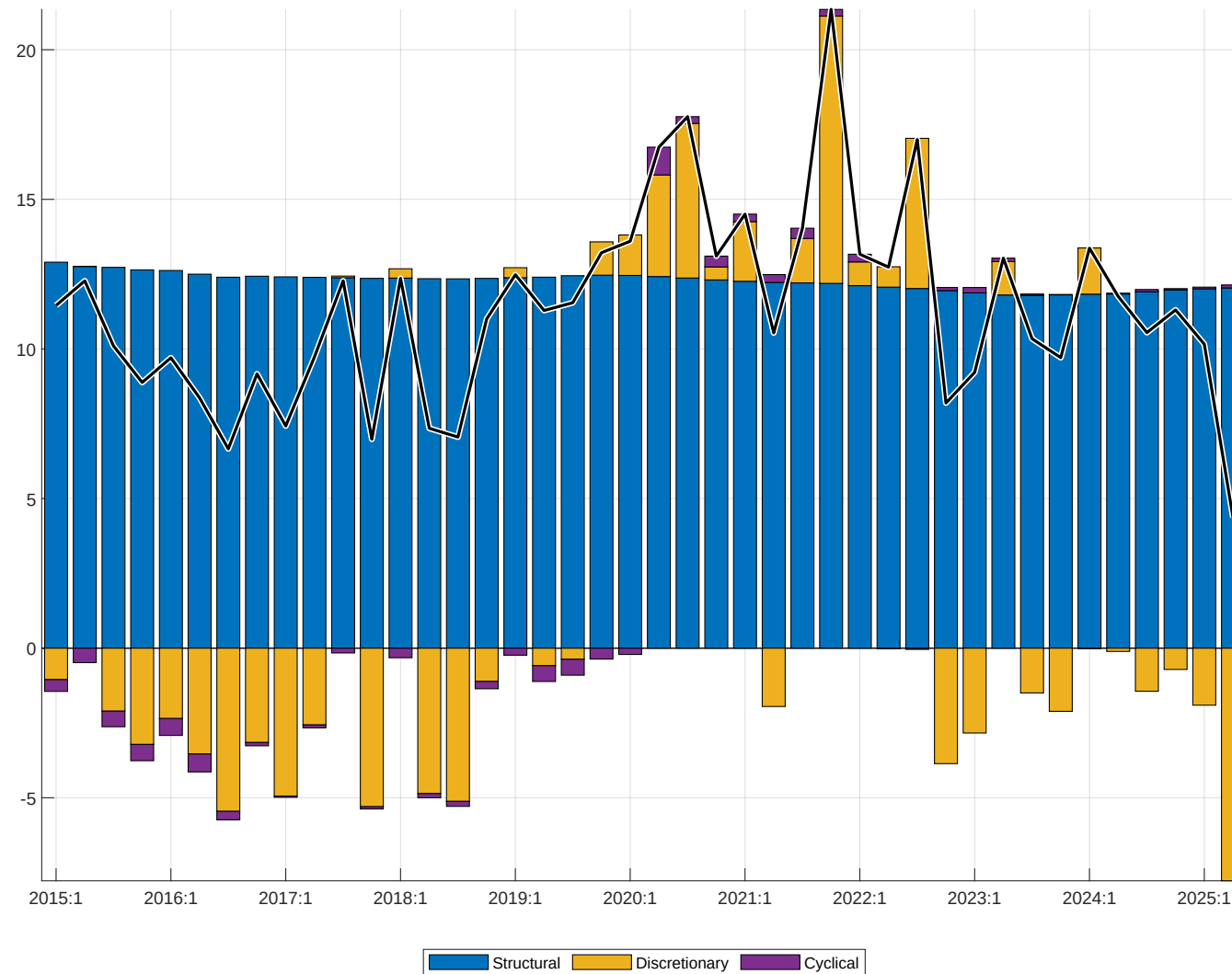
Discr. deficit, % of GDP [def\_y\_discr]

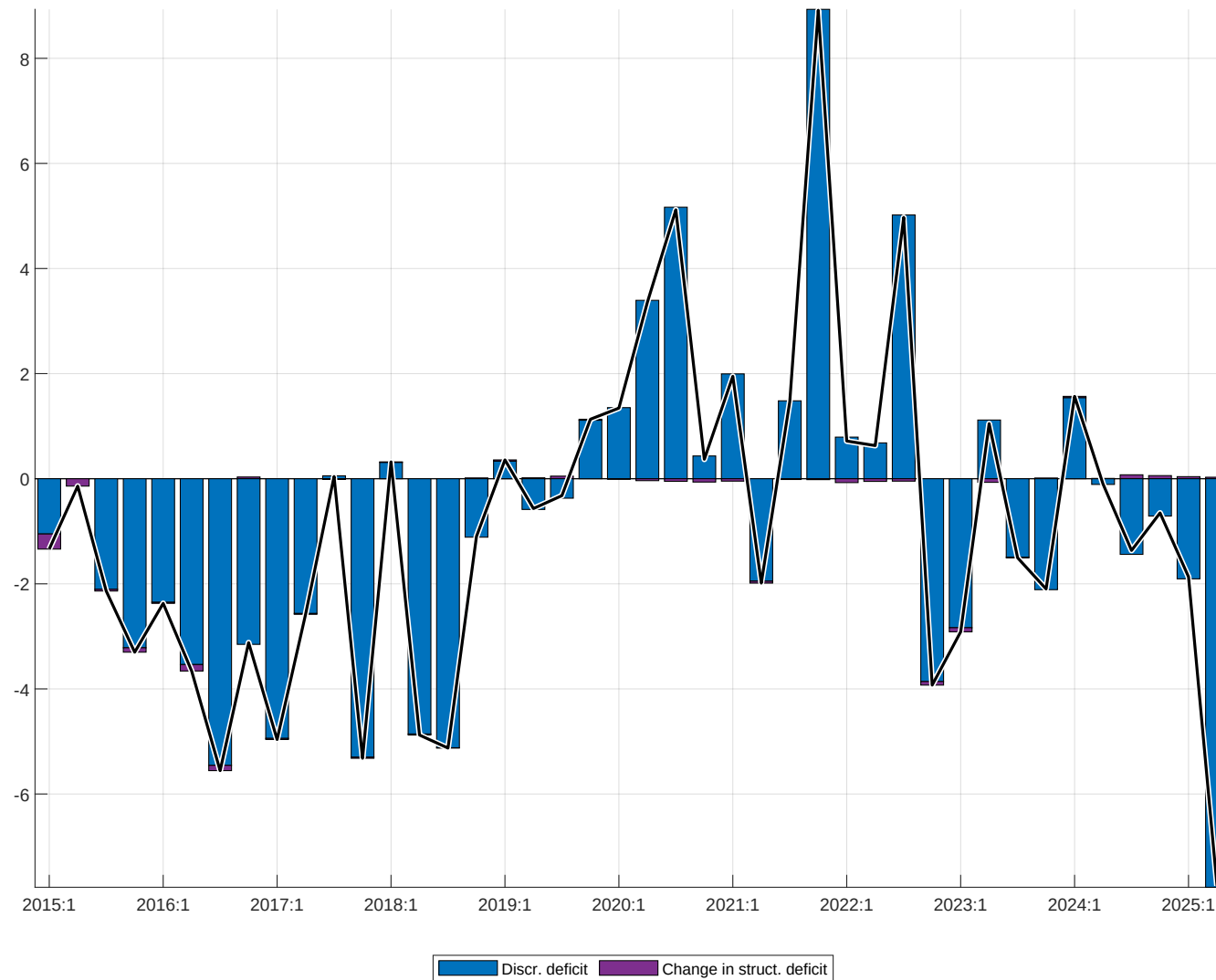


## Struct. deficit, % of GDP [def\_y\_str]

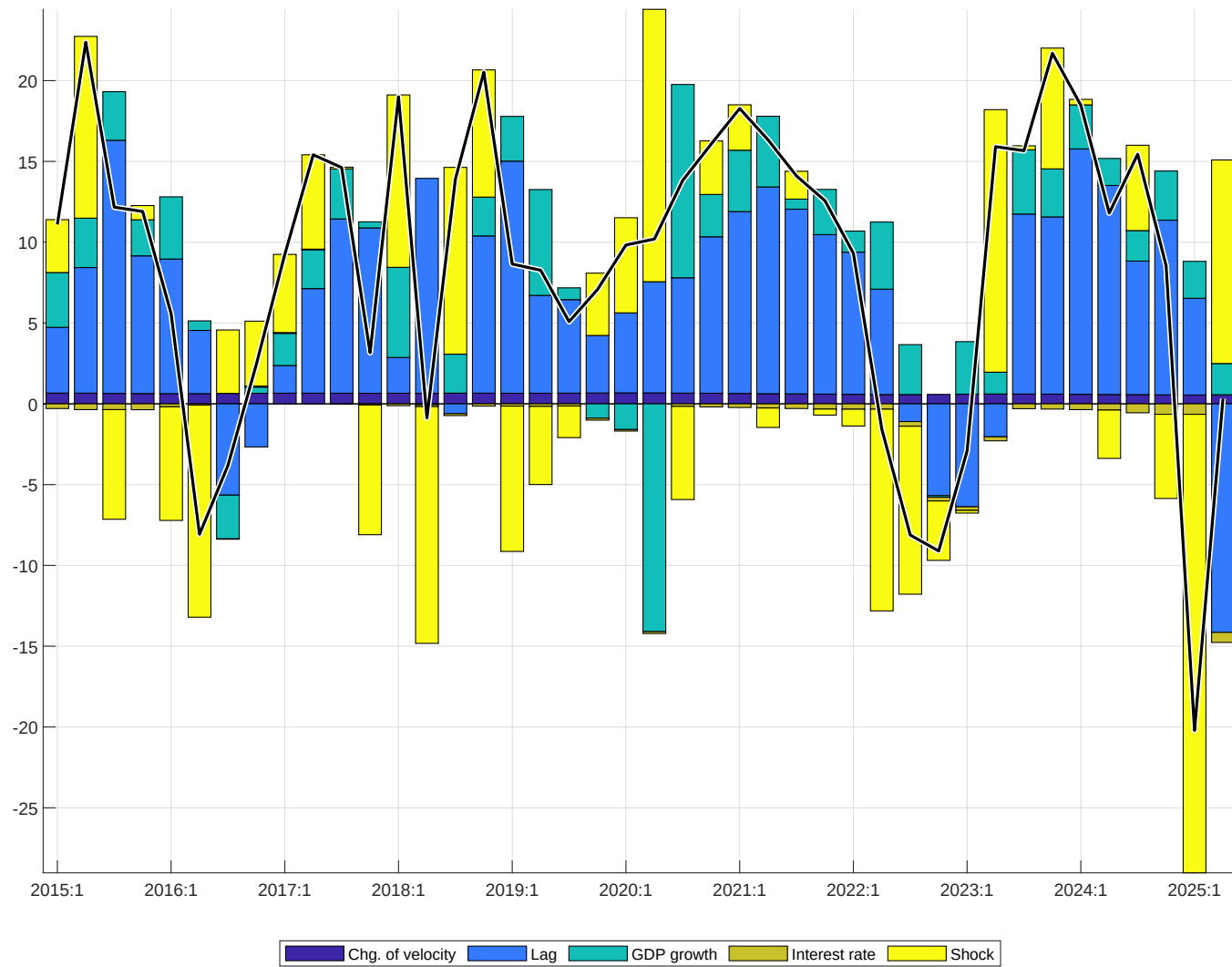


## Budget deficit, % of GDP [def\_y\_scd]



**Fiscal impulse, % of GDP [fisc\_imp]**

## Real money, ann. QQ % [dl\_rmd]



## 4 Decomposition of equations: tables



## Consumption gap, % [l\_cons\_gap]

|             | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|-------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Lag         | 1.05   | -1.27  | -0.13  | -7.85  | -2.03  | -3.17  | -0.43  | -5.83  | -5.89  | -3.54  | -5.04  | 1.56   | 0.88   | 0.74   | -4.21  | -2.15  | -0.14  | 0.31   | 0.29   | -0.04  | -0.53  | -0.01  |
| Expectation | -0.00  | 0.00   | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | 0.00   | 0.00   | -0.00  | -0.00  | -0.00  | 0.00   | 0.00   | 0.00   | 0.00   | -0.00  | -0.00  | -0.00  |
| RIR gap     | 0.13   | 0.00   | -0.35  | -0.01  | -0.27  | -0.07  | -0.02  | -0.16  | -0.01  | 0.12   | 0.32   | 0.22   | 0.11   | 0.23   | 0.41   | 0.60   | 0.56   | 0.46   | 0.50   | 0.47   | 0.54   | 0.49   |
| REER gap    | -0.00  | -0.00  | -0.00  | -0.00  | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   |
| Income      | 1.12   | 0.46   | -2.36  | -0.84  | -0.90  | -0.74  | -0.43  | -0.69  | -0.53  | -0.60  | -0.21  | -0.03  | -0.40  | -0.23  | -0.37  | -0.08  | 0.02   | 0.08   | -0.04  | -0.14  | -0.06  | -0.01  |
| Fisc. imp.  | 0.28   | 0.34   | 0.84   | 1.28   | 0.09   | 0.49   | -0.50  | 0.37   | 2.23   | 0.18   | 0.16   | 1.24   | -0.98  | -0.73  | 0.26   | -0.38  | -0.52  | 0.39   | -0.03  | -0.34  | -0.16  | -0.01  |
| Shock       | -4.54  | 0.27   | -10.08 | 4.30   | -1.77  | 2.83   | -7.59  | -2.74  | -1.25  | -3.92  | 7.17   | -1.64  | 1.53   | -6.49  | 0.60   | 1.79   | 0.56   | -0.80  | -0.79  | -0.77  | 0.05   | 0.00   |
| Total       | -1.96  | -0.20  | -12.08 | -3.13  | -4.87  | -0.66  | -8.97  | -9.06  | -5.44  | -7.76  | 2.40   | 1.35   | 1.13   | -6.48  | -3.32  | -0.22  | 0.48   | 0.45   | -0.06  | -0.82  | -0.17  | 0.00   |

**Investment gap, % [l\_inv\_gap]**

|             | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|-------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Lag         | 23.13  | 23.57  | 15.85  | -14.88 | -36.97 | -3.23  | -22.51 | 17.75  | 16.83  | -7.33  | -2.04  | 2.89   | -1.06  | 8.61   | -2.02  | -27.22 | -16.67 | -10.39 | -6.59  | -4.35  | -3.09  | -1.75  |
| Expectation | 0.00   | 0.00   | -0.00  | -0.00  | -0.00  | -0.00  | 0.00   | 0.00   | -0.00  | -0.00  | 0.00   | -0.00  | 0.00   | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  |
| RIR gap     | 0.13   | 0.00   | -0.35  | -0.01  | -0.27  | -0.07  | -0.02  | -0.16  | -0.01  | 0.12   | 0.32   | 0.22   | 0.11   | 0.23   | 0.41   | 0.60   | 0.56   | 0.46   | 0.50   | 0.47   | 0.54   | 0.49   |
| REER gap    | -0.00  | -0.00  | -0.00  | -0.00  | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   |
| Income      | 1.12   | 0.46   | -2.36  | -0.84  | -0.90  | -0.74  | -0.43  | -0.69  | -0.53  | -0.60  | -0.21  | -0.03  | -0.40  | -0.23  | -0.37  | -0.08  | 0.02   | 0.08   | -0.04  | -0.14  | -0.06  | -0.06  |
| Fisc. imp.  | 0.17   | 0.20   | 0.50   | 0.77   | 0.06   | 0.29   | -0.30  | 0.22   | 1.34   | 0.11   | 0.09   | 0.75   | -0.59  | -0.44  | 0.16   | -0.23  | -0.31  | 0.24   | -0.02  | -0.20  | -0.10  | -0.07  |
| Shock       | 11.70  | 0.16   | -36.54 | -41.92 | 33.12  | -30.88 | 50.57  | 8.77   | -28.90 | 4.55   | 6.29   | -5.47  | 15.19  | -11.28 | -40.05 | 1.27   | 0.42   | -0.53  | -0.55  | -0.53  | 0.05   | 0.49   |
| Total       | 36.26  | 24.39  | -22.89 | -56.88 | -4.97  | -34.63 | 27.31  | 25.89  | -11.28 | -3.14  | 4.45   | -1.64  | 13.25  | -3.11  | -41.87 | -25.65 | -15.98 | -10.14 | -6.69  | -4.76  | -2.66  | -1.75  |

Gov. demand gap, % [l\_gdem\_gap]

|               | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|---------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Output gap    | 5.58   | 2.29   | -11.78 | -4.18  | -4.49  | -3.68  | -2.16  | -3.45  | -2.65  | -2.99  | -1.04  | -0.13  | -2.00  | -1.13  | -1.83  | -0.41  | 0.12   | 0.42   | -0.19  | -0.72  | -0.29  | -0.01  |
| Cyclical      | -0.54  | -0.21  | 0.96   | 0.33   | 0.37   | 0.30   | 0.18   | 0.30   | 0.21   | 0.24   | 0.09   | 0.01   | 0.23   | 0.11   | 0.14   | 0.04   | -0.01  | -0.04  | 0.02   | 0.07   | 0.03   | 0.00   |
| Discretionary | -3.01  | 1.18   | 12.16  | 16.21  | 12.17  | 11.47  | 9.36   | 7.53   | 17.03  | 13.37  | 4.61   | 7.48   | -22.55 | -3.04  | 19.69  | 1.15   | -8.33  | -5.38  | -2.77  | -5.87  | -9.68  | -10.00 |
| Approx. error | 0.32   | -0.05  | 1.11   | 2.13   | 0.97   | 0.80   | 0.40   | 0.18   | 2.39   | 1.24   | -0.05  | 0.14   | 5.12   | 0.25   | 3.48   | -0.06  | 1.09   | 0.57   | 0.23   | 0.62   | 1.31   | 3.00   |
| Total         | 2.35   | 3.20   | 2.46   | 14.49  | 9.02   | 8.90   | 7.78   | 4.56   | 16.98  | 11.86  | 3.61   | 7.50   | -19.19 | -3.82  | 21.48  | 0.73   | -7.13  | -4.42  | -2.71  | -5.90  | -8.63  | -11.00 |

**Export gap, % [l\_exp\_gap]**

|                | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|----------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Lag            | 14.14  | 14.57  | 5.47   | -19.86 | 8.93   | -17.49 | -17.11 | -8.57  | -15.81 | -7.36  | -3.74  | -1.45  | -2.44  | -3.11  | 4.39   | 3.71   | 4.25   | 3.88   | 2.51   | 1.48   | 0.97   | 0.97   |
| Expectation    | 0.00   | 0.00   | -0.00  | 0.00   | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   |
| RIR gap        | 0.00   | 0.00   | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | -0.00  | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   |
| REER gap       | -0.00  | -0.29  | -0.79  | -0.63  | 0.22   | 0.61   | 1.05   | 1.18   | 1.05   | 0.64   | -0.39  | -1.22  | -1.86  | -1.87  | -1.14  | -0.30  | 0.45   | 0.89   | 0.79   | 1.04   | 1.20   | 1.20   |
| Foreign demand | 0.57   | 0.09   | -0.29  | -0.58  | -0.77  | -0.80  | -0.78  | -0.73  | -0.65  | -0.43  | -0.24  | -0.08  | 0.04   | 0.15   | 0.23   | 0.27   | 0.29   | 0.24   | 0.19   | 0.15   | 0.10   | 0.10   |
| Shock          | 7.71   | -5.96  | -34.94 | 34.81  | -35.29 | -8.64  | 3.66   | -16.21 | 4.10   | 1.39   | 2.15   | -1.01  | -0.52  | 11.59  | 2.22   | 2.86   | 0.98   | -1.15  | -1.22  | -1.17  | 0.15   | 0.15   |
| Total          | 22.42  | 8.41   | -30.56 | 13.74  | -26.91 | -26.32 | -13.19 | -24.33 | -11.32 | -5.76  | -2.23  | -3.76  | -4.78  | 6.76   | 5.70   | 6.54   | 5.97   | 3.86   | 2.28   | 1.50   | 2.43   | 2.43   |

**Import gap, % [l\_imp\_gap]**

|                | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 |
|----------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Cons. gap      | -0.87  | -0.09  | -5.38  | -1.39  | -2.17  | -0.29  | -4.00  | -4.04  | -2.43  | -3.46  | 1.07   | 0.60   | 0.50   | -2.89  | -1.48  | -0.10  | 0.22   | 0.20   | -0.02  | -0.37  | -0.07  |
| Invest. gap    | 5.66   | 3.80   | -3.57  | -8.87  | -0.78  | -5.40  | 4.26   | 4.04   | -1.76  | -0.49  | 0.69   | -0.26  | 2.07   | -0.48  | -6.53  | -4.00  | -2.49  | -1.58  | -1.04  | -0.74  | -0.41  |
| Gov.demand gap | 0.65   | 0.88   | 0.68   | 4.00   | 2.49   | 2.46   | 2.15   | 1.26   | 4.69   | 3.27   | 1.00   | 2.07   | -5.30  | -1.05  | 5.93   | 0.20   | -1.97  | -1.22  | -0.75  | -1.63  | -2.38  |
| Export gap     | 2.74   | 1.03   | -3.74  | 1.68   | -3.29  | -3.22  | -1.61  | -2.97  | -1.38  | -0.70  | -0.27  | -0.46  | -0.58  | 0.83   | 0.70   | 0.80   | 0.73   | 0.47   | 0.28   | 0.18   | 0.30   |
| REER gap       | 0.00   | 0.44   | 1.18   | 0.94   | -0.33  | -0.91  | -1.57  | -1.78  | -1.57  | -0.95  | 0.59   | 1.82   | 2.80   | 2.80   | 1.71   | 0.44   | -0.67  | -1.33  | -1.19  | -1.56  | -1.80  |
| Shock          | 5.05   | 8.42   | -12.11 | 10.46  | -5.48  | 0.61   | -5.11  | -4.16  | 0.26   | 2.42   | 3.75   | 1.38   | -0.47  | -5.89  | 4.17   | -0.05  | -0.06  | -0.03  | 0.01   | -0.01  | -0.01  |
| Total          | 13.23  | 14.48  | -22.94 | 6.82   | -9.56  | -6.76  | -5.88  | -7.65  | -2.20  | 0.09   | 6.83   | 5.16   | -0.99  | -6.69  | 4.50   | -2.71  | -4.25  | -3.49  | -2.72  | -4.12  | -4.38  |

**Output gap, % [l\_y\_gap]**

|                | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 |
|----------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Cons. gap      | -1.53  | -0.16  | -9.42  | -2.44  | -3.80  | -0.51  | -7.00  | -7.06  | -4.25  | -6.05  | 1.87   | 1.06   | 0.88   | -5.06  | -2.59  | -0.17  | 0.38   | 0.35   | -0.04  | -0.64  | -0.13  |
| Invest. gap    | 4.71   | 3.17   | -2.98  | -7.39  | -0.65  | -4.50  | 3.55   | 3.37   | -1.47  | -0.41  | 0.58   | -0.21  | 1.72   | -0.40  | -5.44  | -3.33  | -2.08  | -1.32  | -0.87  | -0.62  | -0.35  |
| Gov.demand gap | 0.54   | 0.74   | 0.56   | 3.33   | 2.08   | 2.05   | 1.79   | 1.05   | 3.91   | 2.73   | 0.83   | 1.72   | -4.41  | -0.88  | 4.94   | 0.17   | -1.64  | -1.02  | -0.62  | -1.36  | -1.99  |
| Export gap     | 4.71   | 1.77   | -6.42  | 2.89   | -5.65  | -5.53  | -2.77  | -5.11  | -2.38  | -1.21  | -0.47  | -0.79  | -1.00  | 1.42   | 1.20   | 1.37   | 1.25   | 0.81   | 0.48   | 0.31   | 0.51   |
| Import gap     | -4.63  | -5.07  | 8.03   | -2.39  | 3.35   | 2.36   | 2.06   | 2.68   | 0.77   | -0.03  | -2.39  | -1.81  | 0.34   | 2.34   | -1.57  | 0.95   | 1.49   | 1.22   | 0.95   | 1.44   | 1.53   |
| Shock          | 1.78   | 1.84   | -1.56  | 1.82   | 0.19   | 2.45   | 0.21   | 1.63   | 0.76   | 1.98   | -1.46  | -0.10  | 0.47   | 1.44   | 1.64   | 0.61   | 0.72   | 0.37   | -0.08  | 0.14   | 0.13   |
| Total          | 5.58   | 2.29   | -11.78 | -4.18  | -4.49  | -3.68  | -2.16  | -3.45  | -2.65  | -2.99  | -1.04  | -0.13  | -2.00  | -1.13  | -1.83  | -0.41  | 0.12   | 0.42   | -0.19  | -0.72  | -0.29  |

## GDP tnd., ann. QQ % [dl\_y\_tnd]

|            | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Cons.      | 4.38   | 4.89   | 5.34   | 5.45   | 5.62   | 5.49   | 5.26   | 4.91   | 4.64   | 4.57   | 4.72   | 4.97   | 5.19   | 5.31   | 5.34   | 5.47   | 5.57   | 5.64   | 5.69   | 5.72   | 5.74   | 5.76   |
| Invest.    | 0.68   | 0.75   | 0.81   | 0.81   | 0.83   | 0.79   | 0.73   | 0.64   | 0.57   | 0.54   | 0.54   | 0.57   | 0.60   | 0.61   | 0.61   | 0.64   | 0.66   | 0.68   | 0.70   | 0.72   | 0.73   | 0.74   |
| Gov.demand | 1.46   | 1.81   | 2.13   | 2.18   | 2.29   | 2.16   | 1.96   | 1.66   | 1.41   | 1.32   | 1.38   | 1.53   | 1.65   | 1.69   | 1.67   | 1.74   | 1.79   | 1.82   | 1.84   | 2.22   | 2.19   | 2.19   |
| Export     | 2.57   | 2.83   | 3.06   | 3.09   | 3.14   | 3.02   | 2.84   | 2.57   | 2.35   | 2.27   | 2.32   | 2.44   | 2.54   | 2.57   | 2.52   | 2.55   | 2.56   | 2.55   | 2.53   | 2.50   | 2.46   | 2.46   |
| Import     | -2.75  | -2.39  | -2.08  | -2.03  | -1.94  | -2.06  | -2.27  | -2.57  | -2.84  | -2.97  | -2.96  | -2.87  | -2.81  | -2.82  | -2.87  | -2.83  | -2.80  | -2.77  | -2.75  | -2.73  | -2.72  | -2.72  |
| Total      | 6.33   | 7.88   | 9.27   | 9.49   | 9.95   | 9.40   | 8.52   | 7.22   | 6.12   | 5.73   | 6.02   | 6.65   | 7.16   | 7.37   | 7.27   | 7.57   | 7.78   | 7.92   | 8.00   | 8.42   | 8.40   | 8.40   |

## Core CPI, ann. QQ % [dl\_cpi\_core]

|             | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|-------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Lag         | 1.27   | 2.43   | 2.28   | 3.27   | 2.13   | -1.68  | 0.57   | 0.33   | 0.95   | 2.37   | 4.32   | 6.51   | 4.75   | 5.46   | 3.71   | 1.31   | 3.02   | 2.13   | 0.74   | 2.21   | 1.59   | 2.7    |
| Expectation | 4.55   | 3.70   | 0.43   | 3.15   | -0.62  | 1.45   | 1.84   | 1.22   | 3.14   | 4.63   | 6.95   | 5.06   | 4.39   | 3.62   | 3.92   | 5.53   | 4.67   | 3.35   | 4.36   | 3.96   | 4.81   | 3.7    |
| Direct      | 0.17   | 0.03   | -0.03  | 0.46   | 0.61   | 0.47   | 0.49   | 0.27   | 0.20   | 0.20   | -0.10  | -0.14  | 0.13   | 0.52   | 0.91   | 1.28   | 1.05   | 0.55   | 0.22   | 0.47   | 0.56   | 0.3    |
| Output gap  | 0.89   | 0.37   | -1.88  | -0.67  | -0.72  | -0.59  | -0.35  | -0.55  | -0.42  | -0.48  | -0.17  | -0.02  | -0.32  | -0.18  | -0.29  | -0.06  | 0.02   | 0.07   | -0.03  | -0.12  | -0.05  | -0.0   |
| REER gap    | -0.00  | -0.06  | -0.16  | -0.13  | 0.04   | 0.12   | 0.21   | 0.24   | 0.21   | 0.13   | -0.08  | -0.24  | -0.37  | -0.37  | -0.23  | -0.06  | 0.09   | 0.18   | 0.16   | 0.21   | 0.24   | 0.3    |
| Shock       | 0.06   | 0.06   | 8.72   | -0.02  | -6.24  | 1.85   | -1.81  | 1.20   | 2.69   | 5.48   | 7.68   | 2.40   | 7.02   | 1.54   | -4.28  | 0.63   | -2.75  | -4.17  | 0.85   | -2.20  | 0.80   | -0.7   |
| Total       | 6.93   | 6.52   | 9.36   | 6.08   | -4.79  | 1.62   | 0.95   | 2.71   | 6.77   | 12.34  | 18.60  | 13.57  | 15.60  | 10.59  | 3.74   | 8.62   | 6.10   | 2.11   | 6.30   | 4.54   | 7.95   | 6.3    |



Dir. ext. eff. in core, ann. QQ % [dl\_cpi\_core\_direct]

|                  | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 |
|------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Foreign CPI      | −0.33  | −2.05  | −1.08  | 4.21   | 6.65   | 7.54   | 8.75   | 3.94   | 1.20   | 2.57   | −2.41  | −5.62  | −4.26  | 0.30   | 4.81   | 8.07   | 6.59   | 2.29   | 0.44   | 2.86   | 1.70   |
| Exchange rate    | 5.41   | 4.48   | 2.26   | 6.88   | 7.47   | 3.74   | 3.01   | 3.67   | 5.04   | 3.69   | 2.63   | 5.00   | 9.16   | 12.55  | 15.93  | 19.98  | 16.87  | 11.17  | 6.43   | 9.03   | 11.81  |
| Trend adjustment | −1.75  | −1.73  | −1.73  | −1.82  | −1.89  | −1.96  | −2.04  | −2.12  | −2.18  | −2.20  | −2.22  | −2.23  | −2.28  | −2.35  | −2.45  | −2.52  | −2.54  | −2.52  | −2.47  | −2.42  | −2.38  |
| Total            | 3.33   | 0.70   | −0.55  | 9.28   | 12.24  | 9.31   | 9.72   | 5.49   | 4.07   | 4.06   | −2.00  | −2.86  | 2.63   | 10.49  | 18.29  | 25.52  | 20.92  | 10.94  | 4.40   | 9.46   | 11.13  |

Food CPI, ann. QQ % [dl\_cpi\_food]

|             | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|-------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Lag         | 5.06   | 13.20  | 8.40   | -4.63  | 6.60   | 0.78   | -5.65  | -2.00  | -1.54  | -3.74  | 13.07  | 13.31  | 13.87  | 14.29  | 14.72  | 4.45   | 5.10   | -0.54  | -5.14  | 2.49   | 3.73   | 6.00   |
| Expectation | 11.15  | 8.27   | 1.46   | 8.09   | 3.44   | -1.24  | 1.11   | 1.70   | 1.25   | 11.31  | 9.15   | 9.90   | 9.26   | 9.75   | 3.64   | 4.18   | 0.44   | -3.09  | 2.36   | 4.24   | 5.84   | 2.00   |
| Direct      | 1.14   | 0.58   | -2.04  | 1.25   | 3.87   | 4.01   | 3.63   | -0.75  | 0.75   | 5.16   | 2.73   | -5.48  | 0.03   | 2.42   | 0.82   | -0.31  | 1.85   | 1.35   | 0.63   | -0.92  | 2.26   | 0.00   |
| RER gap     | -0.83  | -1.29  | -1.46  | -1.62  | -0.71  | 0.69   | 1.74   | 1.66   | 2.12   | 2.48   | 2.21   | -0.15  | -1.17  | -1.61  | -1.73  | -2.17  | -1.67  | -0.96  | -0.98  | -1.48  | -1.37  | -1.00  |
| AgrOutp gap | -1.19  | 2.42   | 3.01   | 0.57   | 0.35   | -2.31  | -2.25  | -2.09  | -1.00  | 1.76   | 0.44   | 2.13   | 1.81   | 6.23   | 7.09   | 4.81   | 4.24   | 2.52   | 4.63   | 6.44   | 8.69   | 8.00   |
| Shock       | 22.39  | 0.82   | -22.60 | 15.21  | -11.33 | -18.05 | -4.29  | -2.92  | -12.28 | 20.38  | 10.44  | 19.91  | 17.02  | 10.98  | -11.82 | 3.60   | -11.50 | -13.97 | 5.60   | -0.10  | -0.97  | -10.00 |
| Total       | 37.72  | 24.01  | -13.23 | 18.87  | 2.21   | -16.13 | -5.70  | -4.39  | -10.69 | 37.34  | 38.04  | 39.62  | 40.82  | 42.05  | 12.71  | 14.56  | -1.55  | -14.70 | 7.10   | 10.66  | 18.19  | 6.00   |

Dir. ext. eff. in food, ann. QQ % [dl\_cpi\_food\_direct]

|                        | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 |
|------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Foreign food inflation | 8.20   | 4.02   | −19.51 | 9.54   | 35.76  | 41.28  | 38.41  | −6.14  | 7.22   | 52.06  | 28.11  | −57.11 | −6.67  | 13.54  | −5.98  | −21.28 | 3.13   | 3.38   | 0.35   | −18.21 | 10.00  |
| Exchange rate          | 5.41   | 4.48   | 2.26   | 6.88   | 7.47   | 3.74   | 3.01   | 3.67   | 5.04   | 3.69   | 2.63   | 5.00   | 9.16   | 12.55  | 15.93  | 19.98  | 16.87  | 11.17  | 6.43   | 9.03   | 11.00  |
| Trend adjustment       | −2.22  | −2.67  | −3.17  | −3.95  | −4.56  | −4.93  | −5.10  | −5.03  | −4.74  | −4.16  | −3.44  | −2.72  | −2.19  | −1.89  | −1.77  | −1.78  | −1.49  | −1.02  | −0.48  | −0.05  | 0.00   |
| Total                  | 11.39  | 5.83   | −20.42 | 12.47  | 38.67  | 40.09  | 36.32  | −7.51  | 7.53   | 51.60  | 27.29  | −54.83 | 0.30   | 24.21  | 8.18   | −3.08  | 18.51  | 13.53  | 6.30   | −9.23  | 22.00  |

Energy CPI, ann. QQ % [dl\_cpi\_ener]

|             | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|-------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Lag         | 2.61   | 2.15   | 4.22   | 1.53   | -2.44  | 2.25   | -0.50  | -0.64  | 2.76   | 1.67   | 6.40   | 6.47   | 2.89   | 3.17   | 0.10   | -0.82  | 0.60   | 2.21   | 0.82   | 1.05   | 0.53   | -1.3   |
| Expectation | 4.55   | 4.99   | 1.31   | 0.32   | 3.65   | 2.32   | 2.86   | 4.89   | 4.01   | 7.86   | 7.37   | 3.76   | 3.95   | 2.27   | 2.60   | 5.03   | 6.11   | 4.40   | 4.73   | 4.57   | 3.62   | 4.3    |
| Direct      | 0.09   | -1.66  | -3.64  | 2.37   | 0.34   | 2.30   | 0.85   | 0.39   | 0.63   | 1.49   | 1.01   | -1.11  | -0.83  | -0.37  | -0.05  | 1.20   | -0.01  | 0.18   | 0.50   | -0.38  | -0.24  | 0.4    |
| RER gap     | 0.13   | -0.87  | -2.75  | -1.47  | -1.39  | -0.22  | 0.23   | 0.32   | 0.57   | 1.06   | 1.30   | 0.63   | 0.09   | -0.10  | -0.09  | 0.49   | 0.39   | 0.45   | 0.66   | 0.45   | 0.38   | 0.6    |
| Shock       | 1.23   | 12.29  | 6.97   | -12.51 | 8.83   | -8.64  | -6.00  | 6.09   | -1.28  | 13.51  | 9.78   | 1.80   | 6.60   | -4.57  | -5.85  | -3.52  | 1.77   | -3.97  | -2.51  | -3.55  | -9.63  | -5.9   |
| Total       | 8.62   | 16.90  | 6.12   | -9.76  | 8.99   | -1.99  | -2.56  | 11.05  | 6.69   | 25.59  | 25.87  | 11.54  | 12.70  | 0.40   | -3.29  | 2.38   | 8.85   | 3.26   | 4.19   | 2.13   | -5.33  | -1.7   |

Dir. ext. eff. in energy, ann. QQ % [dl\_cpi\_ener\_direct]

|                          | 2019Q4 | 2020Q1 | 2020Q2  | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 |
|--------------------------|--------|--------|---------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Foreign energy inflation | 3.74   | -82.23 | -178.13 | 118.65 | 17.47  | 119.88 | 48.58  | 25.07  | 35.15  | 78.85  | 55.06  | -54.55 | -45.56 | -27.38 | -15.81 | 41.92  | -16.89 | -2.78  | 17.29  | -30.38 | -1.03  |
| Exchange rate            | 5.41   | 4.48   | 2.26    | 6.88   | 7.47   | 3.74   | 3.01   | 3.67   | 5.04   | 3.69   | 2.63   | 5.00   | 9.16   | 12.55  | 15.93  | 19.98  | 16.87  | 11.17  | 6.43   | 9.03   | -1.03  |
| Trend adjustment         | -4.68  | -5.35  | -6.15   | -7.15  | -8.00  | -8.65  | -9.01  | -9.04  | -8.74  | -8.09  | -7.15  | -6.03  | -4.88  | -3.78  | -2.73  | -1.72  | -0.64  | 0.45   | 1.49   | 2.41   | -1.03  |
| Total                    | 4.47   | -83.09 | -182.02 | 118.38 | 16.94  | 114.97 | 42.58  | 19.70  | 31.45  | 74.45  | 50.53  | -55.58 | -41.28 | -18.61 | -2.61  | 60.18  | -0.65  | 8.84   | 25.21  | -18.94 | -1.03  |

Agric. output gap, % [l\_y\_agr\_gap]

|       | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 | 2025Q2 |
|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Lag   | 0.78   | 0.32   | -0.65  | -0.80  | -0.15  | -0.09  | 0.62   | 0.60   | 0.56   | 0.27   | -0.47  | -0.12  | -0.57  | -0.48  | -1.66  | -1.89  | -1.28  | -1.13  | -0.67  | -1.23  | -1.72  | -2.32  | -2.92  |
| Shock | 0.01   | -1.93  | -1.36  | 0.42   | -0.08  | 1.63   | 0.88   | 0.79   | 0.11   | -1.44  | 0.17   | -1.30  | -0.64  | -3.67  | -3.06  | -1.32  | -1.54  | -0.55  | -2.41  | -3.06  | -4.08  | -3.35  | -2.62  |
| Total | 0.79   | -1.62  | -2.01  | -0.38  | -0.23  | 1.54   | 1.50   | 1.39   | 0.67   | -1.18  | -0.30  | -1.42  | -1.21  | -4.15  | -4.72  | -3.21  | -2.82  | -1.68  | -3.09  | -4.29  | -5.79  | -5.67  | -5.54  |

Policy rate (IB rate used), % [i]

|                    | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 |
|--------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Lag                | 4.26   | 4.27   | 4.31   | 4.17   | 4.11   | 4.09   | 4.04   | 4.05   | 4.05   | 4.03   | 4.13   | 4.29   | 4.70   | 5.30   | 5.68   | 5.97   | 6.15   | 6.34   | 6.37   | 6.37   | 5.61   |
| Neutral rate       | 1.23   | 1.21   | 1.20   | 1.24   | 1.28   | 1.31   | 1.36   | 1.40   | 1.44   | 1.48   | 1.52   | 1.56   | 1.63   | 1.72   | 1.83   | 1.94   | 2.02   | 2.07   | 2.11   | 2.14   | 2.18   |
| Inflation reaction | 0.70   | 0.31   | −0.94  | 0.26   | −0.45  | −0.21  | 0.02   | −0.28  | 0.13   | 0.49   | 0.65   | 0.28   | −0.15  | −0.06  | 0.40   | 0.67   | 0.43   | 0.10   | 0.31   | 0.34   | 0.41   |
| Output gap         | 0.22   | 0.09   | −0.47  | −0.17  | −0.18  | −0.15  | −0.09  | −0.14  | −0.11  | −0.12  | −0.04  | −0.01  | −0.08  | −0.05  | −0.07  | −0.02  | 0.00   | 0.02   | −0.01  | −0.03  | −0.01  |
| FX target          | 0.00   | −0.00  | −0.00  | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | −0.00  | −0.00  | −0.00  | −0.00  | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   |
| Shock              | −1.08  | −0.49  | 1.12   | −0.37  | 0.36   | 0.01   | −0.27  | 0.03   | −0.47  | −0.73  | −0.89  | −0.26  | 0.53   | 0.19   | −0.37  | −0.87  | −0.68  | −0.57  | −0.80  | −1.83  | −1.62  |
| Total              | 5.34   | 5.39   | 5.22   | 5.14   | 5.11   | 5.05   | 5.06   | 5.06   | 5.04   | 5.16   | 5.36   | 5.87   | 6.62   | 7.10   | 7.46   | 7.69   | 7.93   | 7.96   | 7.97   | 7.01   | 6.56   |

Exchange rate, QQ annualized % [dl\_s]

|                 | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|-----------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Expectation     | 4.56   | 4.63   | 4.55   | 5.28   | 4.65   | 3.57   | 3.05   | 3.53   | 4.18   | 4.31   | 5.38   | 6.97   | 8.88   | 9.59   | 9.59   | 10.09  | 8.82   | 7.17   | 6.20   | 6.58   | 7.23   | 7.94   |
| IR differential | -1.87  | -2.13  | -2.54  | -2.51  | -2.51  | -2.50  | -2.52  | -2.51  | -2.49  | -2.43  | -2.14  | -1.58  | -1.26  | -1.21  | -1.16  | -1.15  | -1.28  | -1.32  | -1.32  | -0.96  | -1.04  | -1.12  |
| Premium         | 0.88   | 0.88   | 0.88   | 0.90   | 0.92   | 0.93   | 0.95   | 0.96   | 0.97   | 0.97   | 0.98   | 0.98   | 0.98   | 0.99   | 1.00   | 1.03   | 1.04   | 1.05   | 1.06   | 1.07   | 1.09   | 1.11   |
| Target          | 1.48   | 2.31   | 4.13   | 3.21   | 1.97   | 1.06   | 0.08   | 0.04   | 0.13   | 0.76   | 2.80   | 4.77   | 6.48   | 6.57   | 4.93   | 3.17   | 1.82   | 1.06   | 1.08   | 0.56   | 0.20   | 0.00   |
| Shock           | 0.37   | -1.20  | -4.75  | 0.01   | 2.45   | 0.67   | 1.45   | 1.64   | 2.26   | 0.08   | -4.39  | -6.14  | -5.91  | -3.38  | 1.58   | 6.84   | 6.47   | 3.21   | -0.59  | 1.78   | 4.33   | 5.94   |
| Total           | 5.41   | 4.48   | 2.26   | 6.88   | 7.47   | 3.74   | 3.01   | 3.67   | 5.04   | 3.69   | 2.63   | 5.00   | 9.16   | 12.55  | 15.93  | 19.98  | 16.87  | 11.17  | 6.43   | 9.03   | 11.81  | 12.96  |



## Exchange rate target, QQ % [dl\_s\_tar]

|               | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|---------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Lag           | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   |
| RER trend     | 1.75   | 1.73   | 1.73   | 1.82   | 1.89   | 1.96   | 2.04   | 2.12   | 2.18   | 2.20   | 2.22   | 2.23   | 2.28   | 2.35   | 2.45   | 2.52   | 2.54   | 2.52   | 2.47   | 2.42   | 2.38   | 2.38   |
| CPI target    | 4.60   | 4.59   | 4.61   | 4.68   | 4.74   | 4.78   | 4.83   | 4.86   | 4.89   | 4.89   | 4.89   | 4.89   | 4.92   | 4.98   | 5.07   | 5.18   | 5.24   | 5.28   | 5.29   | 5.32   | 5.34   | 5.34   |
| Core RP trend | -0.55  | -0.54  | -0.52  | -0.45  | -0.38  | -0.34  | -0.30  | -0.27  | -0.26  | -0.27  | -0.28  | -0.27  | -0.24  | -0.17  | -0.07  | 0.04   | 0.11   | 0.13   | 0.14   | 0.15   | 0.17   | 0.17   |
| Foreign CPI   | -1.98  | -1.98  | -1.98  | -1.98  | -1.98  | -1.98  | -1.98  | -1.98  | -1.98  | -1.98  | -1.98  | -1.98  | -1.98  | -1.98  | -1.98  | -1.98  | -1.98  | -1.98  | -1.98  | -1.98  | -1.98  | -1.98  |
| CPI deviation | -0.88  | -0.39  | 1.18   | -0.32  | 0.57   | 0.27   | -0.03  | 0.36   | -0.16  | -0.62  | -0.81  | -0.36  | 0.19   | 0.08   | -0.50  | -0.84  | -0.53  | -0.12  | -0.38  | -0.43  | -0.52  | -0.52  |
| RER gap       | 0.01   | 1.25   | 3.35   | 2.67   | -0.95  | -2.58  | -4.44  | -5.03  | -4.46  | -2.70  | 1.67   | 5.16   | 7.92   | 7.94   | 4.85   | 1.26   | -1.89  | -3.77  | -3.37  | -4.41  | -5.09  | -5.09  |
| Shock         | 0.01   | -0.03  | -0.11  | 0.00   | 0.06   | 0.02   | 0.03   | 0.04   | 0.05   | 0.00   | -0.10  | -0.14  | -0.14  | -0.08  | 0.04   | 0.16   | 0.15   | 0.07   | -0.01  | 0.04   | 0.10   | 0.10   |
| Total         | 2.95   | 4.63   | 8.26   | 6.42   | 3.94   | 2.13   | 0.15   | 0.08   | 0.26   | 1.53   | 5.60   | 9.53   | 12.96  | 13.13  | 9.86   | 6.35   | 3.63   | 2.13   | 2.16   | 1.11   | 0.40   | 0.40   |

RIR (policy) trend, % [r\_tnd]

|                      | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 |
|----------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Foreign RIR          | -1.33  | -1.41  | -1.49  | -1.55  | -1.59  | -1.60  | -1.57  | -1.51  | -1.41  | -1.27  | -1.09  | -0.89  | -0.67  | -0.43  | -0.20  | 0.03   | 0.25   | 0.45   | 0.63   | 0.80   | 0.97   |
| Premium              | 1.76   | 1.75   | 1.76   | 1.80   | 1.83   | 1.86   | 1.89   | 1.91   | 1.93   | 1.94   | 1.95   | 1.96   | 1.96   | 1.98   | 2.00   | 2.05   | 2.08   | 2.11   | 2.13   | 2.14   | 2.15   |
| Exp. RER trend depr. | 1.66   | 1.64   | 1.64   | 1.73   | 1.79   | 1.86   | 1.94   | 2.01   | 2.07   | 2.09   | 2.10   | 2.12   | 2.16   | 2.24   | 2.32   | 2.40   | 2.42   | 2.39   | 2.35   | 2.30   | 2.25   |
| Total                | 2.09   | 1.98   | 1.91   | 1.98   | 2.04   | 2.13   | 2.26   | 2.42   | 2.59   | 2.77   | 2.97   | 3.19   | 3.46   | 3.78   | 4.13   | 4.48   | 4.75   | 4.95   | 5.11   | 5.24   | 5.37   |

Govt. rev., % of GDP [grev\_y]

|               | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|---------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Structural    | 17.89  | 17.93  | 17.97  | 18.00  | 18.03  | 18.05  | 18.08  | 18.09  | 18.11  | 18.12  | 18.12  | 18.12  | 18.12  | 18.12  | 18.13  | 18.14  | 18.15  | 18.16  | 18.18  | 18.20  | 18.22  | 18.23  |
| Cyclical      | 0.17   | 0.13   | -0.52  | -0.08  | -0.21  | -0.12  | -0.19  | -0.22  | -0.13  | -0.15  | 0.05   | 0.04   | -0.03  | -0.14  | -0.05  | -0.03  | -0.02  | -0.01  | -0.02  | -0.05  | -0.03  | -0.01  |
| Discretionary | 1.18   | -0.90  | 2.00   | 1.69   | 1.73   | -0.11  | 3.09   | 0.09   | 1.30   | 3.04   | -0.42  | -0.65  | -1.36  | 0.95   | -0.91  | -0.73  | -1.56  | -2.09  | 0.43   | 0.95   | -0.38  | -1.00  |
| Total         | 19.24  | 17.16  | 19.45  | 19.61  | 19.56  | 17.82  | 20.99  | 17.95  | 19.27  | 21.01  | 17.75  | 17.51  | 16.73  | 18.93  | 17.17  | 17.38  | 16.56  | 16.06  | 18.59  | 19.10  | 17.81  | 16.23  |

**Cyc. govt. rev., % of GDP [grev\_y\_cyc]**

|           | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|-----------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Cons. gap | -0.02  | -0.00  | -0.14  | -0.04  | -0.06  | -0.01  | -0.11  | -0.11  | -0.07  | -0.09  | 0.03   | 0.02   | 0.01   | -0.08  | -0.04  | -0.00  | 0.01   | 0.01   | -0.00  | -0.01  | -0.00  | 0.00   |
| Imp. gap  | 0.08   | 0.09   | -0.14  | 0.04   | -0.06  | -0.04  | -0.04  | -0.05  | -0.01  | 0.00   | 0.04   | 0.03   | -0.01  | -0.04  | 0.03   | -0.02  | -0.03  | -0.02  | -0.02  | -0.02  | -0.03  | -0.03  |
| Ouput gap | 0.11   | 0.05   | -0.24  | -0.08  | -0.09  | -0.07  | -0.04  | -0.07  | -0.05  | -0.06  | -0.02  | -0.00  | -0.04  | -0.02  | -0.04  | -0.01  | 0.00   | 0.01   | -0.00  | -0.01  | -0.01  | -0.01  |
| Shock     | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   |
| Total     | 0.17   | 0.13   | -0.52  | -0.08  | -0.21  | -0.12  | -0.19  | -0.22  | -0.13  | -0.15  | 0.05   | 0.04   | -0.03  | -0.14  | -0.05  | -0.03  | -0.02  | -0.01  | -0.02  | -0.05  | -0.03  | -0.04  |

Struct. govt. rev., % of GDP [grev\_y\_str]

|              | 2019Q 4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q 1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q 1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|--------------|---------|--------|--------|--------|--------|---------|--------|--------|--------|---------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Lag          | 17.67   | 17.71  | 17.75  | 17.79  | 17.82  | 17.85   | 17.87  | 17.90  | 17.91  | 17.92   | 17.94  | 17.93  | 17.93  | 17.93  | 17.94  | 17.95  | 17.95  | 17.96  | 17.98  | 18.00  | 18.02  | 18.04  |
| Steady state | 0.21    | 0.21   | 0.21   | 0.21   | 0.21   | 0.21    | 0.21   | 0.21   | 0.21   | 0.21    | 0.21   | 0.21   | 0.21   | 0.21   | 0.21   | 0.21   | 0.21   | 0.21   | 0.21   | 0.21   | 0.21   | 0.21   |
| Shock        | 0.01    | 0.00   | 0.01   | 0.00   | -0.00  | -0.01   | -0.01  | -0.02  | -0.01  | -0.02   | -0.03  | -0.03  | -0.03  | -0.02  | -0.02  | -0.02  | -0.02  | -0.01  | -0.01  | -0.01  | -0.01  | -0.01  |
| Total        | 17.89   | 17.93  | 17.97  | 18.00  | 18.03  | 18.05   | 18.08  | 18.09  | 18.11  | 18.12   | 18.12  | 18.12  | 18.12  | 18.12  | 18.13  | 18.14  | 18.15  | 18.16  | 18.18  | 18.20  | 18.22  | 18.24  |

Govt. demand (G&S), % GDP [gdem\_y]

|               | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|---------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Structural    | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.58  | 24.65  | 24.72  |
| Cyclical      | -0.13  | -0.05  | 0.27   | 0.10   | 0.10   | 0.08   | 0.05   | 0.08   | 0.06   | 0.07   | 0.02   | 0.00   | 0.05   | 0.03   | 0.04   | 0.01   | -0.00  | -0.01  | 0.00   | 0.02   | 0.01   | 0.00   |
| Discretionary | -0.71  | 0.29   | 3.43   | 4.76   | 3.41   | 3.19   | 2.54   | 2.00   | 5.04   | 3.79   | 1.19   | 1.98   | -4.52  | -0.72  | 6.02   | 0.29   | -1.88  | -1.25  | -0.66  | -1.36  | -2.18  | -3.00  |
| Total         | 23.66  | 24.74  | 28.20  | 29.36  | 28.01  | 27.77  | 27.09  | 26.58  | 29.60  | 28.36  | 25.71  | 26.48  | 20.03  | 23.80  | 30.56  | 24.79  | 22.61  | 23.24  | 23.84  | 23.23  | 22.48  | 20.72  |

Discr. govt. demand (G&S), % GDP [gdem\_y\_discr]

|       | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 | 2025Q2 |
|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Lag   | -1.41  | -0.50  | 0.20   | 2.40   | 3.33   | 2.39   | 2.23   | 1.77   | 1.40   | 3.53   | 2.65   | 0.83   | 1.39   | -3.16  | -0.51  | 4.21   | 0.20   | -1.32  | -0.87  | -0.46  | -0.95  | -1.52  | -1.00  |
| Shock | 0.70   | 0.79   | 3.23   | 2.36   | 0.08   | 0.80   | 0.31   | 0.23   | 3.64   | 0.26   | -1.47  | 1.15   | -5.90  | 2.44   | 6.52   | -3.93  | -2.08  | 0.07   | 0.21   | -0.90  | -1.22  | -2.42  | -1.00  |
| Total | -0.71  | 0.29   | 3.43   | 4.76   | 3.41   | 3.19   | 2.54   | 2.00   | 5.04   | 3.79   | 1.19   | 1.98   | -4.52  | -0.72  | 6.02   | 0.29   | -1.88  | -1.25  | -0.66  | -1.36  | -2.18  | -3.94  | -1.00  |

Cyc. govt. demand (G&S), % GDP [gdem\_y\_cyc]

|            | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Lag        | −0.00  | −0.00  | −0.00  | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | −0.00  | −0.00  | 0.00   | 0.00   | 0.00   |
| Output gap | −0.13  | −0.05  | 0.27   | 0.10   | 0.10   | 0.08   | 0.05   | 0.08   | 0.06   | 0.07   | 0.02   | 0.00   | 0.05   | 0.03   | 0.04   | 0.01   | −0.00  | −0.01  | 0.00   | 0.02   | 0.01   | 0.00   |
| Shock      | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   |
| Total      | −0.13  | −0.05  | 0.27   | 0.10   | 0.10   | 0.08   | 0.05   | 0.08   | 0.06   | 0.07   | 0.02   | 0.00   | 0.05   | 0.03   | 0.04   | 0.01   | −0.00  | −0.01  | 0.00   | 0.02   | 0.01   | 0.00   |



Struct. govt. demand (G&S), % GDP [gdem\_y\_str]

|              | 2019Q 4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|--------------|---------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Lag          | 23.28   | 23.28  | 23.28  | 23.27  | 23.27  | 23.27  | 23.28  | 23.27  | 23.27  | 23.27  | 23.28  | 23.28  | 23.28  | 23.27  | 23.27  | 23.28  | 23.28  | 23.27  | 23.28  | 23.27  | 23.35  | 23.35  |
| Steady state | 1.30    | 1.30   | 1.30   | 1.30   | 1.30   | 1.30   | 1.30   | 1.30   | 1.30   | 1.30   | 1.30   | 1.30   | 1.30   | 1.30   | 1.30   | 1.30   | 1.30   | 1.30   | 1.30   | 1.30   | 1.30   | 1.30   |
| Shock        | -0.08   | -0.07  | -0.08  | -0.07  | -0.08  | -0.07  | -0.07  | -0.07  | -0.07  | -0.08  | -0.07  | -0.07  | -0.08  | -0.07  | -0.08  | -0.07  | -0.07  | -0.07  | -0.07  | 0.00   | 0.00   | -0.01  |
| Total        | 24.50   | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.58  | 24.65  | 24.65  |

## Other govt. exp., % to GDP [oexp\_y]

|               | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|---------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Structural    | 5.86   | 5.88   | 5.89   | 5.87   | 5.84   | 5.81   | 5.80   | 5.80   | 5.80   | 5.73   | 5.68   | 5.63   | 5.57   | 5.50   | 5.43   | 5.43   | 5.45   | 5.49   | 5.52   | 5.54   | 5.54   | 5.54   |
| Cyclical      | -0.07  | -0.03  | 0.14   | 0.05   | 0.05   | 0.04   | 0.03   | 0.04   | 0.03   | 0.04   | 0.01   | 0.00   | 0.02   | 0.01   | 0.02   | 0.00   | -0.00  | -0.01  | 0.00   | 0.01   | 0.00   | 0.00   |
| Discretionary | 3.00   | 0.16   | 1.97   | 2.10   | -1.25  | -1.30  | -1.39  | -0.43  | 5.19   | 0.04   | -0.92  | 2.39   | -0.70  | -1.17  | -5.81  | -2.51  | -1.79  | 0.70   | 0.98   | 0.87   | 1.09   | 0.87   |
| Total         | 8.79   | 6.02   | 8.00   | 8.02   | 4.64   | 4.56   | 4.44   | 5.41   | 11.02  | 5.81   | 4.77   | 8.03   | 4.89   | 4.34   | -0.35  | 2.93   | 3.66   | 6.19   | 6.50   | 6.42   | 6.63   | 5.54   |

Cyc. other govt. exp., % to GDP [oexp\_y\_cyc]

|            | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Lag        | −0.00  | −0.00  | −0.00  | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | −0.00  | −0.00  | 0.00   | 0.00   | 0.00   |
| Output gap | −0.07  | −0.03  | 0.14   | 0.05   | 0.05   | 0.04   | 0.03   | 0.04   | 0.03   | 0.04   | 0.01   | 0.00   | 0.02   | 0.01   | 0.02   | 0.00   | −0.00  | −0.01  | 0.00   | 0.01   | 0.00   | 0.00   |
| Shock      | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   |
| Total      | −0.07  | −0.03  | 0.14   | 0.05   | 0.05   | 0.04   | 0.03   | 0.04   | 0.03   | 0.04   | 0.01   | 0.00   | 0.02   | 0.01   | 0.02   | 0.00   | −0.00  | −0.01  | 0.00   | 0.01   | 0.00   | 0.00   |

**Struct. other govt. exp., % to GDP [oexp\_y\_str]**

|              | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|--------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Lag          | 5.51   | 5.56   | 5.59   | 5.59   | 5.58   | 5.55   | 5.52   | 5.51   | 5.51   | 5.51   | 5.45   | 5.40   | 5.35   | 5.29   | 5.22   | 5.16   | 5.16   | 5.18   | 5.22   | 5.24   | 5.26   | 5.27   |
| Steady state | 0.30   | 0.30   | 0.30   | 0.30   | 0.30   | 0.30   | 0.30   | 0.30   | 0.30   | 0.30   | 0.30   | 0.30   | 0.30   | 0.30   | 0.30   | 0.30   | 0.30   | 0.30   | 0.30   | 0.30   | 0.30   | 0.30   |
| Shock        | 0.05   | 0.02   | -0.00  | -0.02  | -0.04  | -0.03  | -0.02  | -0.01  | -0.01  | -0.07  | -0.07  | -0.06  | -0.09  | -0.09  | -0.09  | -0.03  | -0.01  | 0.01   | 0.00   | -0.01  | -0.02  | -0.02  |
| Total        | 5.86   | 5.88   | 5.89   | 5.87   | 5.84   | 5.81   | 5.80   | 5.80   | 5.80   | 5.73   | 5.68   | 5.63   | 5.57   | 5.50   | 5.43   | 5.43   | 5.45   | 5.49   | 5.52   | 5.54   | 5.54   | 5.54   |

**Budget deficit, % of GDP [def\_y]**

|                    | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 |
|--------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Govt. Demand       | 23.66  | 24.74  | 28.20  | 29.36  | 28.01  | 27.77  | 27.09  | 26.58  | 29.60  | 28.36  | 25.71  | 26.48  | 20.03  | 23.80  | 30.56  | 24.79  | 22.61  | 23.24  | 23.84  | 23.23  | 22.48  |
| Other expenditures | 8.79   | 6.02   | 8.00   | 8.02   | 4.64   | 4.56   | 4.44   | 5.41   | 11.02  | 5.81   | 4.77   | 8.03   | 4.89   | 4.34   | -0.35  | 2.93   | 3.66   | 6.19   | 6.50   | 6.42   | 6.63   |
| Revenues           | -19.24 | -17.16 | -19.45 | -19.61 | -19.56 | -17.82 | -20.99 | -17.95 | -19.27 | -21.01 | -17.75 | -17.51 | -16.73 | -18.93 | -17.17 | -17.38 | -16.56 | -16.06 | -18.59 | -19.10 | -17.81 |
| Total              | 13.22  | 13.60  | 16.74  | 17.76  | 13.10  | 14.51  | 10.54  | 14.04  | 21.35  | 13.17  | 12.74  | 17.00  | 8.19   | 9.22   | 13.03  | 10.34  | 9.71   | 13.37  | 11.76  | 10.55  | 11.31  |

Cyc. deficit, % of GDP [def\_y\_cyc]

|                    | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 |
|--------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Demand             | −0.13  | −0.05  | 0.27   | 0.10   | 0.10   | 0.08   | 0.05   | 0.08   | 0.06   | 0.07   | 0.02   | 0.00   | 0.05   | 0.03   | 0.04   | 0.01   | −0.00  | −0.01  | 0.00   | 0.02   | 0.01   |
| Other expenditures | −0.07  | −0.03  | 0.14   | 0.05   | 0.05   | 0.04   | 0.03   | 0.04   | 0.03   | 0.04   | 0.01   | 0.00   | 0.02   | 0.01   | 0.02   | 0.00   | −0.00  | −0.01  | 0.00   | 0.01   | 0.00   |
| Revenues           | −0.17  | −0.13  | 0.52   | 0.08   | 0.21   | 0.12   | 0.19   | 0.22   | 0.13   | 0.15   | −0.05  | −0.04  | 0.03   | 0.14   | 0.05   | 0.03   | 0.02   | 0.01   | 0.02   | 0.05   | 0.03   |
| Total              | −0.36  | −0.21  | 0.93   | 0.23   | 0.36   | 0.25   | 0.26   | 0.34   | 0.22   | 0.26   | −0.01  | −0.04  | 0.10   | 0.18   | 0.11   | 0.04   | 0.01   | −0.01  | 0.03   | 0.07   | 0.04   |

**Discr. deficit, % of GDP [def\_y\_discr]**

|                    | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 |
|--------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Demand             | -0.71  | 0.29   | 3.43   | 4.76   | 3.41   | 3.19   | 2.54   | 2.00   | 5.04   | 3.79   | 1.19   | 1.98   | -4.52  | -0.72  | 6.02   | 0.29   | -1.88  | -1.25  | -0.66  | -1.36  | -2.18  |
| Other expenditures | 3.00   | 0.16   | 1.97   | 2.10   | -1.25  | -1.30  | -1.39  | -0.43  | 5.19   | 0.04   | -0.92  | 2.39   | -0.70  | -1.17  | -5.81  | -2.51  | -1.79  | 0.70   | 0.98   | 0.87   | 1.09   |
| Revenues           | -1.18  | 0.90   | -2.00  | -1.69  | -1.73  | 0.11   | -3.09  | -0.09  | -1.30  | -3.04  | 0.42   | 0.65   | 1.36   | -0.95  | 0.91   | 0.73   | 1.56   | 2.09   | -0.43  | -0.95  | 0.38   |
| Total              | 1.11   | 1.35   | 3.39   | 5.17   | 0.43   | 1.99   | -1.95  | 1.48   | 8.93   | 0.79   | 0.68   | 5.02   | -3.86  | -2.84  | 1.12   | -1.50  | -2.11  | 1.54   | -0.11  | -1.44  | -0.71  |

**Struct. deficit, % of GDP [def\_y\_str]**

|                    | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 |
|--------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Demand             | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.50  | 24.58  | 24.65  |
| Other expenditures | 5.86   | 5.88   | 5.89   | 5.87   | 5.84   | 5.81   | 5.80   | 5.80   | 5.80   | 5.73   | 5.68   | 5.63   | 5.57   | 5.50   | 5.43   | 5.43   | 5.45   | 5.49   | 5.52   | 5.54   | 5.54   |
| Revenues           | -17.89 | -17.93 | -17.97 | -18.00 | -18.03 | -18.05 | -18.08 | -18.09 | -18.11 | -18.12 | -18.12 | -18.12 | -18.12 | -18.12 | -18.13 | -18.14 | -18.15 | -18.16 | -18.18 | -18.20 | -18.22 |
| Total              | 12.47  | 12.46  | 12.42  | 12.37  | 12.31  | 12.26  | 12.22  | 12.21  | 12.19  | 12.12  | 12.07  | 12.02  | 11.95  | 11.87  | 11.80  | 11.79  | 11.81  | 11.83  | 11.84  | 11.91  | 11.97  |



**Budget deficit, % of GDP [def\_y\_scd]**

|               | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|---------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Structural    | 12.47  | 12.46  | 12.42  | 12.37  | 12.31  | 12.26  | 12.22  | 12.21  | 12.19  | 12.12  | 12.07  | 12.02  | 11.95  | 11.87  | 11.80  | 11.79  | 11.81  | 11.83  | 11.84  | 11.91  | 11.97  | 12.03  |
| Discretionary | 1.11   | 1.35   | 3.39   | 5.17   | 0.43   | 1.99   | -1.95  | 1.48   | 8.93   | 0.79   | 0.68   | 5.02   | -3.86  | -2.84  | 1.12   | -1.50  | -2.11  | 1.54   | -0.11  | -1.44  | -0.71  | -1.00  |
| Cyclical      | -0.36  | -0.21  | 0.93   | 0.23   | 0.36   | 0.25   | 0.26   | 0.34   | 0.22   | 0.26   | -0.01  | -0.04  | 0.10   | 0.18   | 0.11   | 0.04   | 0.01   | -0.01  | 0.03   | 0.07   | 0.04   | 0.00   |
| Total         | 13.22  | 13.60  | 16.74  | 17.76  | 13.10  | 14.51  | 10.54  | 14.04  | 21.35  | 13.17  | 12.74  | 17.00  | 8.19   | 9.22   | 13.03  | 10.34  | 9.71   | 13.37  | 11.76  | 10.55  | 11.31  | 10.99  |

**Fiscal impulse, % of GDP [fisc\_imp]**

|                           | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 |
|---------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Discr. deficit            | 1.11   | 1.35   | 3.39   | 5.17   | 0.43   | 1.99   | -1.95  | 1.48   | 8.93   | 0.79   | 0.68   | 5.02   | -3.86  | -2.84  | 1.12   | -1.50  | -2.11  | 1.54   | -0.11  | -1.44  | -0.07  |
| Change in struct. deficit | 0.02   | -0.01  | -0.04  | -0.05  | -0.06  | -0.05  | -0.04  | -0.02  | -0.02  | -0.07  | -0.05  | -0.05  | -0.07  | -0.08  | -0.07  | -0.01  | 0.01   | 0.02   | 0.01   | 0.07   | 0.02   |
| Total                     | 1.13   | 1.34   | 3.36   | 5.12   | 0.37   | 1.95   | -1.99  | 1.47   | 8.91   | 0.72   | 0.63   | 4.97   | -3.93  | -2.91  | 1.05   | -1.51  | -2.10  | 1.57   | -0.10  | -1.37  | -0.05  |

Real money, ann. QQ % [dl\_rmd]

|                  | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 | 2024Q1 | 2024Q2 | 2024Q3 | 2024Q4 | 2025Q1 |
|------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Chg. of velocity | 0.67   | 0.68   | 0.68   | 0.66   | 0.65   | 0.64   | 0.63   | 0.62   | 0.60   | 0.59   | 0.57   | 0.57   | 0.58   | 0.60   | 0.61   | 0.60   | 0.60   | 0.59   | 0.58   | 0.57   | 0.55   | 0.55   |
| Lag              | 3.56   | 4.95   | 6.88   | 7.14   | 9.68   | 11.25  | 12.79  | 11.43  | 9.87   | 8.80   | 6.52   | -1.10  | -5.68  | -6.37  | -2.04  | 11.14  | 10.96  | 15.18  | 12.93  | 8.26   | 10.81  | 10.81  |
| GDP growth       | -0.89  | -1.59  | -14.10 | 11.96  | 2.62   | 3.80   | 4.37   | 0.62   | 2.80   | 1.31   | 4.15   | 3.09   | -0.10  | 3.25   | 1.35   | 3.98   | 2.97   | 2.73   | 1.67   | 1.89   | 3.04   | 3.04   |
| Interest rate    | -0.12  | -0.10  | -0.12  | -0.16  | -0.19  | -0.23  | -0.26  | -0.29  | -0.33  | -0.34  | -0.33  | -0.29  | -0.23  | -0.22  | -0.25  | -0.30  | -0.33  | -0.36  | -0.39  | -0.56  | -0.65  | -0.65  |
| Shock            | 3.86   | 5.89   | 16.86  | -5.76  | 3.31   | 2.81   | -1.21  | 1.72   | -0.37  | -1.04  | -12.48 | -10.39 | -3.68  | -0.16  | 16.25  | 0.25   | 7.47   | 0.34   | -2.99  | 5.28   | -5.21  | -5.21  |
| Total            | 7.07   | 9.83   | 10.20  | 13.83  | 16.08  | 18.27  | 16.33  | 14.10  | 12.57  | 9.31   | -1.57  | -8.12  | -9.11  | -2.92  | 15.91  | 15.66  | 21.69  | 18.48  | 11.80  | 15.44  | 8.55   | -8.55  |

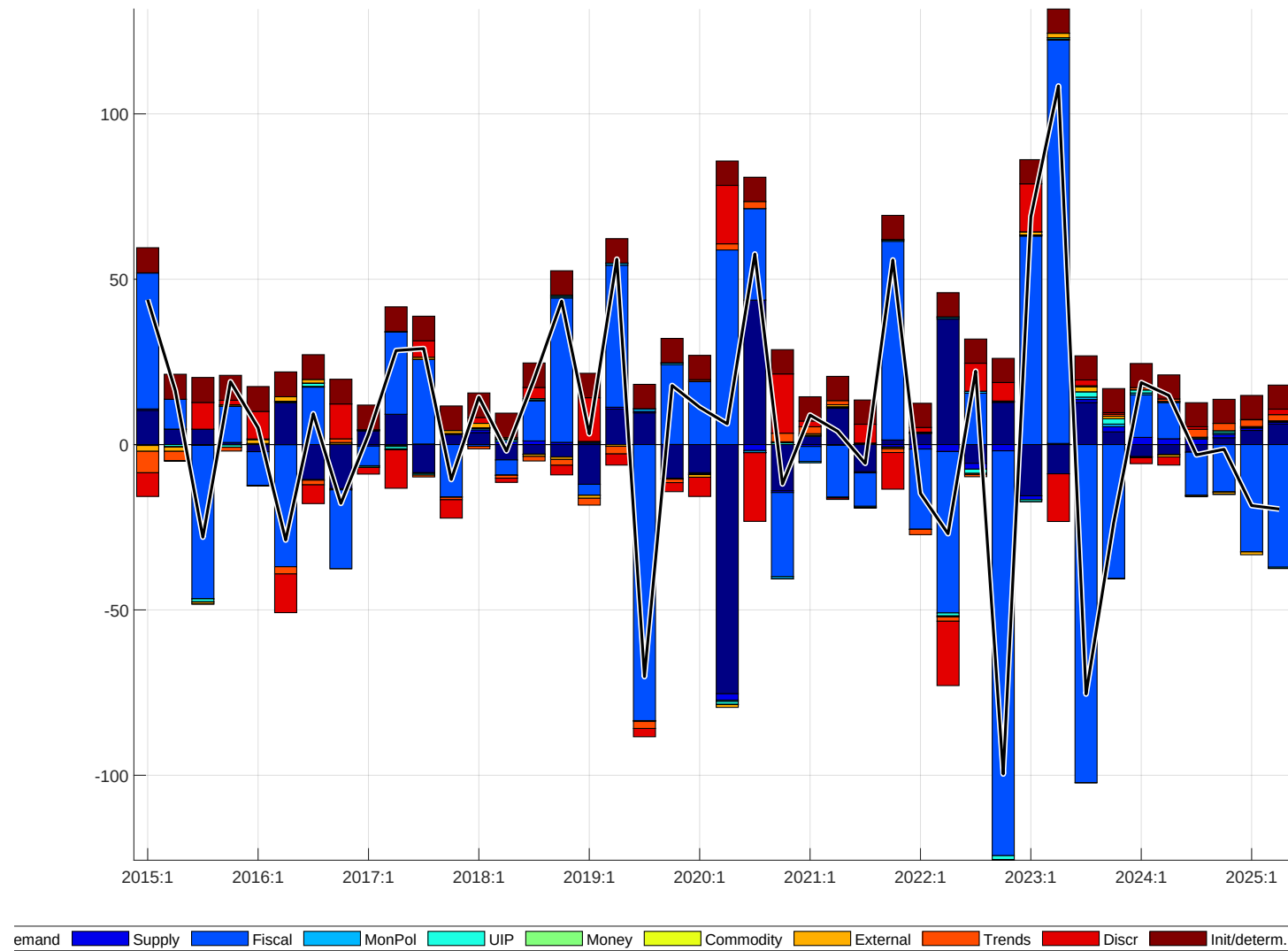
5 Shock decomposition



Figure 1 is a chart showing the decomposition of the real exchange rate (RER) for the US dollar against the Chinese yuan from 2015 to 2025. The chart displays the RER on the y-axis (ranging from -50 to 50) and time on the x-axis (labeled from 2015:1 to 2025:1). The RER is decomposed into 11 components, represented by stacked bars: Supply (dark blue), Fiscal (blue), MonPol (light blue), UIP (cyan), Money (green), Commodity (yellow-green), External (yellow), Trends (orange), Discr (red), and Init/determ. (dark red). A white line represents the total RER, which fluctuates significantly, peaking around 2016 and 2022, and reaching a sharp trough around 2020. The 'Supply' component is the most prominent, especially during the peaks and troughs.



## Gov. demand, ann. QQ % [dl\_gdem]

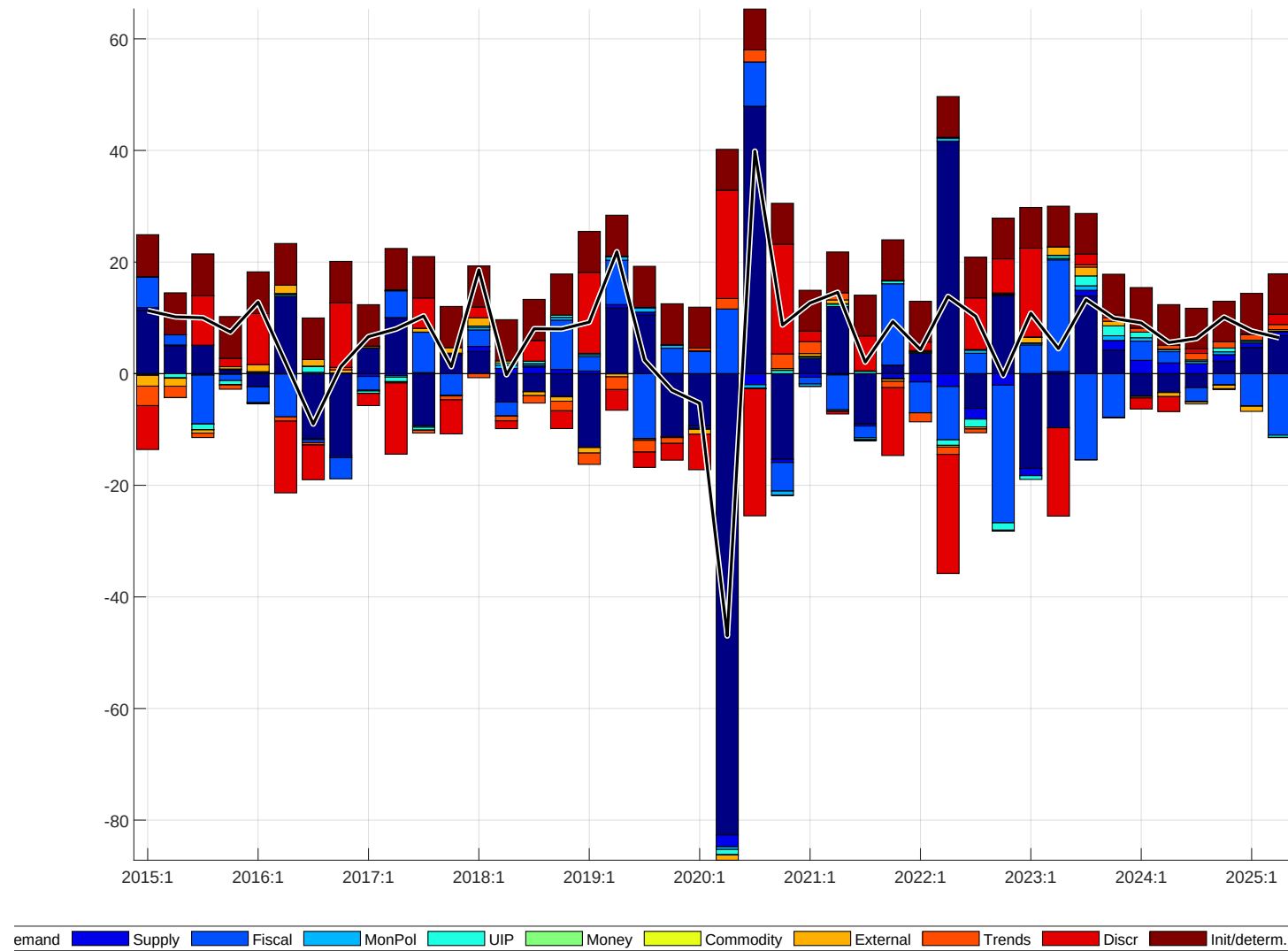




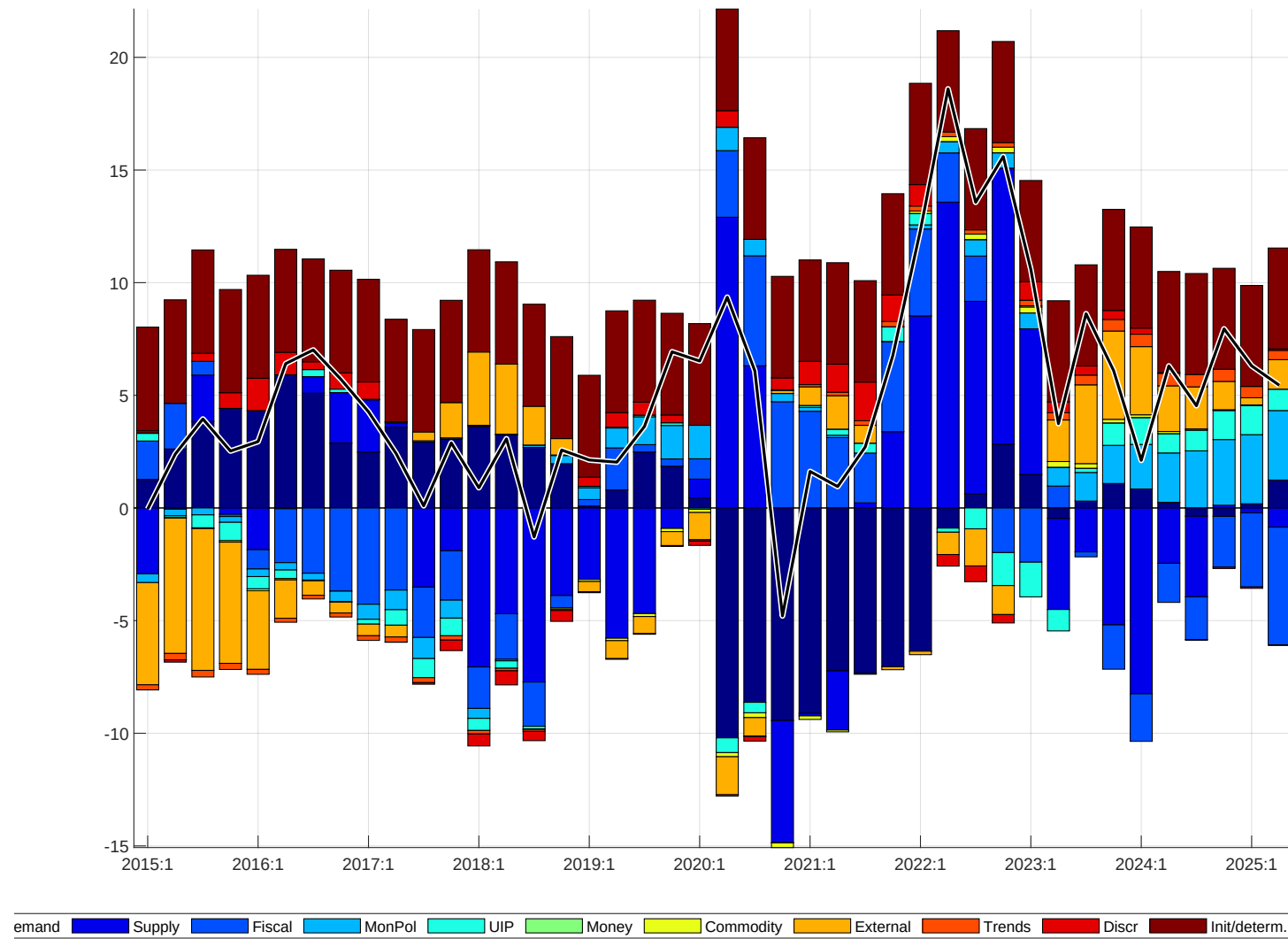


The chart displays the decomposition of the real exchange rate (RER) for the Eurozone against the US Dollar from 2015 to 2025. The y-axis represents the RER value, ranging from -150 to 100. The x-axis shows time in quarters, labeled from 2015:1 to 2025:1. The stacked bars represent different components of the RER, with the total RER shown as a white line. The components are: Supply (dark blue), Fiscal (blue), MonPol (light blue), UIP (cyan), Money (green), Commodity (yellow-green), External (yellow), Trends (orange), Discr (red), and Init/determ. (dark red). The chart shows a significant peak in the RER in early 2021, followed by a sharp decline in early 2020, and a subsequent recovery.

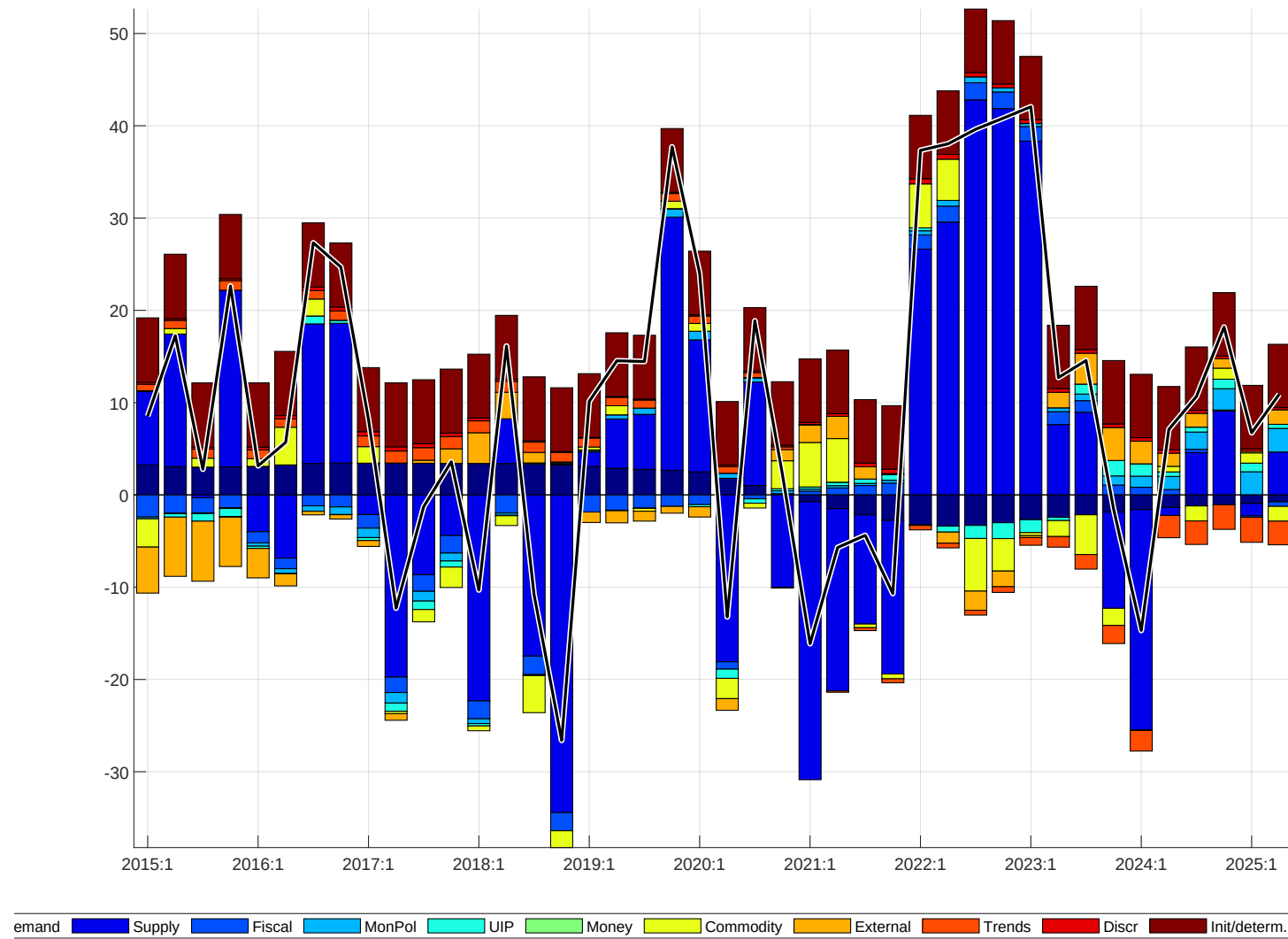
## GDP, ann. QQ % [dl\_y]



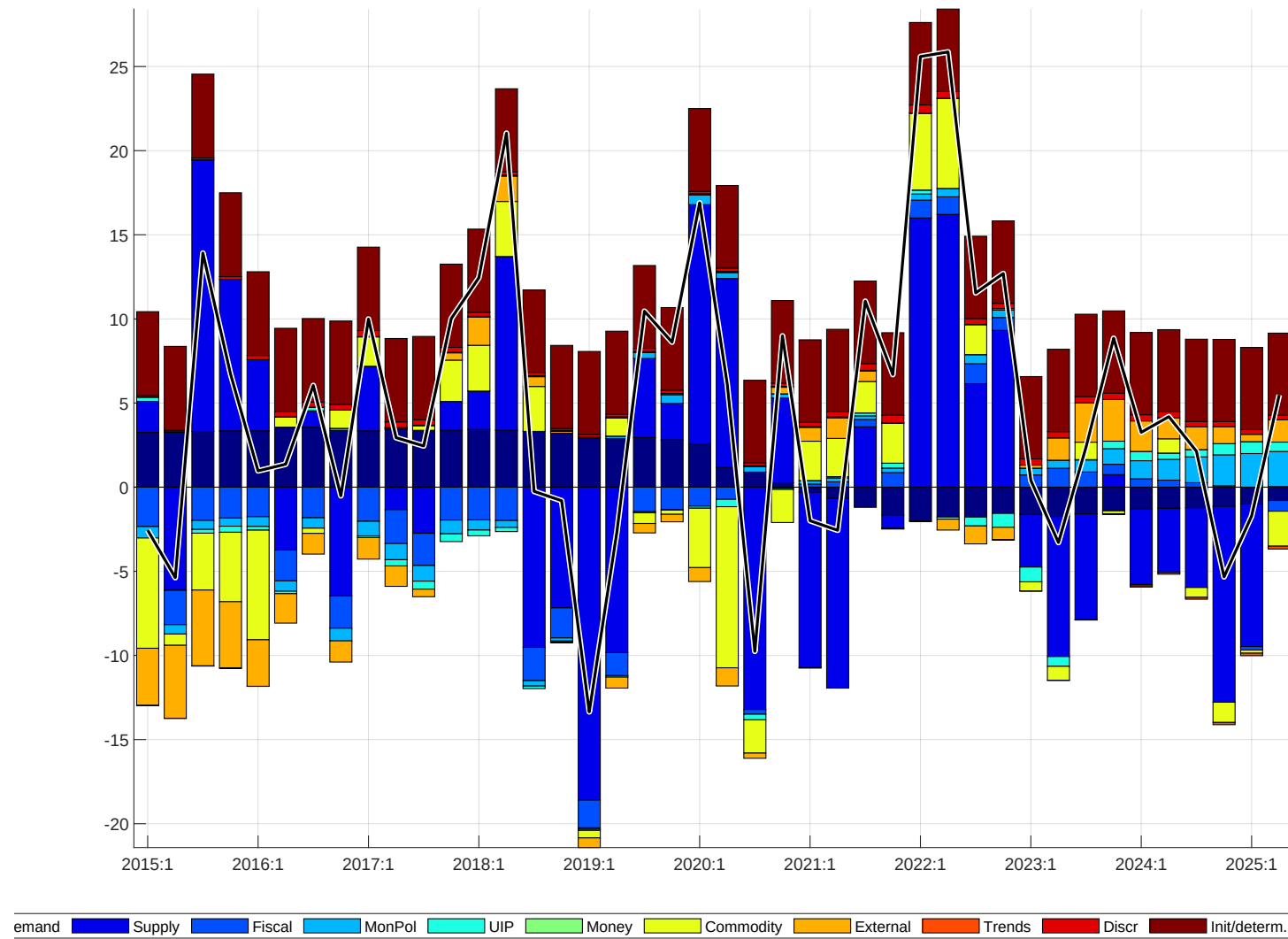
## Core CPI, ann. QQ % [dl\_cpi\_core]



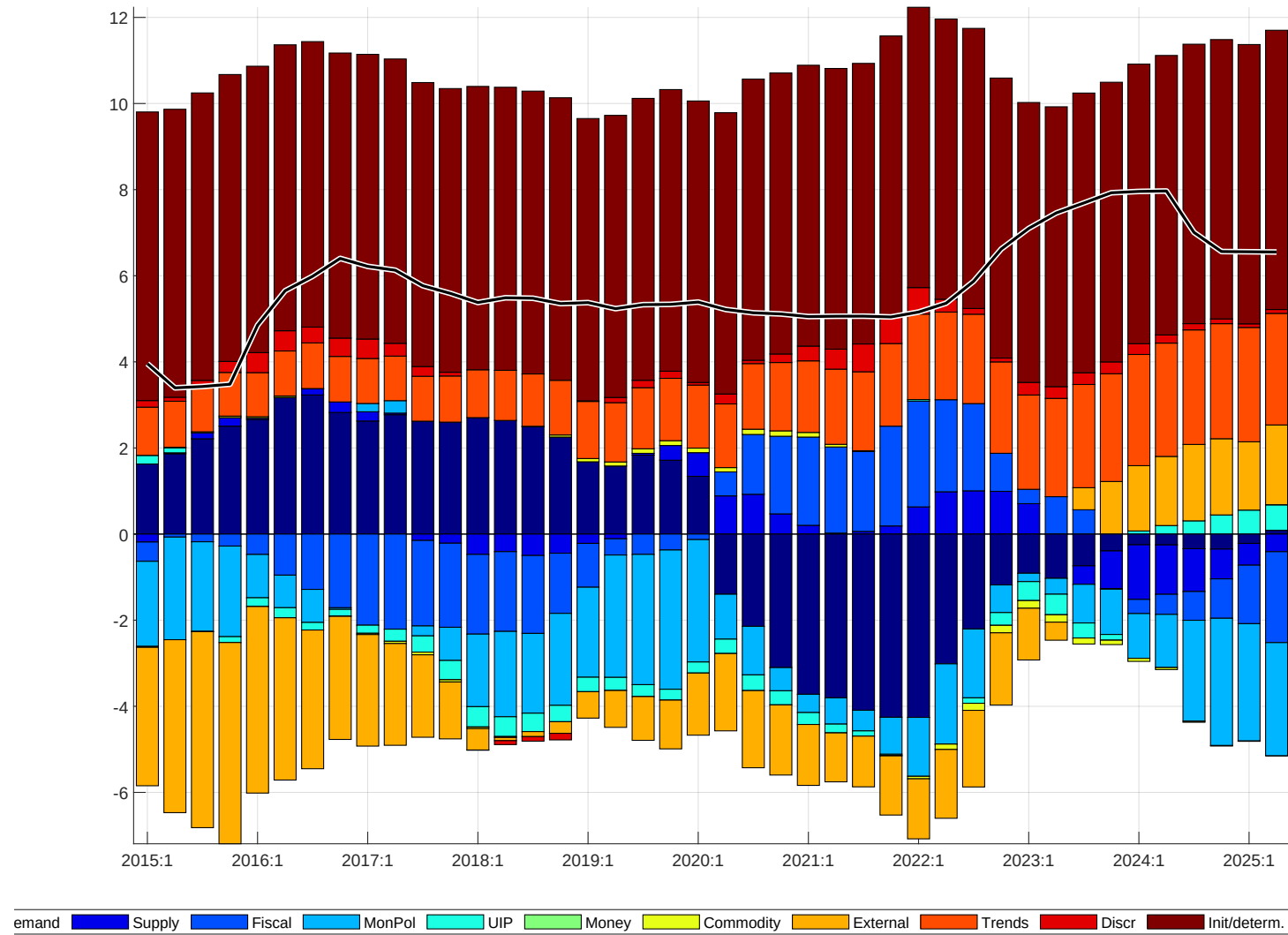
## Food CPI, ann. QQ % [dl\_cpi\_food]



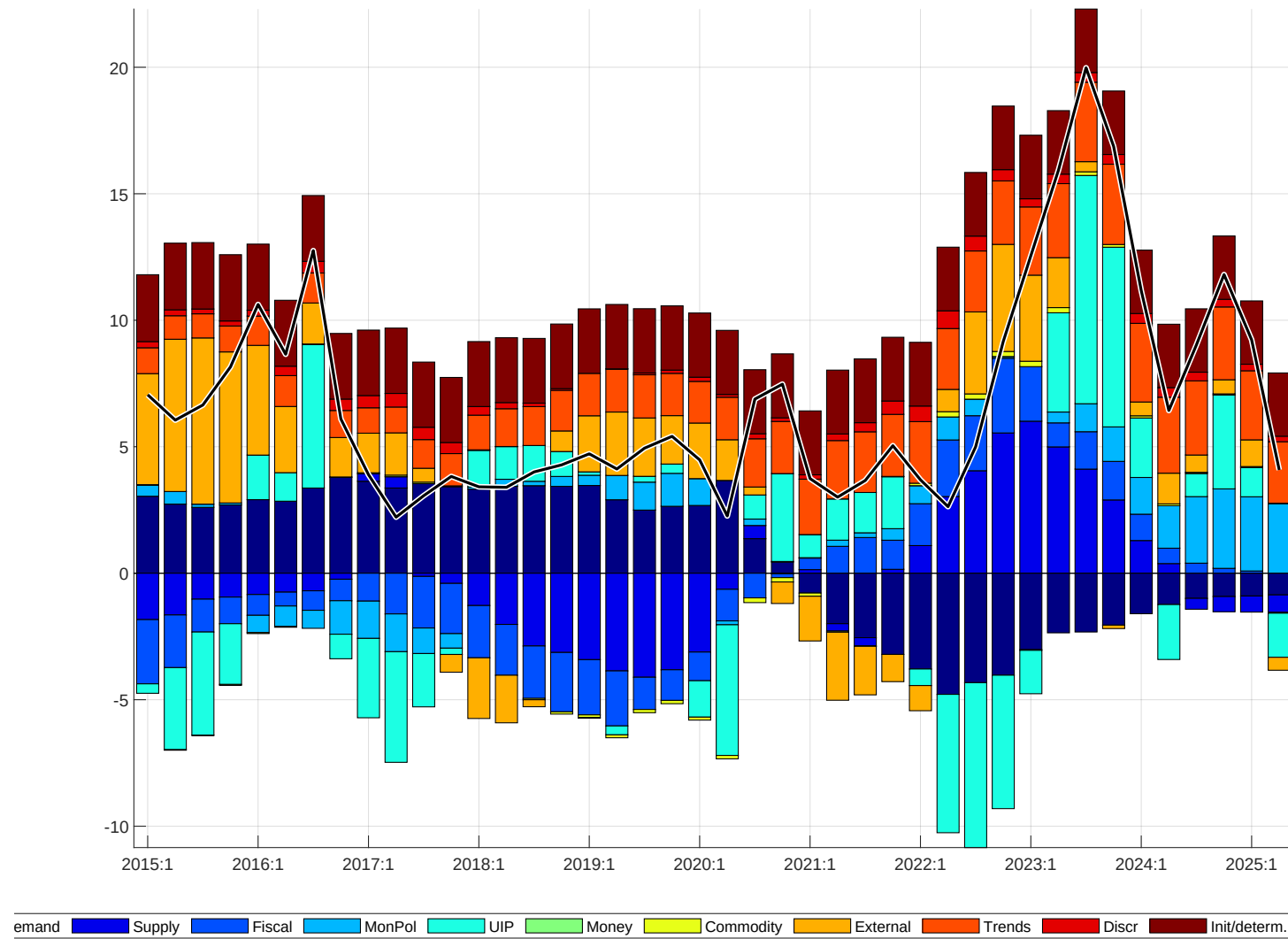
## Energy CPI, ann. QQ % [dl\_cpi\_ener]



## Policy rate (IB rate used), % [i]

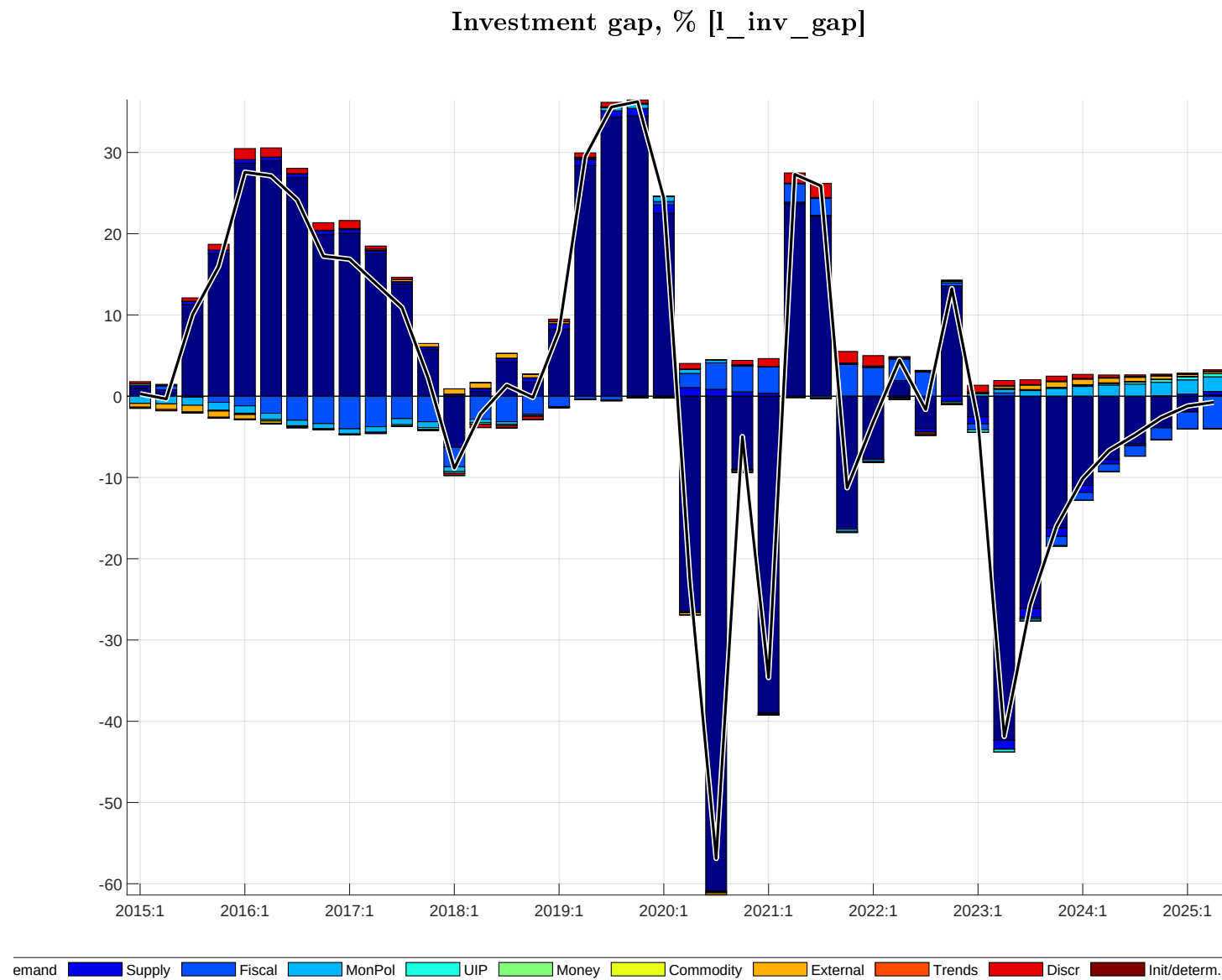


## Exchng. rate, ann. QQ % [dl\_s]

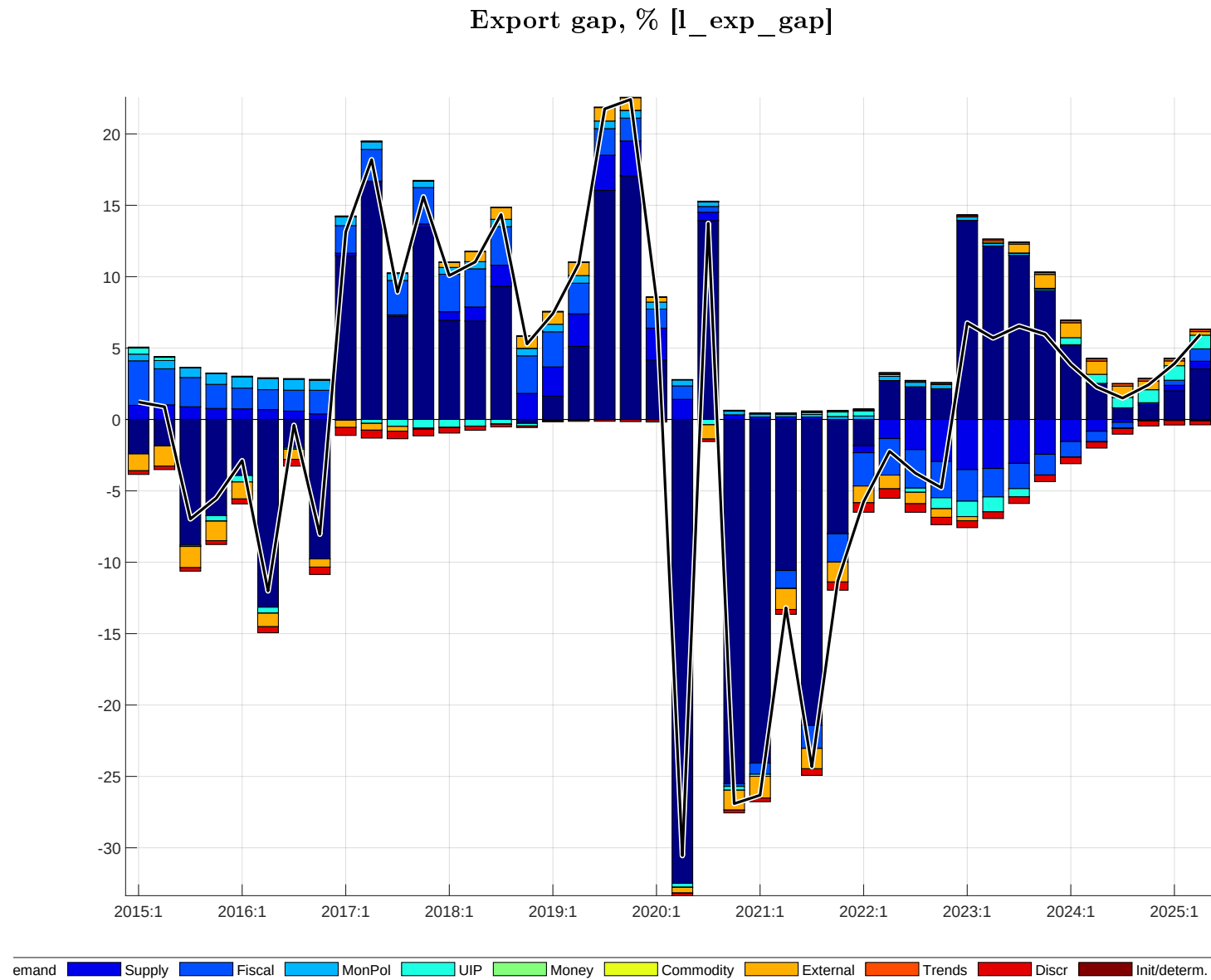




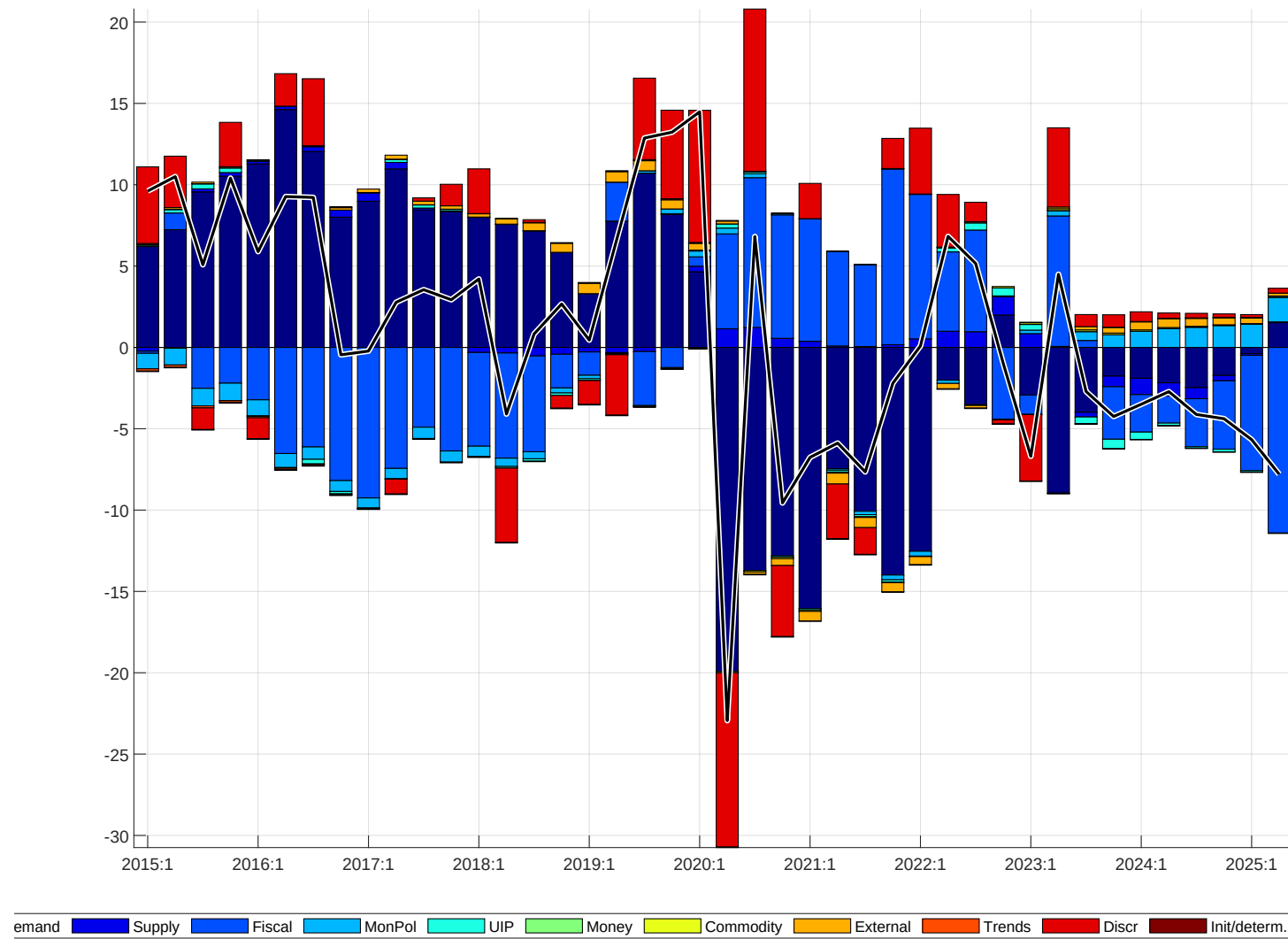


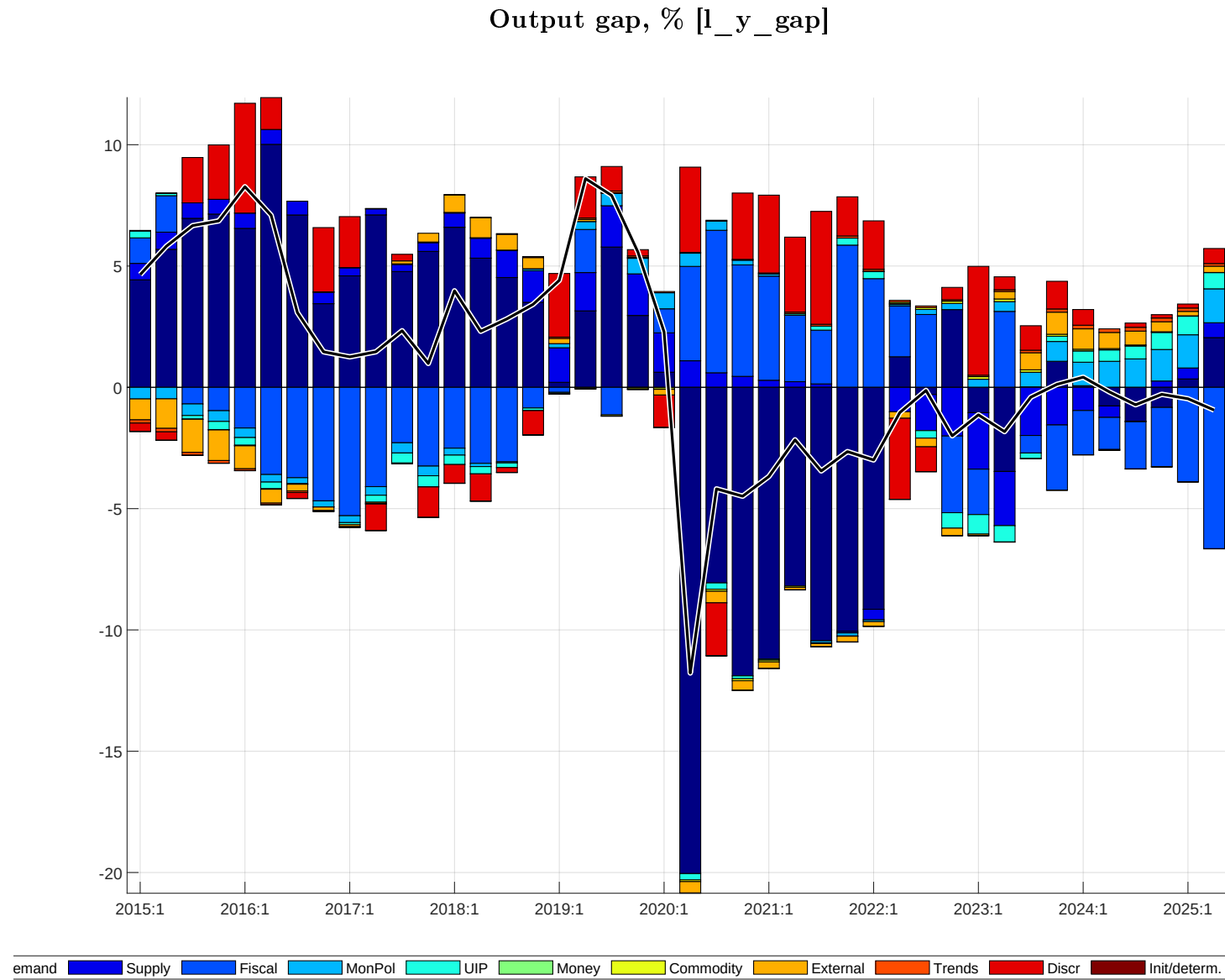




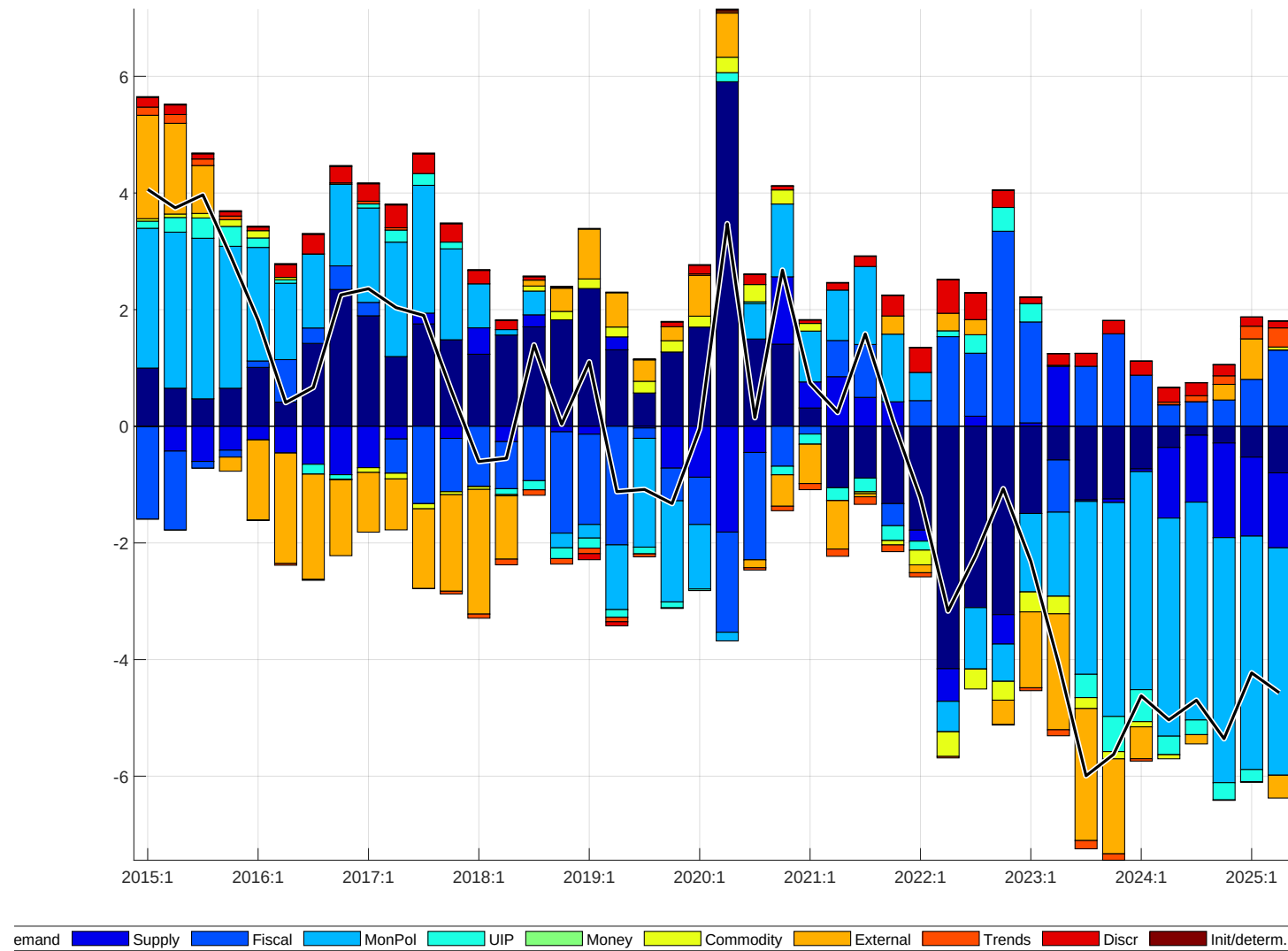


## Import gap, % [l\_imp\_gap]

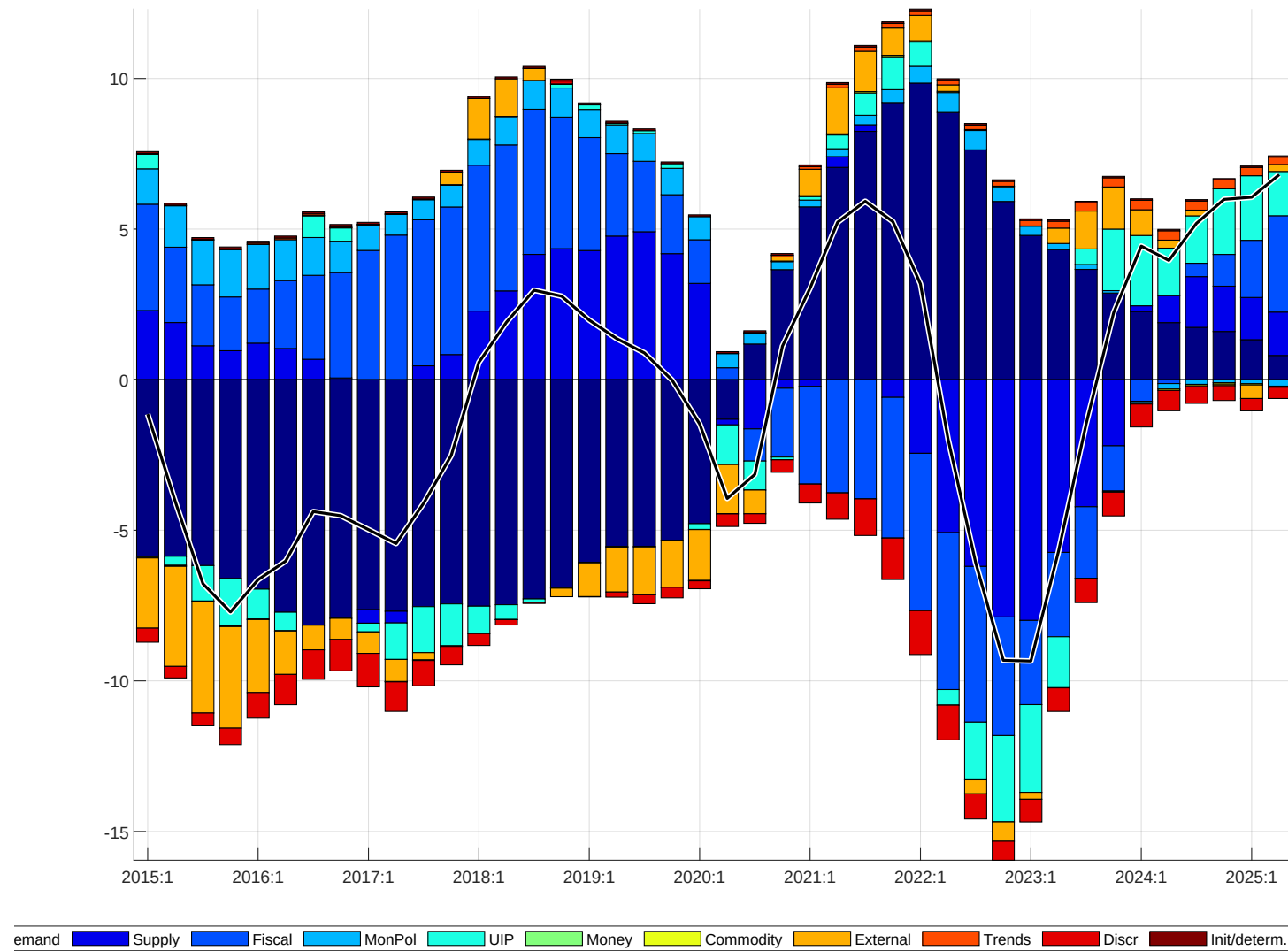




## Exptd. RIR (lending) gap, 4-Q ahead av., % [r4\_gap]



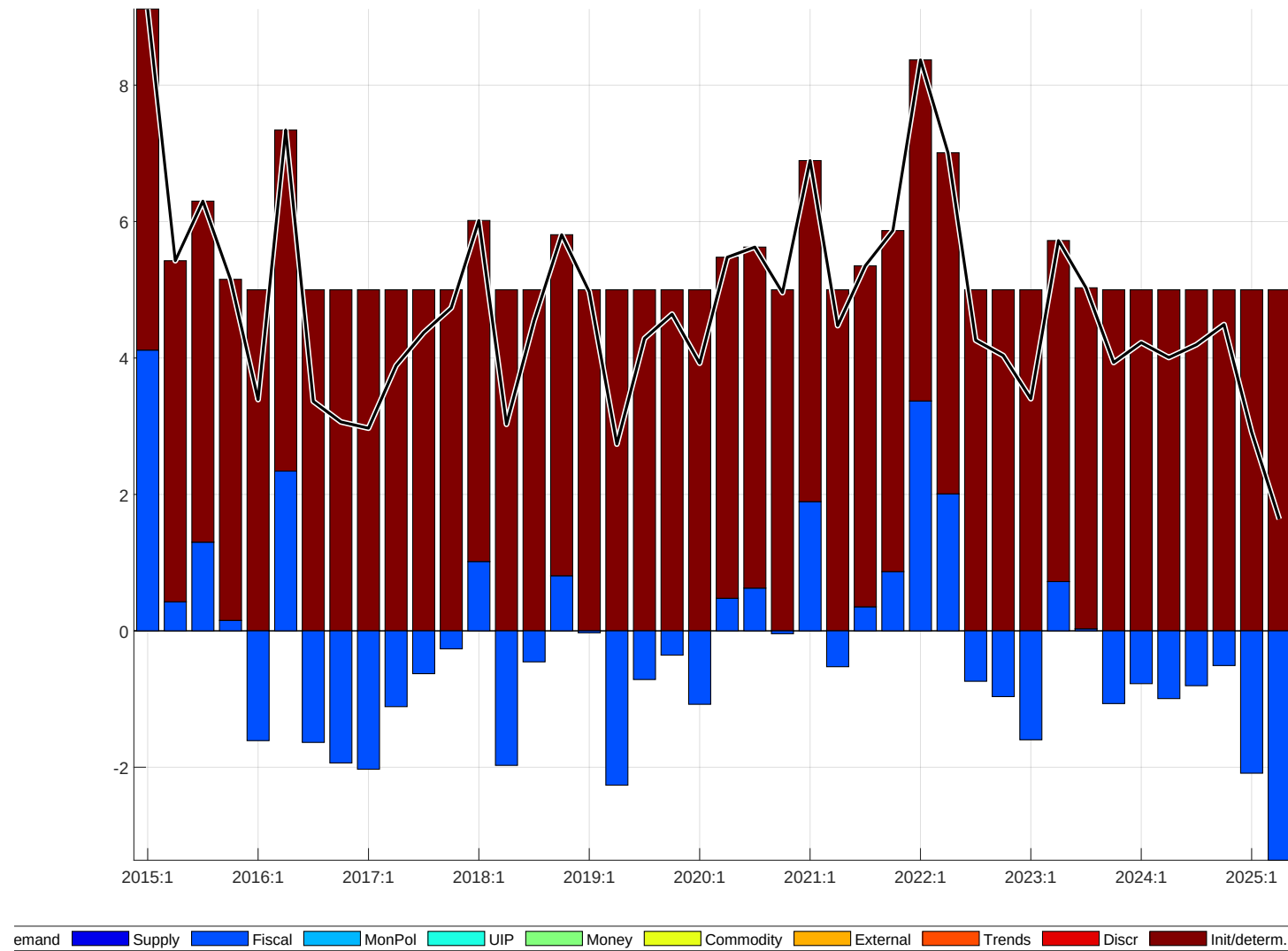
## Real exchng. rate gap, % [l\_z\_gap]







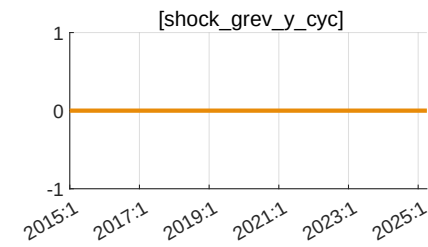
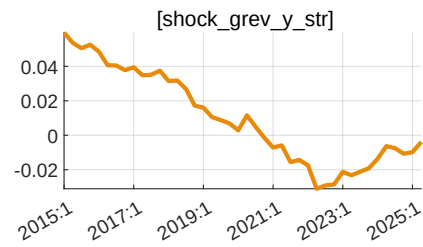
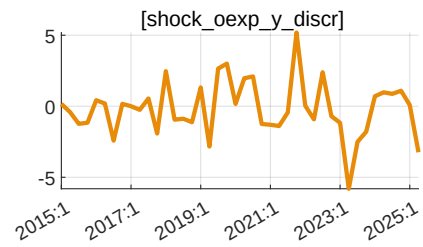
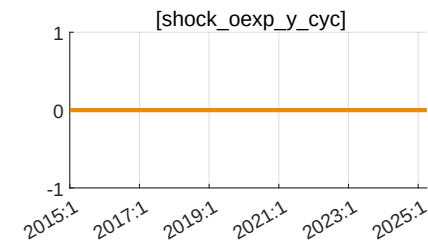
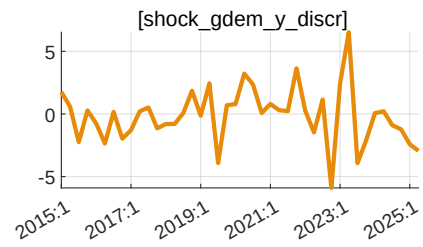
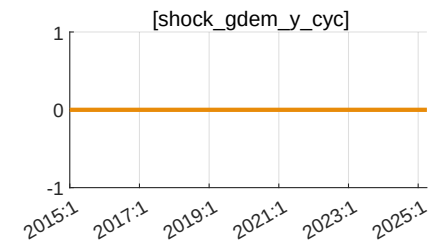
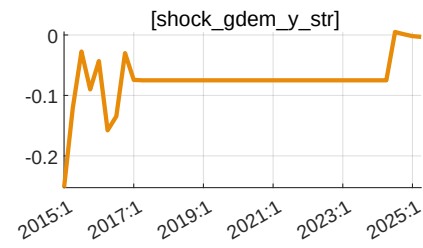
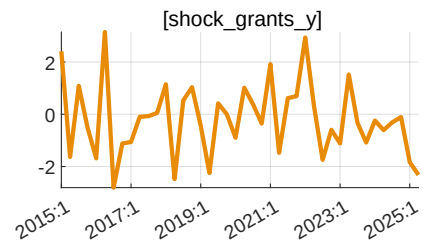
## Fiscal grants, % of GDP [grants\_y]



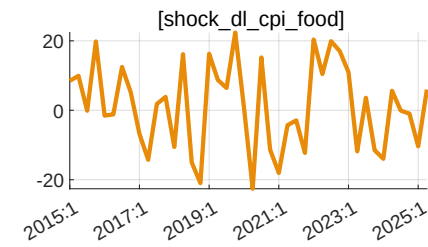
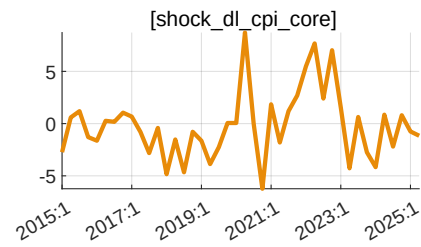
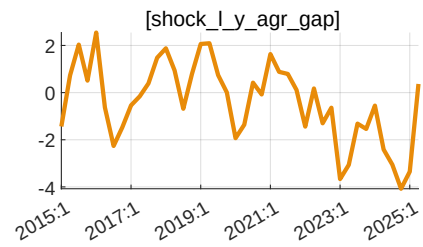
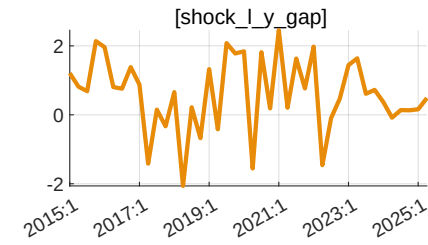
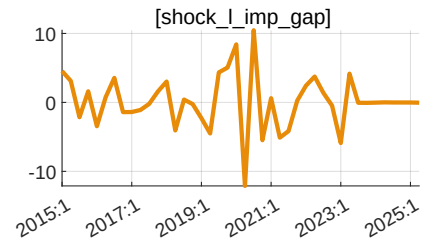
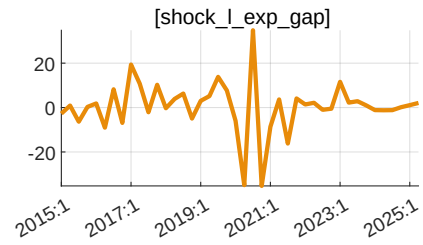
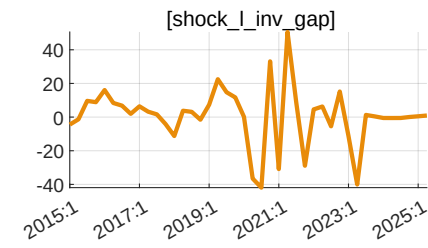
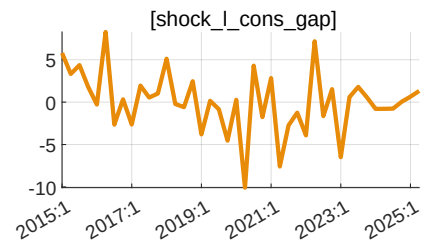
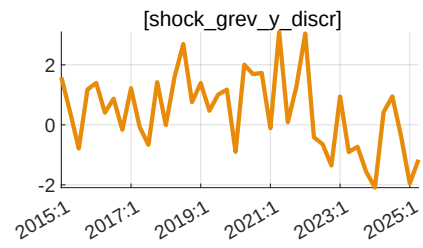
## 6 Shocks



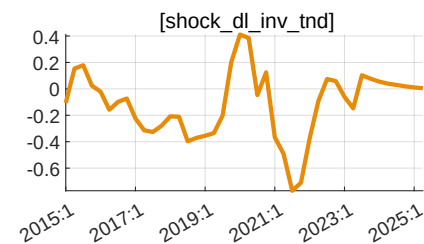
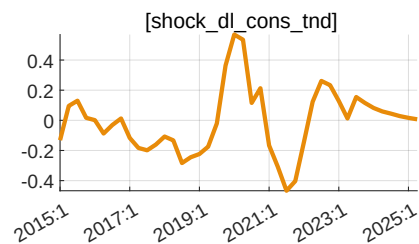
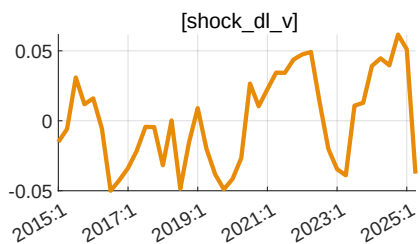
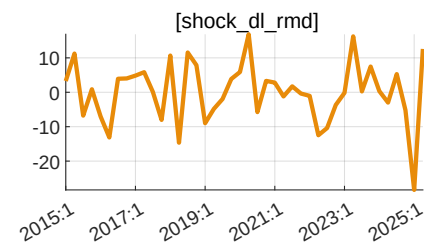
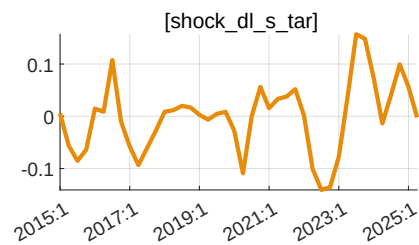
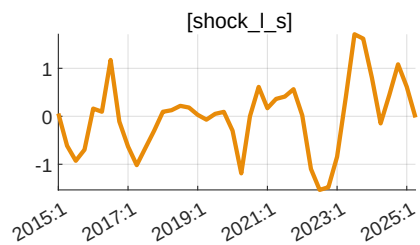
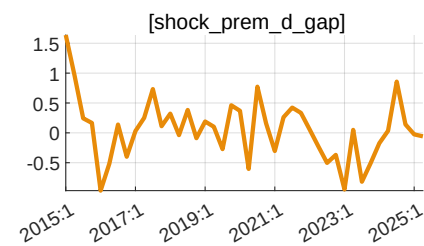
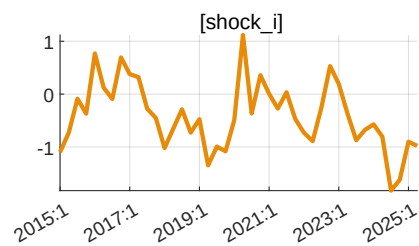
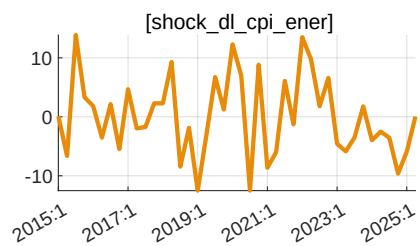
## Shocks (page 1)



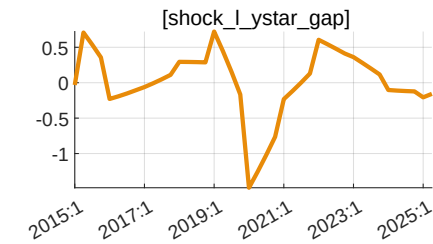
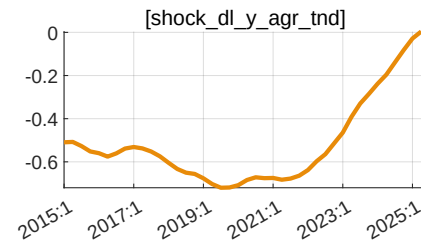
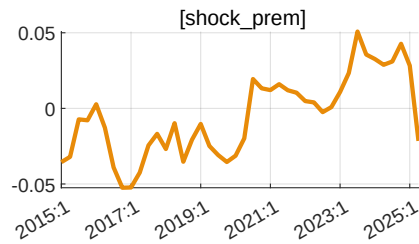
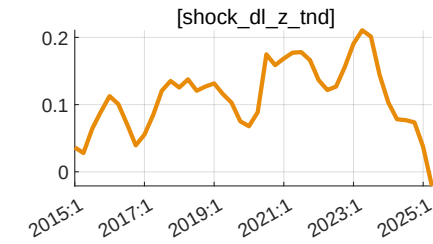
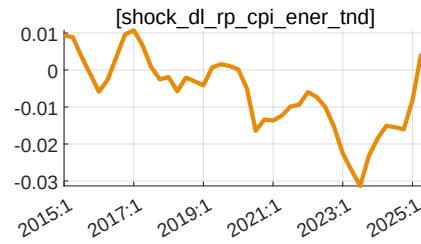
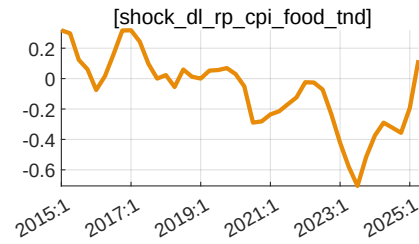
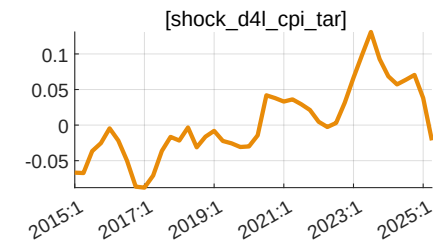
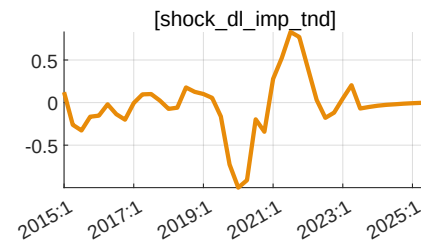
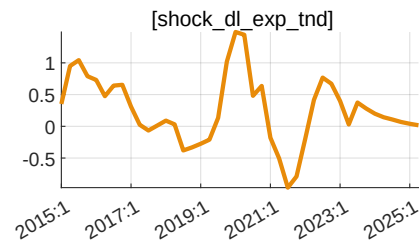
## Shocks (page 2)



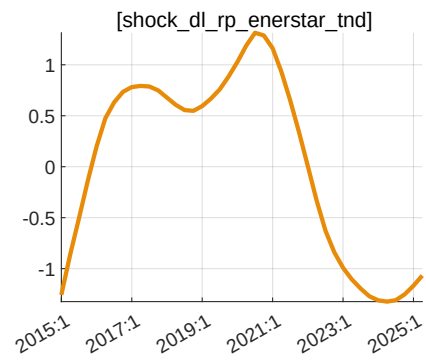
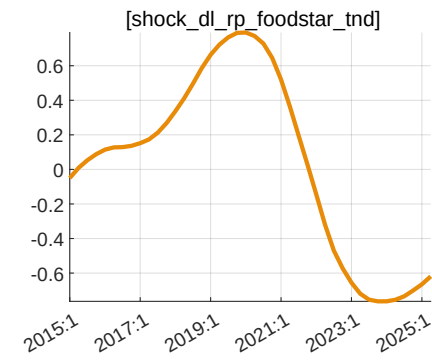
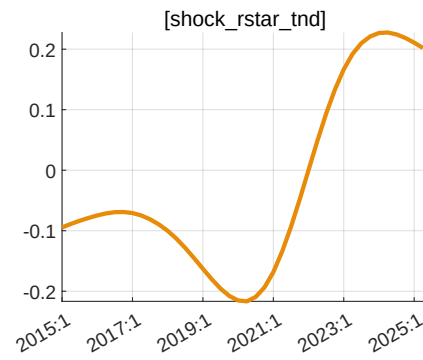
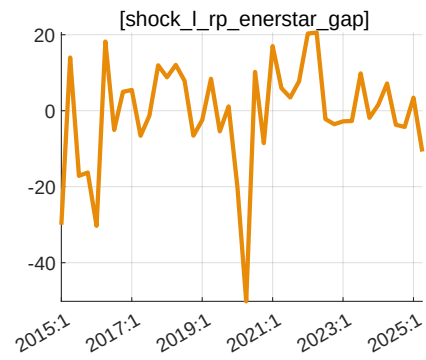
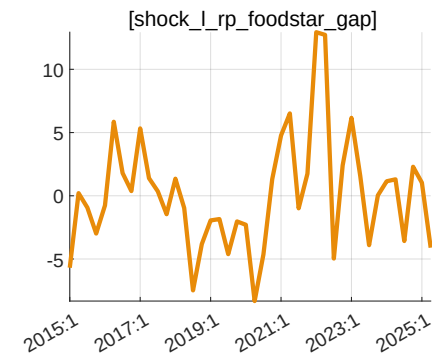
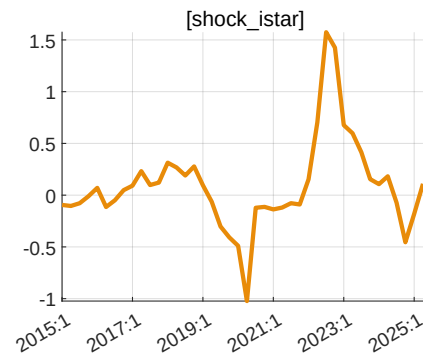
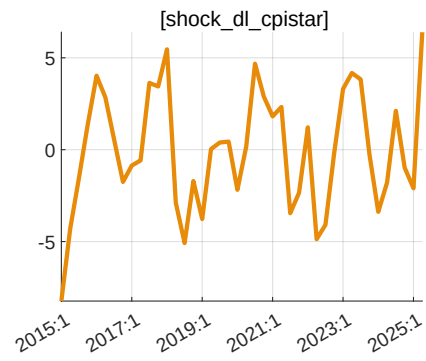
## Shocks (page 3)



## Shocks (page 4)



## Shocks (page 5)





## Shocks

|                          | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 |
|--------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| shock_grants_y           | 0.00   | -0.90  | 1.02   | 0.39   | -0.36  | 1.92   | -1.47  | 0.62   | 0.69   | 2.94   | 0.32   | -1.74  | -0.59  | -1.12  | 1.52   | -0.33  | -1.08  |
| shock_gdem_y_str         | -0.08  | -0.07  | -0.08  | -0.07  | -0.08  | -0.07  | -0.07  | -0.07  | -0.07  | -0.08  | -0.07  | -0.07  | -0.08  | -0.07  | -0.08  | -0.07  | -0.07  |
| shock_gdem_y_cyc         | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   |
| shock_gdem_y_discr       | 0.70   | 0.79   | 3.23   | 2.36   | 0.08   | 0.80   | 0.31   | 0.23   | 3.64   | 0.26   | -1.47  | 1.15   | -5.90  | 2.44   | 6.52   | -3.93  | -2.08  |
| shock_oexp_y_str         | 0.05   | 0.02   | -0.00  | -0.02  | -0.04  | -0.03  | -0.02  | -0.01  | -0.01  | -0.07  | -0.07  | -0.06  | -0.09  | -0.09  | -0.09  | -0.03  | -0.01  |
| shock_oexp_y_cyc         | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   |
| shock_oexp_y_discr       | 3.00   | 0.16   | 1.97   | 2.10   | -1.25  | -1.30  | -1.39  | -0.43  | 5.19   | 0.04   | -0.92  | 2.39   | -0.70  | -1.17  | -5.81  | -2.51  | -1.79  |
| shock_grev_y_str         | 0.01   | 0.00   | 0.01   | 0.00   | -0.00  | -0.01  | -0.01  | -0.02  | -0.01  | -0.02  | -0.03  | -0.03  | -0.03  | -0.02  | -0.02  | -0.02  | -0.02  |
| shock_grev_y_cyc         | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   | 0.00   |
| shock_grev_y_discr       | 1.18   | -0.90  | 2.00   | 1.69   | 1.73   | -0.11  | 3.09   | 0.09   | 1.30   | 3.04   | -0.42  | -0.65  | -1.36  | 0.95   | -0.91  | -0.73  | -1.56  |
| shock_l_cons_gap         | -4.54  | 0.27   | -10.08 | 4.30   | -1.77  | 2.83   | -7.59  | -2.74  | -1.25  | -3.92  | 7.17   | -1.64  | 1.53   | -6.49  | 0.60   | 1.79   | 0.56   |
| shock_l_inv_gap          | 11.70  | 0.16   | -36.54 | -41.92 | 33.12  | -30.88 | 50.57  | 8.77   | -28.90 | 4.55   | 6.29   | -5.47  | 15.19  | -11.28 | -40.05 | 1.27   | 0.42   |
| shock_l_exp_gap          | 7.71   | -5.96  | -34.94 | 34.81  | -35.29 | -8.64  | 3.66   | -16.21 | 4.10   | 1.39   | 2.15   | -1.01  | -0.52  | 11.59  | 2.22   | 2.86   | 0.98   |
| shock_l_imp_gap          | 5.05   | 8.42   | -12.11 | 10.46  | -5.48  | 0.61   | -5.11  | -4.16  | 0.26   | 2.42   | 3.75   | 1.38   | -0.47  | -5.89  | 4.17   | -0.05  | -0.06  |
| shock_l_y_gap            | 1.78   | 1.84   | -1.56  | 1.82   | 0.19   | 2.45   | 0.21   | 1.63   | 0.76   | 1.98   | -1.46  | -0.10  | 0.47   | 1.44   | 1.64   | 0.61   | 0.72   |
| shock_l_y_agr_gap        | 0.01   | -1.93  | -1.36  | 0.42   | -0.08  | 1.63   | 0.88   | 0.79   | 0.11   | -1.44  | 0.17   | -1.30  | -0.64  | -3.67  | -3.06  | -1.32  | -1.54  |
| shock_dl_cpi_core        | 0.06   | 0.06   | 8.72   | -0.02  | -6.24  | 1.85   | -1.81  | 1.20   | 2.69   | 5.48   | 7.68   | 2.40   | 7.02   | 1.54   | -4.28  | 0.63   | -2.75  |
| shock_dl_cpi_food        | 22.39  | 0.82   | -22.60 | 15.21  | -11.33 | -18.05 | -4.29  | -2.92  | -12.28 | 20.38  | 10.44  | 19.91  | 17.02  | 10.98  | -11.82 | 3.60   | -11.50 |
| shock_dl_cpi_ener        | 1.23   | 12.29  | 6.97   | -12.51 | 8.83   | -8.64  | -6.00  | 6.09   | -1.28  | 13.51  | 9.78   | 1.80   | 6.60   | -4.57  | -5.85  | -3.52  | 1.77   |
| shock_i                  | -1.08  | -0.49  | 1.12   | -0.37  | 0.36   | 0.01   | -0.27  | 0.03   | -0.47  | -0.73  | -0.89  | -0.26  | 0.53   | 0.19   | -0.37  | -0.87  | -0.68  |
| shock_prem_d_gap         | 0.46   | 0.37   | -0.60  | 0.77   | 0.16   | -0.30  | 0.26   | 0.42   | 0.33   | 0.06   | -0.23  | -0.50  | -0.37  | -0.95  | 0.05   | -0.82  | -0.51  |
| shock_l_s                | 0.09   | -0.30  | -1.19  | 0.00   | 0.61   | 0.17   | 0.36   | 0.41   | 0.56   | 0.02   | -1.10  | -1.53  | -1.48  | -0.84  | 0.39   | 1.71   | 1.62   |
| shock_dl_s_tar           | 0.01   | -0.03  | -0.11  | 0.00   | 0.06   | 0.02   | 0.03   | 0.04   | 0.05   | 0.00   | -0.10  | -0.14  | -0.14  | -0.08  | 0.04   | 0.16   | 0.15   |
| shock_dl_rmd             | 3.86   | 5.89   | 16.86  | -5.76  | 3.31   | 2.81   | -1.21  | 1.72   | -0.37  | -1.04  | -12.48 | -10.39 | -3.68  | -0.16  | 16.25  | 0.25   | 7.47   |
| shock_dl_v               | -0.05  | -0.04  | -0.03  | 0.03   | 0.01   | 0.02   | 0.03   | 0.03   | 0.04   | 0.05   | 0.05   | 0.01   | -0.02  | -0.03  | -0.04  | 0.01   | 0.01   |
| shock_dl_cons_tnd        | 0.36   | 0.57   | 0.54   | 0.12   | 0.21   | -0.17  | -0.31  | -0.47  | -0.40  | -0.14  | 0.12   | 0.26   | 0.23   | 0.13   | 0.01   | 0.15   | 0.12   |
| shock_dl_inv_tnd         | 0.20   | 0.41   | 0.39   | -0.05  | 0.12   | -0.37  | -0.49  | -0.77  | -0.71  | -0.38  | -0.09  | 0.07   | 0.06   | -0.06  | -0.15  | 0.10   | 0.08   |
| shock_dl_exp_tnd         | 1.02   | 1.49   | 1.44   | 0.48   | 0.64   | -0.18  | -0.51  | -0.97  | -0.79  | -0.20  | 0.42   | 0.77   | 0.67   | 0.40   | 0.03   | 0.37   | 0.28   |
| shock_dl_imp_tnd         | -0.73  | -1.00  | -0.91  | -0.20  | -0.34  | 0.28   | 0.52   | 0.83   | 0.77   | 0.40   | 0.03   | -0.18  | -0.12  | 0.05   | 0.20   | -0.07  | -0.05  |
| shock_d4l_cpi_tar        | -0.03  | -0.03  | -0.01  | 0.04   | 0.04   | 0.03   | 0.04   | 0.03   | 0.02   | 0.00   | -0.00  | 0.00   | 0.03   | 0.07   | 0.10   | 0.13   | 0.09   |
| shock_dl_rp_cpi_food_tnd | 0.07   | 0.03   | -0.05  | -0.29  | -0.28  | -0.24  | -0.21  | -0.17  | -0.12  | -0.02  | -0.03  | -0.07  | -0.24  | -0.42  | -0.58  | -0.71  | -0.51  |
| shock_dl_rp_cpi_ener_tnd | 0.00   | 0.00   | -0.01  | -0.02  | -0.01  | -0.01  | -0.01  | -0.01  | -0.01  | -0.01  | -0.01  | -0.01  | -0.02  | -0.02  | -0.03  | -0.03  | -0.02  |
| shock_dl_z_tnd           | 0.08   | 0.07   | 0.09   | 0.17   | 0.16   | 0.17   | 0.18   | 0.18   | 0.17   | 0.14   | 0.12   | 0.13   | 0.16   | 0.19   | 0.21   | 0.20   | 0.14   |
| shock_prem               | -0.04  | -0.03  | -0.02  | 0.02   | 0.01   | 0.01   | 0.02   | 0.01   | 0.01   | 0.00   | 0.00   | -0.00  | 0.00   | 0.01   | 0.02   | 0.05   | 0.04   |
| shock_dl_y_agr_tnd       | -0.72  | -0.71  | -0.68  | -0.67  | -0.68  | -0.67  | -0.68  | -0.68  | -0.66  | -0.64  | -0.60  | -0.56  | -0.51  | -0.46  | -0.39  | -0.33  | -0.29  |

|                          | 2019Q4 | 2020Q1 | 2020Q2 | 2020Q3 | 2020Q4 | 2021Q1 | 2021Q2 | 2021Q3 | 2021Q4 | 2022Q1 | 2022Q2 | 2022Q3 | 2022Q4 | 2023Q1 | 2023Q2 | 2023Q3 | 2023Q4 |
|--------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| shock_l_ystar_gap        | −0.17  | −1.48  | −1.26  | −1.02  | −0.76  | −0.23  | −0.12  | 0.00   | 0.13   | 0.61   | 0.54   | 0.48   | 0.41   | 0.36   | 0.28   | 0.20   | 0.12   |
| shock_dl_cpistar         | 0.44   | −2.18  | 0.17   | 4.68   | 2.89   | 1.82   | 2.32   | −3.46  | −2.34  | 1.22   | −4.87  | −4.09  | −0.15  | 3.31   | 4.17   | 3.83   | −0.26  |
| shock_istar              | −0.41  | −0.49  | −1.02  | −0.12  | −0.11  | −0.14  | −0.12  | −0.08  | −0.09  | 0.15   | 0.70   | 1.57   | 1.43   | 0.68   | 0.60   | 0.41   | 0.15   |
| shock_l_rp_foodstar_gap  | −2.03  | −2.29  | −8.33  | −4.63  | 1.32   | 4.77   | 6.51   | −0.99  | 1.75   | 12.93  | 12.72  | −4.96  | 2.41   | 6.16   | 1.55   | −3.91  | 0.03   |
| shock_l_rp_enerstar_gap  | 1.13   | −20.06 | −50.17 | 10.22  | −8.53  | 17.02  | 5.94   | 3.49   | 7.73   | 20.31  | 20.58  | −2.20  | −3.55  | −2.81  | −2.70  | 9.82   | −1.87  |
| shock_rstar_tnd          | −0.21  | −0.22  | −0.22  | −0.21  | −0.19  | −0.17  | −0.13  | −0.09  | −0.05  | 0.00   | 0.05   | 0.09   | 0.13   | 0.17   | 0.19   | 0.21   | 0.22   |
| shock_dl_rp_foodstar_tnd | 0.79   | 0.79   | 0.77   | 0.73   | 0.65   | 0.52   | 0.36   | 0.20   | 0.03   | −0.14  | −0.32  | −0.47  | −0.57  | −0.66  | −0.72  | −0.75  | −0.76  |
| shock_dl_rp_enerstar_tnd | 0.89   | 1.03   | 1.19   | 1.31   | 1.29   | 1.16   | 0.93   | 0.65   | 0.34   | 0.00   | −0.33  | −0.63  | −0.84  | −0.99  | −1.11  | −1.20  | −1.27  |